A Conditional Derivation of the Point-Particle 2.5PN Sector from the Unified 4D Toy Model

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Abstract

We present a controlled audit of the first dissipative post-Newtonian gate of the unified 4D toy model, namely the nonspinning two-body 2.5PN sector. Carrying forward the exact projection / leakage / Poisson-hook / worldtube reduction framework together with the already-fixed Newtonian+1PN+2PN conservative hierarchy, we first show that the frozen conservative kernels alone cannot generate a local 2.5PN reaction term. We then use a minimal retarded quadrupole prototype to recover the Burke–Thorne member of the standard Iyer–Will family, which fixes the structural target for any successful completion. The main body of the paper is a channel audit. On the natural compact/passive/outgoing small-body branch, the dangerous lower odd scalar channels are derivative-mediated or super-Ohmic and are pushed above universal point-particle 2.5PN; the carried dipole wake is likewise demoted by the outgoing $\ell = 1$ threshold law and strict small-body scaling. The only surviving universal point-particle route is the orbital/worldtube STF quadrupole. In particular, the solved 2PN P_2 bundle already exhausts the real STF $\ell = 2$ content, the low-frequency $P_2 \to$ STF matching is scalar under isotropy, the passive outgoing $\ell = 2$ branch carries the correct positive $+i\omega^5$ parity/sign, and the static overlap gate passes on the natural branch. The strongest honest result is therefore a *conditional* 2.5PN theorem within a declared closure hierarchy: if the completed moving-throat PDE realizes this passive/outgoing quadrupole branch, then the reduced nonspinning two-body dynamics contains the standard local quadrupolar Burke–Thorne / Iyer–Will reaction term, with all other currently derived odd channels entering only as higher-order finite-size corrections. The remaining serious open problem is no longer generic dissipative physics, but the final passive/outgoing quadrupole normalization, $\hat{m}_0^2\Gamma_5 = 2G/(5c^5)$, on the actual moving-throat branch.

1 Introduction and scope

1.1 Motivation: from full conservative 2PN to the first dissipative gate

The earlier papers in this program already established a strong conservative backbone. On the reduction side, the unified 4D framework supplies the exact projection identities, the projected continuity law with leakage, the controlled Poisson hook, and the worldtube/coherent-defect reduction used to define the brane-facing point-particle sector. On the post-Newtonian side, the carried lower-order ledger fixes the Newtonian and conservative 1PN and 2PN blocks within a declared closure hierarchy, with the surviving viability conditions

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\text{add}} = \frac{1}{2}, \qquad \kappa_{\text{PV}} = \frac{3}{2}, \qquad \beta_{1\text{PN}} = 3, \qquad \frac{L}{a} \approx 1.85.$$

Just as important, the conservative 2PN assembly sharpened the ontology of the particle itself. The defect can no longer be treated as a pure scalar monopole with small corrections; the solved conservative cross sector already looks like a small throat-response theory with a carried odd dipole wake, a genuine $P_0 \oplus P_2$ mouth/support layer, and a separate geometry-closure channel.

The natural next pressure test is therefore not another conservative coefficient match, but the first order at which a GR-like theory must become genuinely retarded and time-asymmetric. In the standard nonspinning two-body problem, that gate is 2.5PN. It is the first place where one must decide whether the model can support a universal radiation-reaction sector rather than only a conservative near-zone ledger. For the present program this question is even sharper, because the fully dynamical moving-throat PDE is not yet solved. The 2.5PN problem is therefore also a diagnostic: it tells us which low-frequency retarded structures the missing PDE must realize if the model is to remain compatible with the GR-like point-particle target.

That is the purpose of the present paper. We do not ask whether one can write some dissipative correction by hand. We ask a more constrained question: whether the current stack, together with the natural compact/passive/outgoing small-body branch suggested by the audits, can support the correct *type* of point-particle 2.5PN sector. Concretely, the theorem gate is whether the lower odd channels are absent or demoted, whether the universal orbital/worldtube STF quadrupole survives, and whether the completed PDE can supply the final normalization of its retarded coefficient.

A first structural consequence is immediate. The frozen conservative hierarchy is real-even in the low-frequency sense, so by itself it cannot generate a nonzero local 2.5PN reaction term. Any successful completion must therefore contain a genuinely retarded sector. The decisive constructive benchmark is the minimal quadrupole odd term

$$K^{\mathrm{ret}}(\omega) = K^{\mathrm{even}}(\omega) + i\,\Gamma_5\omega^5 + O(\omega^7),$$

whose local reduction reproduces the standard Burke–Thorne / Iyer–Will radiation-reaction structure. The point of the present paper is not yet to claim that the full PDE has been solved and this coefficient derived from first principles. The point is to show that the entire 2.5PN problem has now been narrowed to a sharply stated retarded channel audit and, after that audit, to a single remaining normalization problem.

This paper therefore has a deliberately intermediate but strong scope. It is narrower than the full research record because the main text is organized around the smallest theorem chain needed for the current 2.5PN claim. It is stronger than a notebook-level target match because that chain is already sufficient to show, on the natural small-body passive/outgoing branch, that the universal point-particle 2.5PN route is quadrupolar and that the surviving unresolved question is no longer generic dissipative physics, but the final far-zone normalization of that branch.

A second purpose of the paper is bookkeeping clarity. In the 4D framework, projection is not a harmless relabeling; it naturally produces exchange, response, and closure terms. At 2.5PN the same caution becomes even more important, because the argument mixes carried exact identities, reduced-regime small-body statements, protocol-dependent retarded closures, and final theorem consequences. The reader should be able to see, line by line, which parts of the argument are inherited exactly, which parts depend on the declared low-frequency/small-body hierarchy, and which parts remain conditional on the completed PDE. That separation is part of the result.

1.2 What this paper claims

The main claims of the paper are the following.

1. The exact 4D projection / leakage / Poisson-hook / worldtube backbone from the earlier papers can be carried forward unchanged as the reduction framework of the present 2.5PN analysis.

2. The frozen conservative Newtonian+1PN+2PN hierarchy cannot by itself generate a nonzero local 2.5PN reaction term. A successful completion must therefore come from a genuinely retarded odd kernel rather than from further conservative algebra.

3. A minimal retarded quadrupole prototype already reproduces a standard valid local 2.5PN force, namely the Burke–Thorne member of the usual Iyer–Will family. This fixes the correct structural target for the full theorem.

4. A full channel audit shows that, on the natural compact/passive/outgoing small-body branch, the currently derived scalar and dipole odd channels are not universal point-particle 2.5PN spoilers. The scalar breathing/outlet route is derivative-mediated and the compact mouth route is radiative/super-Ohmic, while the carried dipole wake is pushed above universal point-particle 2.5PN by the outgoing $\ell = 1$ threshold law. The compact internal P_2 branch is real but finite-size, not universal.

5. The only surviving universal point-particle 2.5PN route is the orbital/worldtube STF quadrupole. The solved 2PN P_2 bundle already contains the full real STF $\ell = 2$ representation content, the natural low-frequency $P_2 \to$ STF matching is scalar under isotropy, the passive outgoing $\ell = 2$ branch carries the correct positive $+i\omega^5$ parity/sign, and the feared zero-overlap option fails on the natural branch.

6. The strongest honest result supported by the present stack is therefore a *conditional* 2.5PN theorem. Under the natural compact/passive/outgoing small-body assumptions isolated by the audit, the reduced nonspinning two-body dynamics has the structure

$$\mathbf{a} = \mathbf{a}_{\mathrm{N}} + \mathbf{a}_{1\mathrm{PN}} + \mathbf{a}_{2\mathrm{PN}} + \mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}} + \Delta\mathbf{a}^{>2.5\mathrm{PN}}_{\mathrm{fs}},$$

where $\mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}}$ is the standard local quadrupolar Burke–Thorne / Iyer–Will family member generated by the orbital/worldtube STF quadrupole, and the remaining unresolved question is the final normalization of its retarded coefficient.

Stated differently, the point of the paper is not that the model can be adjusted until a 2.5PN-looking force appears. The point is that, once the channel structure is audited carefully, the full problem narrows to a short list of named theorem gates. Most of those gates have now been passed. The remaining one is the quadrupole normalization supplied by the completed PDE,

$$\Gamma_5 \stackrel{?}{=} \Gamma_{5,\mathrm{GR}},$$

on the actual passive/outgoing moving-throat branch.

1.3 What this paper does not claim

The negative side of the scope is equally important.

First, this paper does *not* claim an assumption-free theorem that begins with a fully solved moving-throat PDE and ends automatically with the complete point-particle 2.5PN dynamics. As at 1PN and 2PN, the particle limit used here is a controlled reduction inside a declared hierarchy. The main text therefore presents a reduced-theory theorem chain, not a finished collective- coordinate theorem of the full bulk PDE.

Second, this paper does *not* claim that every conceivable lower odd channel has been ruled out in every possible PDE completion. The scalar and dipole sectors have been narrowed substantially, but the strongest statements are tied to the natural compact/passive/outgoing small-body branch identified by the audit. A genuinely new direct scalar source beyond the current continuity plus reciprocal-boundary ontology, a noncompact or effectively one-dimensional low-frequency outlet, a breakdown of the small-body hierarchy, or a strongly non-isotropic/singular quadrupole map would fall outside the present theorem.

Third, this paper does *not* claim a completed derivation of the final quadrupole normalization. The present audit fixes the representation content, the lower-channel demotions, the sign of the passive/outgoing quadrupole odd term, and the existence of a nonzero static overlap. What remains open is the exact far-zone normalization of that branch. Without that final step, one cannot honestly state an unconditional GR-like point-particle 2.5PN theorem.

Fourth, the paper is restricted to the nonspinning two-body radiation-reaction gate. It does not attempt a full treatment of spin couplings, strong-field nonperturbative completion, generic many-body dissipative dynamics, or higher-PN radiative sectors. Nor does it attempt to replace the broader appendical or script-side throat-response record by a single closed PDE derivation. The main text is intentionally focused on the first dissipative theorem gate.

These non-claims are not qualifications to hide. They are part of the correct interpretation of the result. The paper should be read as establishing the sharpest honest 2.5PN statement currently supported by the program: a conditional quadrupolar theorem with a narrow remaining normalization gap.

1.4 Claim taxonomy

To keep the argument referee-safe, we retain the same bookkeeping discipline used in the conservative assembly papers, with one adjustment appropriate to the present stage.

Exact. Identities that follow directly from the declared 4D reduction framework, symmetry bookkeeping, low-frequency algebra, or exact source-map manipulations once the carried lower-order ledger has been stated.

Controlled reduction. Steps that require an explicit regime assumption, such as the carried small-body hierarchy, the passive/outgoing branch, the low-frequency near-zone reduction, or the worldtube/orbital limit used to isolate the universal STF quadrupole.

Protocol closure. Statements that depend on a declared response closure rather than on a fully solved moving-throat PDE. In the present paper the most important examples are the compact passive/outgoing outlet assumptions used to demote the lower odd channels.

Conditional theorem. The final theorem consequence after the previous ingredients are inserted. In contrast to the conservative 2PN paper, the endpoint here is not yet a fully normalized assembly theorem, but a conditional statement that the correct 2.5PN architecture follows once the completed PDE realizes the audited passive/outgoing quadrupole branch with the proper final normalization.

This taxonomy is not cosmetic. At 2.5PN it prevents four logically distinct moves from being blurred together: carrying forward exact lower-order structure, using the declared small-body/passive/outgoing branch, invoking the minimal retarded closures required to demote nonuniversal channels, and stating the final theorem consequence that survives after those ingredients are frozen.

1.5 Roadmap

The next section gives an executive summary of the final theorem ledger. The following section records the minimal carry-forward dictionary from the earlier 4D, 1PN, and 2PN papers and states the exact 2.5PN target. The paper then begins with the decisive benchmark and low-frequency framework: the structural no-go for purely real-even conservative kernels and the Burke–Thorne prototype that fixes the correct retarded target. The three channel audits come next. The scalar section records why the naive monopole risk was serious, why several early rescue hopes failed, and why the surviving scalar channels are nevertheless derivative-mediated or finite-size on the natural branch. The dipole section shows why the carried odd wake is not removed by naive center-of-mass or Ward-identity arguments, but is demoted above universal point-particle 2.5PN by the outgoing $\ell = 1$ threshold law. The quadrupole section then isolates the surviving universal branch, proves the STF representation/source-map facts needed for the match, and explains why the remaining issue is normalization rather than existence.

The final main-text sections state the best current conditional theorem, separate what is fixed from what remains symbolic or open, and record the role of the SymPy and Mathematica audit files as a referee archive rather than as a derivation engine. The appendices collect the detailed low-frequency selection-rule algebra and the longer scalar, dipole, quadrupole, and normalization ledgers that would obscure the main theorem if left in the body of the paper.

2 Executive summary of results

This paper is the point at which the 4D program becomes a full audit of the first dissipative post-Newtonian gate rather than a further conservative closure exercise. The earlier papers supply the exact projection/reduction backbone of the unified 4D model together with the frozen Newtonian+1PN+2PN conservative ledger. The present paper then asks the first genuinely radiative question: whether that stack can be extended to a GR-like point-particle 2.5PN sector. The answer is now sharp. The frozen conservative hierarchy alone cannot generate a nonzero local 2.5PN reaction term, but a minimal retarded quadrupole prototype already reproduces the standard Burke–Thorne / Iyer–Will structure, and the full channel audit shows that on the natural compact/passive/outgoing small-body branch the only surviving universal point-particle 2.5PN route is the orbital/worldtube STF quadrupole.

Headline result. Within the stated closure hierarchy, the strongest honest claim supported by the present stack is a *conditional* 2.5PN theorem. If the completed moving-throat PDE realizes the passive/outgoing small-body branch isolated by the audit, then the reduced nonspinning two-body dynamics takes the form

$$\mathbf{a} = \mathbf{a}_{\mathrm{N}} + \mathbf{a}_{1\mathrm{PN}} + \mathbf{a}_{2\mathrm{PN}} + \mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}} + \Delta\mathbf{a}^{>2.5\mathrm{PN}}_{\mathrm{fs}},$$

where $\mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}}$ is the standard local quadrupolar Burke–Thorne / Iyer–Will family member generated by the orbital/worldtube STF quadrupole, while all other currently derived odd channels are pushed above universal point-particle 2.5PN. The remaining unresolved problem is no longer generic dissipative physics; it is the final passive/outgoing quadrupole normalization,

$$\Gamma_5 \stackrel{?}{=} \Gamma_{5,\mathrm{GR}},$$

on the actual moving-throat branch.

Table 1 lists the load-bearing results established in the present paper and the role each plays in the theorem chain.

How the ledger closes. The proof chain has a negative half and a positive half. The negative half is that the conservative hierarchy alone cannot cross the dissipative gate: a local 2.5PN term requires a genuinely retarded odd kernel. The positive half is that once a minimal quadrupole odd term is admitted, the resulting local force already lands on the standard Burke–Thorne / Iyer–Will structure. The rest of the paper is therefore not a search for some arbitrary dissipative correction, but a channel audit asking whether lower odd routes survive and whether the model contains the correct universal quadrupole branch.

That audit is now sharply organized by PN order. Writing

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \frac{a}{r} \sim \epsilon^{\delta}, \qquad \delta > 0,$$

the derivative scalar breathing outlet scales as

$$\epsilon^{5/2+3\delta},$$

Table 1: Load-bearing results of the present 2.5PN paper.

Item	Result	Status
Conservative benchmark	Frozen conservative kernels cannot by themselves generate a local 2.5PN term, while the minimal retarded quadrupole prototype lands on the Burke–Thorne member of the Iyer–Will family.	Settled
Scalar gate	Direct scalar overlap is structurally dangerous, but the continuity-derived breathing outlet is derivative-coupled and the compact mouth route is super-Ohmic / higher-order; on the natural small-body branch the derived scalar odd channels are pushed above universal point-particle 2.5PN.	Conditional rescue
Dipole gate	The carried odd wake is real and not removed by center-of-mass or vector Ward identities, but a clean outgoing $\ell = 1$ completion begins at $i\omega^3$ rather than $i\omega$; in the strict small-body branch the wake is demoted to finite-size status.	Conditional rescue
Quadrupole gate	The solved 2PN P_2 bundle is exactly the full real STF $\ell = 2$ representation; the orbital/worldtube STF source map is explicit; the retarded matching matrix is scalar under isotropy; the passive/outgoing $\ell = 2$ branch gives the correct positive $+i\omega^5$ odd term; and the static overlap gate passes with $m_0 \neq 0$.	Surviving universal branch
Remaining gap	The old one-pole conservative scaffold is already too small to match the outgoing $\ell = 2$ fingerprint; the only serious unresolved question is the final passive/outgoing normalization of the quadrupole coefficient.	Open

the outgoing dipole wake scales in the same way,

$$\epsilon^{5/2+3\delta},$$

and the compact internal P_2 mouth/support branch enters only at

$$\epsilon^{9/2+5\delta}.$$

All of these therefore lie above universal point-particle 2.5PN on the strict small-body branch. By contrast, the orbital/worldtube STF quadrupole carries no extra small-body penalty and enters directly through the standard quadrupolar slot

$$i\,\Gamma_5\omega^5.$$

This is the compact reason the full channel audit ends with a single surviving universal driver.

How the quadrupole side tightens the target. The later normalization package goes beyond the statement that the quadrupole branch merely survives. The exact outgoing $\ell = 2$ fingerprint,

$$\widehat{Y}_2^{\text{out}}(\omega) = 1 + \frac{a^2}{9c_s^2}\omega^2 + \frac{4a^4}{81c_s^4}\omega^4 + i\frac{a^5}{27c_s^5}\omega^5 + \cdots,$$

already rules out the old one-pole conservative scaffold. The smallest admissible isotropic conservative precursor is instead

$$\widehat{Y}_{Q,\min}^{\text{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}, \qquad \Omega_Q = \frac{3c_s}{2a}.$$

In canonical invariant language this forces

$$\overline{K}_4\,\overline{K}_0 = 4\overline{K}_2^2, \qquad \overline{\Gamma}_5 = 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}},$$

once $(\overline{K}_0, \overline{K}_2)$ are known on that branch. So the remaining problem is narrower than "derive a whole new dissipative theory." Future higher-order conservative work only has to determine whether the grouped real P_2 sector realizes exactly this minimal moment pattern; if it does, the odd Burke–Thorne coefficient follows automatically.

Strongest honest verdict. The correct summary statement is therefore not that the full moving-throat PDE has already been solved through 2.5PN. It is that the present stack has reduced the first dissipative PN gate to a sharply constrained theorem chain. The lower odd scalar and dipole channels are no longer generic universal spoilers on the natural compact/passive/outgoing small-body branch. The quadrupole route is not merely plausible; it is already structurally singled out, with the correct representation content, the correct outgoing parity/sign, and a nonvanishing static overlap. The only serious remaining gap is the final passive/outgoing quadrupole normalization on the actual moving-throat branch. In that precise sense, the 2.5PN program is now a normalization problem rather than a whole-model rescue problem.

3 Carry-forward dictionary and theorem target

This section is intentionally downstream of the parent 4D action paper and the full conservative 1PN and 2PN assembly papers. We do not repeat those longer derivations here. Instead, we record only the objects, frozen lower-order values, and branch assumptions that the present 2.5PN audit actually uses. The purpose of the section is threefold: to freeze notation, to make clear what is being imported rather than rederived, and to state the exact theorem target that organizes the channel audits in the sections that follow.

The guiding principle is the same one used in the earlier conservative papers. What is exact in the parent reduction framework should remain visibly exact; what enters only after a declared near-zone, small-body, or passive/outgoing assumption should remain visibly reduced. The present paper is therefore built on an imported dictionary rather than on a fresh rederivation of the full moving-throat PDE.

3.1 Minimal imported 4D reduction dictionary

The parent theory lives on a $(4+1)$-dimensional spacetime with coordinates

$$X^M = (t, x, y, z, w), \qquad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \qquad \mathbf{x} = (x, y, z) \in \mathbb{R}^3.$$

The fundamental bulk objects carried into the present paper are the matter field $\psi(\mathbf{X}, t)$, its density

$$\rho(\mathbf{X}, t) = |\psi(\mathbf{X}, t)|^2,$$

the optional localized gauge field $A_M(\mathbf{X}, t)$ with localization profile $Z(w)$, and the effective throat geometry variables

$$a(t), \qquad L(t).$$

When electric charge is named explicitly in the inherited 4D notation, it is the corrected branch quantity

$$q_* = \eta_Q e_*, \qquad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}}, \qquad Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w)\, dw,$$

not the gravity-side scalar coefficient κ_ρ that enters the post-Newtonian ledger below.

A brane observer does not see the full bulk fields directly. Brane observables are introduced by a fixed normalized projection kernel $W(w)$,

$$\int_{-\infty}^{\infty} W(w)\, dw = 1,$$

through the projection map

$$\mathcal{P}_W[Q](t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w)\, Q(t, \mathbf{x}, w)\, dw.$$

In particular,

$$\rho_{\rm br} = \mathcal{P}_W[\rho], \qquad \mathbf{J}_{\rm br} = \mathcal{P}_W[(j^x, j^y, j^z)], \qquad \mathbf{v}_{\rm br} = \frac{\mathbf{J}_{\rm br}}{\rho_{\rm br}} \quad (\rho_{\rm br} > 0).$$

This projection operation is distinct from *reduction*. Projection defines what a brane observer measures; reduction refers to the further controlled elimination of the extra coordinate under an explicit ansatz or scaling regime.

The exact projected continuity law carried forward from the parent theory is

$$\partial_t \rho_{\rm br} + \nabla_3 \cdot \mathbf{J}_{\rm br} = S_{\rm leak},$$

with leakage source

$$S_{\rm leak} = -\big[W(w) j^w\big]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w) j^w \, dw.$$

After Helmholtz decomposition of the brane velocity,

$$\mathbf{v}_{\rm br} = \nabla_3 \phi_{\rm br} + \mathbf{v}_T, \qquad \nabla_3 \cdot \mathbf{v}_T = 0,$$

one obtains the exact longitudinal identity

$$\rho_{\rm br} \, \nabla_3^2 \phi_{\rm br} = S_{\rm leak} - \partial_t \rho_{\rm br} - (\nabla_3 \rho_{\rm br}) \cdot (\nabla_3 \phi_{\rm br} + \mathbf{v}_T).$$

Under the carried quasi-static near-zone assumptions this reduces to the Poisson hook

$$\nabla_3^2 \phi_{\rm br} \simeq \frac{S_{\rm eff}}{\rho_0}, \qquad \Phi_N = \lambda_\Phi \, \phi_{\rm br},$$

which is the origin of the Newtonian scalar sector used in the lower-order particle reduction.

The second imported reduction step is the controlled worldtube/coherent-defect limit. For a compact defect of size $a \ll r$ in a smooth external field, the center-of-mass reduction gives the usual point-particle force law

$$M_A \, \ddot{\mathbf{X}}_A = -M_A \nabla \Phi_{\rm ext}(\mathbf{X}_A) + O\!\left(\frac{a^2}{r^2} \, \frac{\Phi_{\rm ext}}{r}\right),$$

so that the universal point-particle source at leading order is controlled by the worldline/worldtube multipoles rather than by arbitrary internal details of the defect. That worldtube dictionary is what later allows the universal 2.5PN route to be stated in terms of the orbital STF quadrupole.

3.2 Frozen conservative carry-forward ledger

The present paper imports the conservative Newtonian+1PN+2PN ledger as frozen input, not as open fit parameters. The surviving lower-order viability conditions are

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\rm add} = \frac{1}{2}, \qquad \kappa_{\rm PV} = \frac{3}{2}, \qquad \beta_{\rm 1PN} = 3, \qquad \frac{L}{a} \approx 1.85.$$

The first four values are the scalar mass-dressing, optical/barotropic, added-mass, and adiabatic breathing-response coefficients frozen by the earlier program. The fifth is the resulting carried 1PN ledger value, and the last is the preferred throat/cavity aspect ratio selected by the earlier geometric and spectral analyses.

The parity-even wake data imported from the full conservative 1PN assembly are likewise frozen:

$$\alpha^2 = \frac{3}{4}, \qquad a_H = 0, \qquad K_{\rm vec} = \frac{2}{\pi^2}.$$

These quantities are not retuned in the present paper. They define the carried conservative vector/wake branch against which the later odd dipole audit is performed.

The full conservative 2PN assembly sharpened the particle ontology further. By the start of the present 2.5PN campaign, the defect could no longer be treated as a pure scalar monopole with small corrections. The frozen conservative structure already looked like a small throat-response theory with three distinct pieces:

1. a carried odd dipole wake sector,
2. a genuine even $P_0 \oplus P_2$ mouth/support layer,
3. and a separate geometry-closure channel.

In the notation carried from the 2PN files, the zero-frequency odd wake residues are

$$R_{1\perp} = \frac{7}{2}, \qquad R_{10} = 4,$$

while the frozen zero-frequency even support data are

$$R_0 = R_{20} = R_{21} = R_{22} = 1, \qquad J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0\right).$$

What remains genuinely open from the 2PN side is not the existence of these support channels but their full dynamical completion. In particular, the pole or compliance scales

$$\Omega_{1\perp},\ \Omega_{10},\ \Omega_0,\ \Omega_{20},\ \Omega_{21},\ \Omega_{22},\ \Omega_g$$

are not claimed as already derived from the fully solved moving-throat PDE. This is one reason the present paper continues to speak in terms of a declared closure hierarchy rather than an assumption-free first-principles theorem.

Two structural consequences of this imported ledger are worth stating explicitly. First, the present paper is not allowed to repair a dissipative failure by retuning lower-order conservative constants that were already fixed in the earlier papers. Second, the defect already carries enough internal support structure that the 2.5PN audit must distinguish carefully between universal point-particle channels and finite-size throat-response channels. That distinction is essential later when the compact internal P_2 branch is demoted relative to the universal orbital/worldtube STF quadrupole.

3.3 Natural compact/passive/outgoing small-body branch

The channel audit in the later sections is organized around a single natural low-frequency branch. It is not presented as an assumption-free theorem of the completed PDE. It is presented as the minimal branch that remains compatible with the carried ontology and the standard point-particle 2.5PN target.

The working assumptions of this branch are the following.

First, the point-particle limit is taken as a strict small-body expansion,

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \frac{a}{r} \sim \epsilon^\delta, \qquad \delta > 0,$$

so that finite-size channels may be consistently separated from universal point-particle channels.

Second, low-frequency scalar and vector outlets are taken to be compact and outgoing rather than effectively one-dimensional or directly Ohmic. This is the branch on which the continuity-derived breathing vertex is derivative-mediated, the compact mouth route is radiative/super-Ohmic, and the outgoing $\ell = 1$ dipole completion begins at the cubic odd threshold rather than at a linear one.

Third, the isolated low-frequency defect/worldtube sector is assumed to be rotationally invariant at leading order. This is what later collapses the retarded $P_2 \to$ STF matching problem to a scalar function rather than a full anisotropic matrix problem.

Fourth, the universal quadrupole source is assumed to survive the source map with nonzero static overlap. In the notation used later,

$$\hat{m}_0 = 1 + O\left(\frac{a^2}{r^2}\right),$$

so the orbital/worldtube STF quadrupole is not naturally projected away on the same branch.

These assumptions are strong enough to formulate a theorem target but weak enough to remain honest about what is still open. They do not yet solve the completed PDE. They identify the natural branch on which the current 2.5PN conditional theorem is meant to live.

3.4 Effective source dictionary for the dissipative sector

The worldtube reduction implies that the effective STF source of the two-body system has the structural form

$$I^{\text{eff}}_{ij} = \mu\, x_{\langle i}x_{j\rangle} + Q^{ij}_A + Q^{ij}_B + \cdots\,,$$

where μ is the reduced mass, x_i the relative separation vector, and Q^{ij}_A, Q^{ij}_B denote compact internal finite-size quadrupoles. The first term is the universal orbital/worldtube STF quadrupole; the remaining terms belong to the finite-size throat-support sector.

This is the source split that the rest of the paper uses. The universal point-particle question at 2.5PN is therefore not whether some compact internal quadrupole can radiate. It is whether the universal orbital/worldtube STF piece survives all lower-channel audits and acquires the correct retarded normalization. The internal P_2 sector remains real and important, but in the strict small-body hierarchy it is tracked as part of the finite-size response ledger rather than identified with the universal point-particle source.

3.5 The exact theorem target

The low-frequency odd hierarchy identified in the benchmark framework is

$$K^{\text{odd}}_0(\omega) \sim i\gamma_1\omega, \qquad K^{\text{odd}}_1(\omega) \sim i\gamma_3\omega^3, \qquad K^{\text{odd}}_2(\omega) \sim i\gamma_5\omega^5.$$

The point-particle 2.5PN theorem target is therefore to show that the first two odd channels are absent from the universal point-particle sector on the natural branch, while the third survives with the correct sign, source content, and normalization.

To state that target precisely, introduce a canonical real STF basis E^{ij}_a, $a = 1, \ldots, 5$, and define the canonical orbital STF quadrupole coordinates by

$$q^{\text{can}}_a \equiv I^{\text{orb}}_{ij} E^{ij}_a\,, \qquad I^{\text{orb}}_{ij} = \mu\, x_{\langle i}x_{j\rangle}.$$

The unique local odd quadratic action with this STF structure is

$$L^{(2)}_{\text{rr}} = -\frac{\gamma}{2}\sum_a q^{\text{can}}_{a,-}\left(q^{\text{can}}_{a,+}\right)^{(5)}.$$

In the physical limit this produces the relative acceleration

$$a_i = -\gamma\, x^j \left(I^{\text{orb}}_{ij}\right)^{(5)}.$$

Comparison with the standard Burke–Thorne force,

$$a^{\text{BT}}_i = -\frac{2G}{5c^5}\, x^j \left(I^{\text{orb}}_{ij}\right)^{(5)},$$

fixes the canonical GR normalization target

$$\gamma_{\mathrm{GR}} = \frac{2G}{5c^5}.$$

The toy-model quadrupole sector is not written directly in that canonical basis. If the low-frequency source coordinate used by the worldtube/throat response is normalized as

$$q_a^{\mathrm{toy}} = \hat{m}_0\, q_a^{\mathrm{can}},$$

and if the retarded matching matrix on the natural isotropic branch has the form

$$\mathcal{M}^R_{ab}(\omega) = \Big[\hat{m}_0 + \hat{m}_2\omega^2 + \hat{m}_4\omega^4 + i\Gamma_5\omega^5 + O(\omega^6)\Big]\delta_{ab}, \qquad \hat{m}_0 \neq 0, \qquad \Gamma_5 > 0,$$

then the canonically normalized odd coefficient is controlled only by the product

$$\gamma_{\mathrm{toy}} = \hat{m}_0^2\Gamma_5.$$

Matching to Burke–Thorne is therefore equivalent to the scalar equality

$$\boxed{\hat{m}_0^2\Gamma_5 = \frac{2G}{5c^5}.}$$

This is the sharpest normalization target used in the present paper.

At the level of the reduced equations of motion, the full conditional theorem target is therefore

$$\mathbf{a} = \mathbf{a}_{\mathrm{N}} + \mathbf{a}_{1\mathrm{PN}} + \mathbf{a}_{2\mathrm{PN}} + \mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}} + \Delta\mathbf{a}^{>2.5\mathrm{PN}}_{\mathrm{fs}},$$

where

1. $\mathbf{a}_{\mathrm{N}} + \mathbf{a}_{1\mathrm{PN}} + \mathbf{a}_{2\mathrm{PN}}$ is the frozen conservative ledger imported from the earlier papers,
2. $\mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}}$ is the standard local Burke–Thorne / Iyer–Will quadrupole family member generated by the universal orbital/worldtube STF quadrupole,
3. and $\Delta\mathbf{a}^{>2.5\mathrm{PN}}_{\mathrm{fs}}$ collects all currently derived scalar, dipole, and compact internal P_2 channels that are pushed above universal point-particle 2.5PN on the strict small-body branch.

Equivalently, the universal point-particle theorem fails if any one of the following occurs: a surviving scalar $i\omega$ channel, a surviving universal dipole $i\omega^3$ channel, a vanishing source-map overlap $\hat{m}_0 = 0$, a wrong-sign quadrupole coefficient $\Gamma_5 < 0$, or a final normalization mismatch relative to $2G/(5c^5)$. The rest of the paper is organized precisely as an audit of those gates.

4 Decisive benchmark and low-frequency framework

This section does three jobs that set the logic of the rest of the paper. First, it records the first decisive negative result of the 2.5PN campaign: within the frozen Newtonian+1PN+2PN hierarchy, one cannot obtain a nonzero local radiation-reaction term by further conservative algebra alone. Second, it fixes the constructive target that any successful completion must reproduce, namely the Burke–Thorne quadrupole prototype and its match to the standard Iyer–Will local 2.5PN family. Third, it organizes the remaining problem as a low-frequency odd-channel audit. Once that framework is in place, the rest of the theorem chain becomes a clean decision tree: scalar and dipole odd channels must be shown absent, demoted, or finite-size only, while the quadrupole branch must survive with the correct sign and source map.

4.1 Why the frozen conservative hierarchy cannot generate local 2.5PN

At the start of the 2.5PN push, the imported conservative hierarchy was already strong but it was still, in the low-frequency sense relevant here, a *real-even* theory. For any reduced collective coordinate $q(t)$ or worldtube source variable, the most general local conservative kernel has the form

$$K_{\mathrm{cons}}(\omega) = K_0 + K_2\omega^2 + K_4\omega^4 + \cdots, \qquad K_{2m} \in \mathbb{R}.$$

Passing to the time domain gives

$$K_{\mathrm{cons}}(\partial_t) = K_0 + K_2\partial_t^2 + K_4\partial_t^4 + \cdots.$$

Every local operator produced in this way is time-reversal even. Such terms can renormalize inertial, elastic, or static-response coefficients, but they cannot by themselves generate a genuinely dissipative reaction force. In particular, they cannot produce the first nonconservative near-zone term required by a GR-like nonspinning two-body theory at 2.5PN.

The first decisive no-go statement of the 2.5PN audit is therefore

> The frozen conservative hierarchy alone cannot generate a nonzero local 2.5PN reaction term.

This is not merely a bookkeeping remark. It identifies the precise place where the conservative 2PN success stops. Any valid continuation must add a *genuinely retarded* sector rather than attempt to manufacture dissipation from more conservative algebra.

The smallest admissible escape hatch is equally sharp. If the model is to admit a local quadrupolar radiation-reaction term at all, the reduced kernel must pick up an odd imaginary contribution. At minimal order this means

$$K^{\mathrm{ret}}(\omega) = K^{\mathrm{even}}(\omega) + i\,\lambda_5\omega^5 + O(\omega^7).$$

The odd term is the first time-asymmetric piece, the quintic power is the local signature of a leading quadrupolar radiation-reaction sector, and the sign of λ_5 distinguishes damping from antidamping. If instead a universal point-particle scalar or dipole odd term survived at lower order,

$$i\lambda_1\omega \qquad \text{or} \qquad i\lambda_3\omega^3,$$

it would dominate the quadrupole branch and spoil the GR-like target. This is why the rest of the paper is organized as an odd-channel audit rather than as a single undifferentiated dissipative calculation.

4.2 Burke–Thorne prototype and the standard 2.5PN family

Once the conservative no-go is established, one must still show that the target itself is coherent: if one supplies the *minimal* retarded quadrupole structure suggested above, does the resulting local force actually land on a known correct 2.5PN sector? The answer is yes, and the check is completely explicit.

The first step is the usual two-body STF source decomposition. Let $M = m_1 + m_2$, $\mu = m_1m_2/M$, X_i the center-of-mass coordinate, and x_i the relative coordinate. Then the mass dipole and STF quadrupole are

$$D_i = m_1r_{1i} + m_2r_{2i} = MX_i,$$

$$Q_{ij} = \sum_{A=1}^{2} m_A r_{A\langle i}r_{Aj\rangle} = MX_{\langle i}X_{j\rangle} + \mu x_{\langle i}x_{j\rangle}.$$

So in the center-of-mass frame the only universal point-particle source is the orbital/worldtube STF quadrupole

$$Q_{ij} = \mu\, x_{\langle i}x_{j\rangle}.$$

The minimal constructive benchmark is then the Burke–Thorne local force law

$$a^i_{\rm rel} = -\frac{2G}{5c^5}\, x^j Q^{(5)}_{ij}.$$

Using the Newtonian relative equations of motion to reduce the fifth time derivatives to the standard local basis, one obtains

$$\mathbf{a}_{2.5\rm PN} = \frac{8G^2m^2\nu}{5c^5r^3}\left[A\,\dot r\,\mathbf{n} + B\,\mathbf{v}\right], \qquad \nu \equiv \frac{\mu}{m},$$

with the explicit coefficients

$$A = 18v^2 + \frac{2Gm}{3r} - 25\dot r^2, \qquad B = -6v^2 + \frac{2Gm}{r} + 15\dot r^2.$$

This already has the standard local 2.5PN form. To make the comparison precise, recall the Iyer–Will gauge family,

$$\mathbf{a}_{2.5\rm PN} = \frac{8G^2m^2\nu}{5c^5r^3}\left[A_{\alpha,\beta}\,\dot r\,\mathbf{n} + B_{\alpha,\beta}\,\mathbf{v}\right],$$

with

$$A_{\alpha,\beta} = 3(1+\beta)v^2 + \frac{23+6\alpha-9\beta}{3}\frac{Gm}{r} - 5\beta\dot r^2,$$

$$B_{\alpha,\beta} = -(2+\alpha)v^2 - (2-\alpha)\frac{Gm}{r} + 3(1+\alpha)\dot r^2.$$

Matching the benchmark coefficients gives the unique solution

$$\alpha = 4, \qquad \beta = 5,$$

which is precisely the Burke–Thorne member of the standard 2.5PN family.

This benchmark supplies the positive half of the theorem ledger:

A minimal retarded quadrupole odd term is sufficient to reproduce a valid 2.5PN local force.

Combined with the conservative no-go, it sharply reframes the rest of the program. The question is no longer whether some dissipation can be written down; it is whether the model can derive a retarded completion whose first physical odd term is quadrupolar and whose coefficient has the correct sign and normalization.

4.3 Low-frequency hierarchy and influence-functional bookkeeping

With the constructive benchmark fixed, the remaining theorem problem can be organized entirely in terms of odd low-frequency channels. On the reduced side, the odd part of a retarded influence kernel may be written schematically as

$$K^{\rm odd}(\omega) = i\gamma_1\omega + i\gamma_3\omega^3 + i\gamma_5\omega^5 + \cdots .$$

The time-domain signs are fixed by Fourier convention:

$$i\omega \longleftrightarrow -\partial_t, \qquad i\omega^3 \longleftrightarrow +\partial_t^3, \qquad i\omega^5 \longleftrightarrow -\partial_t^5.$$

So the three leading odd channels correspond to local forces of the form

$$F_1 \sim -\gamma_1 \dot{q}, \qquad F_3 \sim +\gamma_3 q^{(3)}, \qquad F_5 \sim -\gamma_5 q^{(5)}.$$

What matters physically is that each odd term carries a genuine dissipative piece once total derivatives are separated off. The relevant identities are

$$\begin{aligned} \dot{q}(-\dot{q}) &= -\dot{q}^2, \\ \dot{q}\, q^{(3)} &= \frac{d}{dt}(\dot{q}\,\ddot{q}) - \ddot{q}^2, \\ \dot{q}(-q^{(5)}) &= -\frac{d}{dt}(\dot{q}\, q^{(4)} - \ddot{q}\, q^{(3)}) - (q^{(3)})^2. \end{aligned}$$

Thus the wrong odd channel cannot be dismissed as a pure gauge artifact merely because it comes with a higher derivative; after subtraction of the associated Schott term, it still carries positive semidefinite dissipation. The lowest surviving odd channel is therefore the dominant physical one and must be eliminated if it is not the desired GR-like quadrupole branch.

This immediately yields the channel hierarchy used throughout the rest of the paper:

$$\boxed{\begin{array}{c} n = 1 \text{ (scalar / monopole candidate)}, \\ n = 3 \text{ (dipole / vector candidate)}, \\ n = 5 \text{ (quadrupole candidate)}. \end{array}}$$

At the framework level, this is still only a classification rule. Later sections show that it is also the correct physical hierarchy on the natural compact, passive, outgoing small-body branch: the scalar side is the first dangerous odd channel, the carried dipole wake is the second, and the quadrupole branch is the first one compatible with a universal GR-like point-particle 2.5PN theorem.

The same bookkeeping can be written in model-specific language using the already solved 2PN support package. If the scalar support and geometry sectors pick up linear odd coefficients δ_{01} and δ_{g1}, while the quadrupole support sector picks up a quintic odd coefficient δ_{205}, then the natural carried low-frequency combinations are

$$\begin{aligned} \gamma_1^{\text{eff}} &= J_0^2 \delta_{01} - \Delta_{\text{geom}}\, \delta_{g1} = \frac{16}{5}\, \delta_{01} - \frac{281}{80}\, \delta_{g1}, \\ \gamma_5^{\text{eff}} &= J_{20}^2 \delta_{205} = \frac{25}{16}\, \delta_{205}. \end{aligned}$$

These are not yet the final theorem coefficients. Their role is more basic: they translate the abstract odd-channel hierarchy into the carried scalar/geometry and P_2 support data inherited from the earlier conservative papers.

The rest of the paper is therefore a controlled channel audit with a simple logic. A successful completion must show that

1. the scalar $n = 1$ channel is absent, derivative-mediated, or finite-size only,
2. the dipole $n = 3$ channel is absent, demoted, or pushed above universal point-particle 2.5PN,
3. and the quadrupole $n = 5$ channel survives with positive retarded sign and with a nonvanishing source map to the orbital/worldtube STF quadrupole.

Once those three statements are known, the theorem problem is reduced to a single remaining question: the final passive/outgoing normalization of the surviving quadrupole branch.

5 Scalar-sector audit and demotion

This section addresses the earliest and potentially most dangerous odd channel. Once the low-frequency audit is organized as

$$K_0^{\mathrm{odd}}(\omega) \sim i\gamma_1\omega, \qquad K_1^{\mathrm{odd}}(\omega) \sim i\gamma_3\omega^3, \qquad K_2^{\mathrm{odd}}(\omega) \sim i\gamma_5\omega^5,$$

the scalar side becomes the first genuine kill-switch test. If a physical monopole odd term survives in the reduced subsystem, it contaminates the theorem before the desired quadrupole 2.5PN branch even has a chance to matter. The purpose of the present section is therefore not merely to "discuss scalar corrections," but to decide whether the scalar channel is a universal point-particle obstruction or can be demoted to derivative-coupled and/or finite-size status on the natural compact/passive/outgoing small-body branch.

The section proceeds in four steps. First, it isolates the naive scalar danger in the carried 2PN packaging and shows why the simplest continuity and no-static-flux arguments do not kill it. Second, it records the explicit projection-overlap counterexample and the exact projection-locking condition that would be needed for an automatic scalar Ward identity. Third, it shows that the scalar channels actually derived from the current ontology are derivative-coupled or finite-size only. Fourth, it uses the compact reciprocal-mouth analysis to demote the remaining mouth channel above universal point-particle 2.5PN. The net result is a demotion rather than an unconditional deletion: the scalar side is no longer the default kill switch, but a narrower class of non-generic PDE loopholes remains open.

5.1 Why the scalar channel is the first gate

The scalar danger was already explicit in the conservative 2PN throat packaging. On the scalar-like side, the reduced response can be written as

$$S_{\mathrm{eff}}(\omega) = J_0^2 Y_0(\omega) + J_{20}^2 Y_{20}(\omega) - \Delta_{\mathrm{geom}} Y_g(\omega),$$

with fixed static weights

$$J_0^2 = \frac{16}{5}, \qquad J_{20}^2 = \frac{25}{16}, \qquad \Delta_{\mathrm{geom}} = \frac{281}{80}.$$

If the scalar channels pick up low-frequency odd pieces of the form

$$Y_0^{\mathrm{ret}} = Y_0^{\mathrm{even}} + i\delta_{01}\omega + \cdots, \qquad Y_g^{\mathrm{ret}} = Y_g^{\mathrm{even}} + i\delta_{g1}\omega + \cdots,$$

then the effective scalar odd coefficient is already fixed to be

$$\gamma_1^{\mathrm{eff}} = \frac{16}{5}\delta_{01} - \frac{281}{80}\delta_{g1}.$$

So the first scalar theorem target is sharp:

$$\boxed{\gamma_1^{\mathrm{eff}} = 0.}$$

This is not an arbitrary bookkeeping preference. A genuine scalar odd term sits at the bottom of the odd hierarchy and would therefore dominate any later dipole or quadrupole channel.

The danger is also structurally natural. For an outgoing three-dimensional scalar channel,

$$G_R(\omega, r) = \frac{e^{i\omega r/c_s}}{4\pi r} = \frac{1}{4\pi r} + \frac{i\omega}{4\pi c_s} - \frac{\omega^2 r}{8\pi c_s^2} - \frac{i\omega^3 r^2}{24\pi c_s^3} + \cdots .$$

So if the model carries any nonzero effective scalar monopole overlap to a gapless scalar outlet, the linear odd term is generic. This is why the scalar side had to be audited before the dipole and quadrupole branches could be trusted.

5.2 Why naive scalar cancellation is not automatic

The first scalar test was whether the current stack already forced $\gamma_1^{\mathrm{eff}} = 0$ from the exact projected continuity identity together with the later no-static-flux theorem. It does not.

The exact projected continuity law gives

$$\partial_t \rho_{\mathrm{br}} + \nabla_3 \cdot \mathbf{J}_{\mathrm{br}} = S_{\mathrm{leak}},$$

and after integration over all ordinary three-space for an isolated localized defect,

$$\frac{d}{dt} N_{\mathrm{br}}(t) = I_{\mathrm{leak}}(t), \qquad N_{\mathrm{br}}(t) = \int d^3x\, \rho_{\mathrm{br}}, \qquad I_{\mathrm{leak}}(t) = \int d^3x\, S_{\mathrm{leak}}.$$

In frequency space this becomes

$$-i\omega\, \delta N_{\mathrm{br}}(\omega) = \delta I_{\mathrm{leak}}(\omega).$$

This identity is exact and important, but it only tells us what scalar leakage *means*. It does not force the integrated leakage to vanish.

Nor does the later static theorem

$$Z^{\mathrm{eff}}_{\sigma,\mathrm{flux}}(0) = 0$$

close the issue. That theorem kills a DC scalar bleed channel in the frozen conservative theory, but it does not eliminate a genuinely dynamical scalar odd term. It only says that the dangerous scalar channel, if present, must be an authentically retarded/open-system effect rather than something already hidden in the static conservative ledger.

The deeper reason the naive Ward-identity hope fails is that the brane-visible scalar content is a projection overlap rather than the full bulk norm. If one writes a centered breathing family as

$$\rho(\mathbf{x}, w, t) = n(\mathbf{x}) g_a(w), \qquad \int d^3x\, n(\mathbf{x}) = N,$$

then the brane-visible scalar content is

$$N_{\mathrm{br}}(a) = N\, C(a), \qquad C(a) = \int dw\, W(w) g_a(w).$$

The exact integrated leakage identity becomes

$$I_{\mathrm{leak}} = N\, \dot{a}\, \frac{dC}{da}.$$

So scalar leakage vanishes only if the projection overlap is independent of the breathing coordinate. That is not generic.

A fully explicit counterexample is obtained from the Gaussian choice

$$W_\ell(w) = \frac{1}{\sqrt{\pi}\,\ell} e^{-w^2/\ell^2}, \qquad g_a(w) = \frac{1}{\sqrt{\pi}\, a} e^{-w^2/a^2},$$

for which

$$C(a) = \frac{1}{\sqrt{\pi}\sqrt{a^2+\ell^2}}, \qquad \frac{dC}{da} = -\frac{a}{\sqrt{\pi}(a^2+\ell^2)^{3/2}} \neq 0.$$

Hence

$$I_{\mathrm{leak}} = -\frac{N a \dot{a}}{\sqrt{\pi}(a^2+\ell^2)^{3/2}},$$

and for harmonic breathing $a(t) = a_0 + \varepsilon e^{-i\omega t}$,

$$\delta I_{\mathrm{leak}} = i\omega N \varepsilon \frac{a_0}{\sqrt{\pi}(a_0^2+\ell^2)^{3/2}}.$$

The point of this example is not that Gaussians are physically mandatory; it is that the current projection framework admits a centered isolated breathing family with a surviving scalar odd term. The scalar Ward identity is therefore not automatic.

What would rescue the scalar channel at this stage is the exact *projection-locking* condition. Let the isolated defect family be parameterized by collective coordinates q^A. Then

$$I_{\mathrm{leak}}(t) = \dot{q}^A B_A(q), \qquad B_A(q) = \partial_A N_{\mathrm{br}}(q).$$

The leading scalar odd term vanishes if and only if

$$B_A(q_0)v^A = 0$$

for every dynamically allowed scalar tangent vector v^A at the operating point. Equivalently, if the soft scalar tangent space is spanned by modes $\Psi_{(n)}$, then one must have

$$\int d^3x\, dw\, W(w)\Psi_{(n)}(\mathbf{x}, w) = 0 \qquad \text{for every soft scalar mode } \Psi_{(n)}.$$

This is a mathematically clean rescue target, but it is restrictive and is not already implied by the current stack.

5.3 The carried 2PN scalar support/geometry branch is finite-size only

The next step was to ask whether the dangerous scalar odd term might still be harmless because it belonged only to a finite-size 2PN support/geometry branch rather than to the universal point-particle sector. This turns out to be the first important scalar demotion.

For a clean outgoing scalar $\ell = 0$ completion,

$$\Lambda_0(k) = -\frac{1}{a_0} + ik, \qquad k = \frac{\omega}{c_s},$$

so the normalized retarded admittance is

$$Y_{0,\mathrm{norm}}(k) = \frac{1}{1 - ia_0k} = 1 + ia_0k - a_0^2k^2 - ia_0^3k^3 + O(k^4).$$

Applying the same logic to the carried scalar geometry port gives an analogous expansion with length scale a_g. Inserting these into the 2PN scalar combination yields

$$S_{\mathrm{eff}}^{\mathrm{ret}}(k) = \frac{5}{4} + ik\left(\frac{16}{5}a_0 - \frac{281}{80}a_g\right) + O(k^2).$$

The effective scalar odd coefficient on this branch is therefore governed by the single length combination

$$\ell_{\mathrm{eff}} = \frac{16}{5}a_0 - \frac{281}{80}a_g.$$

Exact cancellation would require

$$a_g = \frac{256}{281}a_0,$$

but the key point for the present theorem chain is that this exact tuning is not needed to avoid a universal point-particle obstruction. The reason is that this scalar channel already enters through a 2PN prefactor. On the strict small-body branch,

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \frac{a}{r} \sim \epsilon^{\delta}, \qquad \delta > 0,$$

the odd correction from this 2PN scalar support/geometry branch scales as

$$\epsilon^2\left(\sqrt{\epsilon}\,\frac{\ell_{\mathrm{eff}}}{r}\right) = \epsilon^{5/2+\delta}.$$

For every $\delta > 0$, this lies above universal point-particle 2.5PN. So this branch is demoted from a universal scalar killer into a finite-size correction channel.

5.4 The breathing channel is the sharper scalar danger

After the 2PN scalar support branch was demoted, the sharper scalar danger was seen to come from the older 1PN breathing/PV channel. If that channel were to pick up a direct retarded correction of the form

$$\kappa_{\mathrm{PV}}^{\mathrm{ret}}(\omega) = \kappa_{\mathrm{PV}}(0)\left[1 + i\ell_{\mathrm{PV}}\frac{\omega}{c_s} + O(\omega^2)\right],$$

then, because it enters with a 1PN prefactor, its contribution would scale like

$$\epsilon\left(\sqrt{\epsilon}\,\frac{\ell_{\mathrm{PV}}}{r}\right) = \epsilon^{3/2+\delta},$$

which could sit at or below 2.5PN depending on the small-body scaling. At that stage of the audit, this breathing route was therefore the main scalar threat.

The first important observation was that a direct gapless scalar overlap does in fact generically generate the dangerous $i\omega$ term, while a simple cavity gap does *not* rescue the theorem if the internal mode inherits Ohmic leakage from a gapless outlet. In other words, the danger was real enough that one could not save the breathing channel by hand-waving about discreteness.

The decisive rescue came from deriving the breathing-to-outlet vertex from the exact continuity structure rather than guessing it. Let

$$q(t) = q_0 + \delta q(t), \qquad \rho = \rho_0 + \delta q\,\rho_q^{(1)} + O(\delta q^2).$$

A static displacement along the breathing family is still a static configuration, so the scalar current cannot begin at $O(\delta q)$. The first current is kinematic:

$$j^i = \dot{\delta q}\,\mathcal{J}_q^i + O(\delta q\dot{q}, \dot{q}^2, \ddot{q}).$$

Linearized continuity gives

$$\partial_i\mathcal{J}_q^i = -\rho_q^{(1)},$$

and after projection one finds

$$\delta S_{\mathrm{leak}}(\mathbf{x}, t) = \dot{\delta q}(t)B_q(\mathbf{x}),$$

with

$$B_q(\mathbf{x}) = -\left[W\mathcal{J}_q^w\right]_{-\infty}^{+\infty} + \int W'(w)\mathcal{J}_q^w\,dw.$$

So in frequency space,

$$\boxed{\delta S_{\mathrm{leak}}(\mathbf{x}, \omega) = (-i\omega)\,\delta q(\omega)\,B_q(\mathbf{x}).}$$

This is the load-bearing theorem: the breathing-to-outlet vertex derived from continuity is $\dot{q}$-like or zero, not direct q-like.

That reclassifies the earlier Gaussian counterexample. The counterexample still shows that the derivative-coupled monopole coefficient can be nonzero, but it no longer shows a direct scalar source. Once a low-frequency scalar outlet with retarded kernel

$$\Pi_q^R(\omega) = A + iB\omega + C\omega^2 + iD\omega^3 + \cdots$$

is integrated out, the induced breathing kernel takes the form

$$K_{qq}^R(\omega) = \omega^2\Pi_q^R(\omega) = A\omega^2 + iB\omega^3 + C\omega^4 + iD\omega^5 + \cdots.$$

The first dissipative odd term is therefore cubic rather than linear:

$$\Im K_{qq}^R(\omega) \propto \omega^3.$$

On the natural small-body branch, the resulting derived breathing correction scales as

$$\epsilon^{5/2+3\delta},$$

which lies above universal point-particle 2.5PN for every $\delta > 0$. This is the decisive demotion of the old breathing/PV scalar danger.

5.5 There is no third linear scalar outlet, and the remaining mouth channel is super-Ohmic

Once the continuity-derived breathing vertex was shown to be derivative-coupled, the remaining scalar subproblem was whether some additional linear scalar outlet might still arise elsewhere in the reduced subsystem. The answer, within the current ontology, is no.

Integrating the exact projected continuity law over a moving defect-centered control volume $D(t)$ gives

$$\frac{d}{dt}\int_{D(t)} \rho_{\rm br}\, d^3x = -\oint_{\partial D(t)} (\mathbf{J}_{\rm br} - \rho_{\rm br}\mathbf{u}_b)\cdot \mathbf{n}\, dA + \int_{D(t)} S_{\rm leak}\, d^3x.$$

So the exact scalar-outlet ledger has only two channels:

$$\boxed{\text{worldtube/mouth boundary flux}} \qquad \text{and} \qquad \boxed{\text{projected leakage}}.$$

There is no third independent linear scalar outlet in the current ontology. Moreover, the projected momentum-leakage and projection-induced stress channels are quadratic in velocities, schematically $\rho v_a v_w$ and $\rho v_a v_b$, so around a static background they cannot generate a new linear scalar source. This leaves only one unresolved scalar loophole: *a genuinely dynamic scalar mouth/worldtube boundary admittance.*

If that admittance is parameterized naively as

$$Z_\sigma^{\rm eff}(\omega) = iz_1\omega + z_2\omega^2 + iz_3\omega^3 + \cdots ,$$

then the first genuine dissipative mouth term is the real coefficient $z_2\omega^2$. Dimensional counting would suggest

$$z_2 \sim \frac{a^2}{c_s^3}, \qquad \ell_{\rm mouth} \sim \frac{a^2}{L} \sim O(a),$$

so a 1PN breathing correction built from such a term would scale like

$$\epsilon\left(\sqrt{\epsilon}\,\frac{a}{r}\right) = \epsilon^{3/2+\delta},$$

and could therefore still reach 2.5PN if $\delta = 1$. So the mouth channel remained a real loophole until a stronger theorem was found.

That stronger theorem follows from three assumptions which are exactly the ones picked out by the natural compact branch: the static mouth problem is a compliance problem rather than a static flux law, the linear scalar mouth port is a reciprocal kinematic port

$$j = i\omega\beta q,$$

and the only linear dissipation is compact outgoing scalar radiation. Then the loaded scalar coordinate equation takes the form

$$u_\sigma(\omega) = \Big[K - (M + M_{\rm add})\omega^2 + iR_2\omega^3 + O(\omega^4)\Big]q(\omega),$$

so

$$Z_\sigma^{\rm eff}(\omega) = \frac{j}{u_\sigma} = i\frac{\beta}{K}\omega + i\frac{\beta(M+M_{\rm add})}{K^2}\omega^3 + \frac{\beta R_2}{K^2}\omega^4 + O(\omega^5).$$

Therefore

$$\boxed{z_2 = 0.}$$

The first dissipative mouth term is not $O(\omega^2)$ but $O(\omega^4)$. After subsystem reduction, the unresolved mouth contribution becomes a cubic odd scalar correction of schematic form

$$\delta\kappa_{\rm mouth}^{\rm ret}(\omega) = \delta\kappa_{\rm mouth}^{(0)} + \kappa_2\omega^2 - i\Lambda_3\left(\frac{\omega}{c_s}\right)^3 + O(\omega^4),$$

which in the old 1PN breathing sector scales as

$$\epsilon\left(\sqrt{\epsilon}\,\frac{a}{r}\right)^3 = \epsilon^{5/2+3\delta}.$$

So on any strict small-body branch, the compact mouth loophole is pushed above universal point-particle 2.5PN.

This also sharpens the final scalar no-go. Within the current ontology plus a local reversible finite-throat PDE completion, no derived linear Ohmic scalar mouth bleed appears. A dangerous Ohmic scalar sink would require genuinely new structure outside the natural compact-radiative branch, such as a semi-infinite one-dimensional outlet, an explicitly non-Hamiltonian resistive boundary layer, or a direct q-like scalar overlap source not captured by the continuity and reciprocal-boundary bookkeeping.

5.6 Scalar verdict

The stable outcome of the scalar audit is therefore mixed but substantially more favorable than the early-session picture. Several negative results remain real: projected continuity plus no-static-flux do not by themselves force $\gamma_1^{\text{eff}} = 0$; a naive isolated scalar Ward identity fails because the brane-visible scalar content is a projection overlap rather than the full bulk norm; and a simple cavity gap is not enough if a breathing mode inherits Ohmic leakage from a gapless outlet. So the scalar side is not theorem-closed from first principles.

At the same time, the derived scalar channels are now much more sharply constrained. The carried 2PN scalar support/geometry branch is finite-size only, not a universal point-particle 2.5PN killer. The breathing-to-outlet vertex derived from continuity is derivative-coupled or zero, not direct q-coupled, so the first derived odd breathing correction is cubic rather than linear. The exact scalar subsystem ledger has only two derived linear outlet channels — projected leakage and worldtube/mouth boundary flux — and the compact reciprocal-mouth analysis forces the dangerous dissipative coefficient z_2 to vanish. Within the current ontology plus a reversible finite-throat completion, no derived linear Ohmic scalar mouth bleed appears.

The correct summary statement is therefore

> Within the current derived ontology, scalar outlets are not generically direct $i\omega$ monopole killers, and on the natural compact/passive/outgoing small-body branch the derived scalar odd channels are derivative-coupled or finite-size suppressed above universal point-particle 2.5PN.

This is a demotion rather than a total erasure. The remaining scalar loopholes are narrow and explicit: exact projection-locking has not been proved, exact matter/geometry cancellation has not been derived from first principles, and a future PDE could still reintroduce a dangerous scalar channel by adding new structure outside the present ontology. But the scalar side is no longer the default kill switch of the 2.5PN program. That is why the audit can proceed to the carried dipole wake and then to the surviving quadrupole branch.

6 Dipole/vector-sector audit and demotion

This section addresses the second odd channel in the low-frequency hierarchy. Once the scalar side has been demoted from a generic universal point-particle obstruction to a derivative-coupled and/or finite-size branch, the next serious threat is the carried odd dipole wake already visible in the conservative 2PN packaging. That sector is not hypothetical. The carried zero-frequency residues

$$R_{1\perp} = \frac{7}{2}, \qquad R_{10} = 4,$$

come with explicit body-local ports, so the question is no longer whether some vector-like odd channel *could* exist, but whether this specific wake can survive retarded completion without spoiling the GR-like point-particle 2.5PN theorem. The logic of the section is therefore sharper than in the scalar audit. First, it shows that the carried odd wake is genuinely a relative-sector object rather than a disguised center-of-mass artifact. Second, it records the failure of the obvious algebraic rescues: momentum conservation, translational invariance, vanishing defect-centered dipole, and pure-Schott reinterpretation do not by themselves kill the odd dipole coefficients. Third, it shows that the natural outgoing $\ell = 1$ completion is not Ohmic: the absorptive piece begins at ω^3 rather than ω. Fourth, it proves that if a nonzero dipole wake still survives unsuppressed at the same formal 2.5PN order, then it cannot be absorbed into the standard Iyer–Will gauge family. The final demotion is therefore spectral and scaling-based rather than algebraic: on the clean outgoing small-body branch, the dipole wake is naturally pushed above universal point-particle 2.5PN and becomes a finite-size correction.

6.1 Why the dipole/vector sector is the next gate

After the scalar audit, the next odd channel in the low-frequency hierarchy is the dipole/vector branch. The reason it had to be treated as a genuine theorem gate is that the constructive 2PN packaging already contains a concrete odd wake sector rather than a merely hypothetical coupling. In the instantaneous pair frame,

$$\mathbf{v} = \mathbf{u} + d\,\mathbf{n}, \qquad \mathbf{u}\cdot\mathbf{n} = 0, \qquad \mathbf{n} = \frac{\mathbf{x}}{r}, \qquad d = \dot{r},$$

the carried odd ports are

$$\Pi^{(1\perp)} = \sqrt{\frac{7}{2}}\,\mathbf{u}, \qquad \Pi^{(10)} = 2d.$$

These are velocity-like objects, not static internal moments. So if either port acquires a physical odd retarded completion, it can feed directly into relative dissipation rather than only into a conservative response renormalization. The right theorem question is therefore sharp from the start:

> Is the carried odd dipole wake physically harmless in the reduced two-body problem, or does any retarded odd completion generate a genuine lower-order or same-order relative force?

6.2 Exact CM/relative content of the carried odd wake

The first possible rescue would be that the carried wake is secretly a center-of-mass artifact. It is not. Write

$$\mathbf{x}_A = \mathbf{X} + \eta_B\mathbf{x}, \qquad \mathbf{x}_B = \mathbf{X} - \eta_A\mathbf{x},$$
$$\eta_A = \frac{m_A}{M}, \qquad \eta_B = \frac{m_B}{M}, \qquad M = m_A + m_B.$$

so that

$$\mathbf{v}_A = \mathbf{V} + \eta_B\mathbf{v}, \qquad \mathbf{v}_B = \mathbf{V} - \eta_A\mathbf{v}, \qquad \mathbf{V} = \dot{\mathbf{X}}, \qquad \mathbf{v} = \dot{\mathbf{x}}.$$

Decompose both velocities relative to the orbital line of centers,

$$\mathbf{V} = \mathbf{U} + D\,\mathbf{n}, \qquad \mathbf{v} = \mathbf{u} + d\,\mathbf{n}.$$

Then the body-local odd ports become

$$\Pi_A^{(1\perp)} = \sqrt{\frac{7}{2}}\,(\mathbf{U} + \eta_B\mathbf{u}), \qquad \Pi_B^{(1\perp)} = \sqrt{\frac{7}{2}}\,(\mathbf{U} - \eta_A\mathbf{u}),$$

$$\Pi_A^{(10)} = 2(D + \eta_B d), \qquad \Pi_B^{(10)} = 2(D - \eta_A d).$$

The difference ports are therefore exactly

$$\boxed{\Pi_A^{(1\perp)} - \Pi_B^{(1\perp)} = \sqrt{\frac{7}{2}}\,\mathbf{u},} \qquad \boxed{\Pi_A^{(10)} - \Pi_B^{(10)} = 2d.}$$

This is the first theorem-level result of the dipole audit. The carried odd wake difference channel is pure relative velocity, not pure center-of-mass motion. The harmless mass-dipole identity

$$D_i^{\text{mass}} = MX_i$$

therefore does not rescue the carried wake sector. The dipole side cannot be dismissed by the same reasoning that removes the universal mass dipole from the center-of-mass frame.

6.3 Why the obvious algebraic rescues fail

Before turning to the outgoing spectral law, it is useful to record why the naive algebraic rescues fail. The minimal low-frequency retarded completion of the two carried dipole ports is

$$Y_{1\perp}^{\text{ret}}(\omega) = R_{1\perp} + i\sigma_\perp \omega + \cdots, \qquad Y_{10}^{\text{ret}}(\omega) = R_{10} + i\sigma_\parallel \omega + \cdots.$$

At this stage the point is not that this is the correct completion. The point is to ask whether *any* linear odd term would already be dangerous. In the relative variables, the local odd part of the influence functional reduces to

$$S_{\text{odd,loc}}^{(1)} = \frac{1}{2}\int dt\left[\frac{7}{2}\,\sigma_\perp\,\mathbf{u}_-\cdot\dot{\mathbf{u}}_+ + 4\,\sigma_\parallel\,d_-\dot{d}_+\right].$$

The Euler–Lagrange derivative then gives the relative jerk-type force

$$F_x = -\frac{7\sigma_\perp}{4}x^{(3)}, \qquad F_y = -\frac{7\sigma_\perp}{4}y^{(3)}, \qquad F_z = -2\sigma_\parallel z^{(3)}.$$

So any nonzero linear odd coefficient in either dipole channel produces a genuine relative dissipative force.

None of the obvious kinematic rescues removes this term. At the body-action level,

$$L_{\text{body}} = a\,(\dot{x}_{A-} - \dot{x}_{B-})(\ddot{x}_{A+} - \ddot{x}_{B+})$$

gives

$$F_A = -a\,(x_A^{(3)} - x_B^{(3)}), \qquad F_B = +a\,(x_A^{(3)} - x_B^{(3)}),$$

so that

$$F_A + F_B = 0.$$

Thus total momentum conservation is perfectly compatible with a nonzero odd dipole wake in the relative sector. In CM/relative variables the same term reduces to

$$L_{\text{body}} = a\,\dot{x}\,\ddot{x},$$

with no dependence on the center-of-mass coordinate at all, so translational invariance also does not kill the effect. Nor does the old vanishing worldtube-dipole theorem apply here: the carried odd ports are velocity-like auxiliary channels rather than static first moments of the defect profile.

The pure-Schott reinterpretation also fails generically. The power associated with the jerk-type force is

$$P = -\frac{d}{dt}\left[\frac{7\sigma_\perp}{4}(\dot{x}\,\ddot{x} + \dot{y}\,\ddot{y}) + 2\sigma_\parallel\dot{z}\,\ddot{z}\right] + \frac{7\sigma_\perp}{4}(\ddot{x}^2 + \ddot{y}^2) + 2\sigma_\parallel\ddot{z}^2.$$

Unless the odd coefficients vanish, there is a genuine non-total-derivative dissipative piece. So the carried odd wake is not generically "just Schott." At this stage of the audit the dipole sector therefore looked like a potentially fatal obstruction.

6.4 Outgoing $\ell = 1$ threshold law and the same-order obstruction

The first major rescue came from abandoning the generic Ohmic auxiliary picture and asking instead what happens if the dipole ports arise from a clean outgoing three-dimensional partial-wave channel. For a genuine outgoing $\ell = 1$ branch, the Dirichlet-to-Neumann eigenvalue behaves at low frequency as

$$\Lambda_1(k) = -\frac{2}{a} + ak^2 + ia^2k^3 + O(k^4), \qquad k = \frac{\omega}{c}.$$

So the absorptive part begins at

$$\Im\Lambda_1 \sim k^3,$$

not at k. After inversion and normalization to a fixed static residue, this becomes

$$Y_{1,\text{norm}}(k) = R\left[1 + \frac{1}{2}(ak)^2 + i\frac{1}{2}(ak)^3 + O((ak)^4)\right].$$

Hence the first odd dipole term is

$$\boxed{\Im Y_1^{\text{ret}}(\omega) \propto \left(\frac{a\omega}{c}\right)^3,}$$

with no linear $O(\omega)$ part. This materially changes the meaning of the earlier jerk-type failure: it becomes a warning about generic Ohmic auxiliary completion, not the necessary behavior of a clean outgoing dipole channel.

The rescue is real, but it is not complete. Because the ports are velocity-like, an $i\omega^3$ odd term still produces a fifth-derivative local force,

$$F_x = -\frac{7\chi_\perp}{4}x^{(5)}, \qquad F_y = -\frac{7\chi_\perp}{4}y^{(5)}, \qquad F_z = -2\chi_\parallel z^{(5)}.$$

So the dipole channel is no longer an earlier-than-2.5PN killer, but it still remains a possible *same-order* modifier of the local 2.5PN sector. The right next question is therefore not whether the outgoing law helps — it does — but whether a surviving same-order dipole term can be absorbed into the standard local 2.5PN family.

6.5 Same-order survival is incompatible with the standard 2.5PN family

After Newtonian order reduction, the isotropic fifth-derivative dipole structure sits in the same local basis as the standard 2.5PN force,

$$\{\dot{r}\,\mathbf{n}, \mathbf{v}\}.$$

Let $\lambda_\perp$ and $\lambda_\parallel$ denote the normalized transverse and longitudinal dipole coefficients after factoring out the standard overall 2.5PN prefactor. The dipole correction can then be written as

$$\Delta\mathbf{a}_{\text{dip}} = \frac{8G^2M^2\nu}{5c^5r^3}\left[\Delta A\,\dot{r}\,\mathbf{n} + \Delta B\,\mathbf{v}\right],$$

with

$$\Delta A = (9\lambda_\perp + 36\lambda_\parallel)v^2 + (-8\lambda_\perp - 22\lambda_\parallel)\frac{GM}{r} + (-45\lambda_\perp - 60\lambda_\parallel)\dot{r}^2,$$

$$\Delta B = -9\lambda_\perp v^2 + 8\lambda_\perp\frac{GM}{r} + 45\lambda_\perp\dot{r}^2.$$

The direct matching problem is then

$$\mathbf{a}_{\text{BT}} + \Delta\mathbf{a}_{\text{dip}} = s\,\mathbf{a}_{\text{IW}}(\alpha, \beta),$$

where $\mathbf{a}_{\mathrm{BT}}$ is the Burke–Thorne benchmark and $\mathbf{a}_{\mathrm{IW}}(\alpha, \beta)$ is the standard Iyer–Will two-parameter local family. The symbolic solve gives the unique solution

$$\boxed{\alpha = 4, \qquad \beta = 5, \qquad s = 1, \qquad \lambda_{\perp} = 0, \qquad \lambda_{\parallel} = 0.}$$

This is stronger than saying that the dipole term "merely shifts the coefficients a little." It means that if a nonzero dipole wake survives at the same formal 2.5PN order, it cannot be absorbed into the standard GR local 2.5PN gauge family at all. A same-order surviving dipole contribution would therefore represent a genuine universality-breaking deformation of the GR-like point-particle sector.

Two physical readings are especially clean. First, a nonzero transverse piece $\lambda_{\perp}$ changes the circular-orbit tangential damping coefficient itself, so it is immediately observable and not a gauge reshuffle. Second, even if the circular limit happened to look less dramatic, a nonzero longitudinal piece $\lambda_{\parallel}$ still breaks the generic-orbit structure and cannot be absorbed into the remaining Iyer–Will freedom. The dipole gate is therefore fully sharp: same-order survival is fatal for GR-like universality.

6.6 Small-body demotion and the conditional verdict

After the matching test, there is no longer any gauge-theoretic rescue route for a same-order dipole contribution. The only remaining rescue is scaling suppression. This rescue is available on the natural clean outgoing small-body branch.

The carried odd dipole wake sits on top of a conservative 1PN cross-sector ingredient, so it carries one explicit factor

$$\epsilon \equiv \left(\frac{v}{c}\right)^2.$$

The outgoing $\ell = 1$ odd factor contributes

$$\left(\frac{a\omega}{c}\right)^3.$$

Using orbital near-zone scaling,

$$\omega \sim \frac{v}{r} \sim \frac{c\sqrt{\epsilon}}{r},$$

one obtains

$$\frac{a\omega}{c} \sim \sqrt{\epsilon}\,\frac{a}{r}.$$

Hence the dipole odd correction scales as

$$\epsilon \left(\sqrt{\epsilon}\frac{a}{r}\right)^3 = \epsilon^{5/2} \left(\frac{a}{r}\right)^3.$$

If the small-body family obeys

$$\frac{a}{r} \sim \epsilon^{\delta}, \qquad \delta > 0,$$

then the dipole term enters at

$$\boxed{\epsilon^{5/2+3\delta},}$$

which lies strictly above universal point-particle 2.5PN. Representative cases are

$$\frac{a}{r} \sim \epsilon^{1/2} \Rightarrow 4\text{PN}, \qquad \frac{a}{r} \sim \epsilon \Rightarrow 5.5\text{PN}.$$

This is the main dipole rescue theorem of the audit:

$$\boxed{\text{With a clean outgoing } \ell = 1 \text{ completion, the carried dipole wake is naturally a finite-size correction rather than a universal point-particle 2.5PN obstruction.}}$$

The verdict is therefore conditional but meaningful. The dipole wake is real and structurally dangerous if treated naively. It is not removed by center-of-mass bookkeeping, momentum conservation, translational invariance, or other algebraic Ward-type arguments. But on the clean outgoing branch the absorptive part begins at ω^3, not ω, and in the strict small-body family the resulting correction is demoted above universal point-particle 2.5PN. What is *not* proven here is equally important: the current stack does not yet show from first principles that the moving-throat PDE enforces this clean outgoing $\ell = 1$ branch, excludes non-outgoing or Ohmic auxiliary completions, or makes the dipole coefficients vanish exactly rather than merely pushing them above the point-particle sector. The dipole/vector side is thus conditionally rescued, not yet fully first-principles closed.

7 Quadrupole sector: the surviving universal branch

This section treats the only odd channel left naturally in the universal point-particle 2.5PN slot after the scalar and dipole audits. Its task is not to invent quadrupole physics from scratch, but to decide which part of the model's already visible P_2 structure actually carries the theorem: the literal compact internal mouth/support quadrupole, or the orbital/worldtube STF source quadrupole. The logic is by now quite sharp. First, the solved 2PN P_2 ports already exhaust the full real STF $\ell = 2$ representation content. Second, a clean outgoing $\ell = 2$ completion automatically produces the correct positive $+i\omega^5$ retarded parity/sign. Third, the compact internal P_2 route is real but finite-size suppressed above universal point-particle 2.5PN. Fourth, the worldtube source decomposition identifies $\mu x_{\langle i}x_{j\rangle}$ as the universal source while the internal P_2 sector survives only as a finite-size correction. Fifth, rotational invariance collapses the low-frequency $P_2 \to$ STF matching problem to one scalar function, and the feared zero-overlap option fails on the natural small-body branch. The net result is that the quadrupole side is no longer a speculative escape hatch. It is the actual surviving universal theorem branch, with the remaining gap reduced to the final passive/outgoing normalization.

7.1 Why the quadrupole branch is now the central route

By the end of the scalar and dipole audits, the lower odd channels had been narrowed substantially but not erased from first principles. The scalar side no longer looked like a generic immediate killer once the breathing/outlet channel was shown to be continuity-mediated and derivative-coupled, and once the compact mouth channel was shown to be radiative rather than Ohmic. The carried dipole wake also no longer looked like an unavoidable same-order spoiler once the clean outgoing $\ell = 1$ threshold law demoted it in the strict small-body branch. Neither sector was mathematically finished, but both had been pushed out of the position of default unavoidable failure mode. That left the quadrupole channel as the only branch still naturally sitting in the right place to produce a universal point-particle 2.5PN reaction term.

This is also exactly what one would want physically. The whole purpose of the 2.5PN test was to see whether the model could reproduce the standard pattern: conservative closure through 2PN and then the first universal dissipative reaction at quadrupole order. Once the lower odd channels are narrowed or pushed into finite-size status, the remaining question is not whether the model contains some quadrupole-like decoration, but whether its own P_2 sector can actually support that universal branch.

The quadrupole audit therefore begins with one necessary separation. The direct 2PN readout already contains a concrete real P_2 multiplet,

$$\Pi^{(20)}, \qquad \Pi^{(21c)}, \qquad \Pi^{(21s)}, \qquad \Pi^{(22c)}, \qquad \Pi^{(22s)},$$

with solved even zero-frequency support residues

$$R_{20} = R_{21} = R_{22} = 1,$$

and a separate geometry-closure channel which is scalar rather than quadrupolar. So the question was never whether the model shows any sign of $\ell = 2$ at all. The real question is how that P_2 content should be interpreted. The rest of the section shows that two distinct routes are hiding behind the same apparent quadrupole structure:

1. a literal compact internal mouth/support quadrupole,
2. the universal orbital/worldtube STF source quadrupole.

The first route is real but finite-size; the second is the one that survives in the universal point-particle theorem.

7.2 The solved P_2 bundle already is the full real STF $\ell = 2$ content

In the instantaneous pair frame write

$$\mathbf{v} = \mathbf{u} + d\,\mathbf{n}, \qquad \mathbf{u}\cdot\mathbf{n} = 0, \qquad \mathbf{u} = (u_x, u_y, 0), \qquad \mathbf{v} = (u_x, u_y, d).$$

The solved real P_2 ports are then

$$\Pi^{(20)} = \frac{1}{2}(3d^2 - v^2), \qquad \Pi^{(21c)} = \sqrt{2}\,du_x, \qquad \Pi^{(21s)} = \sqrt{2}\,du_y,$$

$$\Pi^{(22c)} = \frac{u_x^2 - u_y^2}{2\sqrt{2}}, \qquad \Pi^{(22s)} = \frac{2u_xu_y}{2\sqrt{2}}.$$

Now form the STF tensor

$$Q_{ij} = v_iv_j - \frac{1}{3}\delta_{ij}v^2.$$

In a real orthonormal STF basis $E_{20}, E_{21c}, E_{21s}, E_{22c}, E_{22s}$, the exact pair-frame decomposition is

$$q_{20} = \sqrt{\frac{2}{3}}\,\Pi^{(20)}, \qquad q_{21c} = \Pi^{(21c)}, \qquad q_{21s} = \Pi^{(21s)},$$

$$q_{22c} = 2\Pi^{(22c)}, \qquad q_{22s} = 2\Pi^{(22s)}.$$

Equivalently,

$$Q_{ij} = \sqrt{\frac{2}{3}}\,\Pi^{(20)}E_{20,ij} + \Pi^{(21c)}E_{21c,ij} + \Pi^{(21s)}E_{21s,ij} + 2\Pi^{(22c)}E_{22c,ij} + 2\Pi^{(22s)}E_{22s,ij}.$$

So the solved 2PN P_2 package is not merely quadrupole-like; it is exactly the full real STF $\ell = 2$ representation content.

This is the first theorem-level positive result on the quadrupole side. The missing PDE does not need to create an $\ell = 2$ representation from scratch. That structure is already present. What remains is to determine how the already isolated real P_2 content matches onto the actual source/radiative quadrupole of the reduced two-body dynamics.

7.3 Outgoing $\ell = 2$ completion gives the correct parity and sign, while the compact internal route is finite-size only

For a clean outgoing three-dimensional quadrupole partial wave,

$$\Lambda_2(k) = -\frac{3}{a} + \frac{a}{3}k^2 + \frac{a^3}{9}k^4 + i\frac{a^4}{9}k^5 + O(k^6), \qquad k = \frac{\omega}{c_s}.$$

After inversion and normalization to fixed static residue R_2, the retarded admittance is

$$Y_2^{\mathrm{ret}}(k) = R_2\left[1 + \frac{a^2k^2}{9} + \frac{4a^4k^4}{81} + i\frac{a^5k^5}{27} + O(k^6)\right].$$

Therefore

$$\Im Y_2^{\mathrm{ret}}(\omega) = \frac{R_2 a^5}{27c_s^5}\,\omega^5 + O(\omega^7),$$

with *positive* sign on the outgoing branch. This is exactly the desired quadrupole parity law. Just as the scalar and dipole threshold analyses gave odd terms beginning at ω and ω^3 respectively, the clean outgoing $\ell = 2$ branch begins at ω^5, which is precisely the universal local 2.5PN quadrupolar slot.

The sign is equally important. For one quadrupole component with local odd kernel

$$L_{\mathrm{rr}}^{(2)} = -\frac{\Gamma}{2}\,q_- q_+^{(5)}, \qquad \Gamma > 0,$$

one finds

$$F = -\frac{\Gamma}{2}q^{(5)}, \qquad P = F\dot{q} = \frac{dE_{\mathrm{Schott}}}{dt} - \frac{\Gamma}{2}(q^{(3)})^2.$$

So the positive $+i\omega^5$ coefficient corresponds to damping rather than antidamping. By the time one reaches this point of the audit, the quadrupole side has already passed three major gates: the representation exists, the odd parity is correct, and the sign is dissipative.

At the same time, this calculation also clarifies what the literal compact internal P_2 mouth/support route can and cannot do. If one treats that quadrupole as an internal support mode of size a, then the conservative support sector already carries a 2PN prefactor,

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \epsilon^2,$$

and the outgoing threshold factor contributes

$$\left(\frac{a\omega}{c_s}\right)^5 \sim \left(\sqrt{\epsilon}\,\frac{a}{r}\right)^5.$$

The total scaling is therefore

$$\epsilon^2\left(\sqrt{\epsilon}\,\frac{a}{r}\right)^5 = \epsilon^{9/2}\left(\frac{a}{r}\right)^5,$$

which lies above universal point-particle 2.5PN even before any additional small-body suppression is imposed. The compact internal P_2 route is thus real, but it is too small to be the universal theorem driver. It belongs to the finite-size throat-response sector rather than to the universal point-particle branch.

This is not a defeat for the quadrupole program. It is a clarification. The project does not need the compact internal P_2 route to carry the theorem. Indeed the theorem is cleaner if it does not. The internal P_2 bundle can remain as a higher-order finite-size signature while the universal point-particle 2.5PN term is carried instead by the orbital/worldtube STF quadrupole.

7.4 The universal source is the orbital/worldtube STF quadrupole

For one compact defect A, in defect-centered coordinates $\boldsymbol{\xi}$, the source quadrupole is

$$I_{ij}^{(A)} = \int d^3\xi\, \rho_A(\boldsymbol{\xi}, t)\big(X_A^i + \xi^i\big)\big(X_A^j + \xi^j\big)^{\mathrm{STF}}.$$

Expanding gives

$$I_{ij}^{(A)} = m_A X_A^{\langle i} X_A^{j\rangle} + 2X_A^{\langle i} d_A^{j\rangle} + Q_A^{ij},$$

with

$$d_A^i \equiv \int d^3\xi\, \rho_A \xi^i, \qquad Q_A^{ij} \equiv \int d^3\xi\, \rho_A \xi^{\langle i}\xi^{j\rangle}.$$

The carried worldtube reduction already imposes vanishing defect-centered dipole, so

$$d_A^i = 0.$$

Hence

$$I_{ij}^{(A)} = m_A X_A^{\langle i} X_A^{j\rangle} + Q_A^{ij},$$

and after summing the two bodies and going to the center-of-mass frame one gets

$$I_{ij}^{\mathrm{eff}} = \mu x_{\langle i} x_{j\rangle} + Q_A^{ij} + Q_B^{ij} + O(c^{-2}, a^3\nabla^3\Phi).$$

This is the decisive source split:

$$\boxed{I_{ij}^{\mathrm{orb}} = \mu x_{\langle i} x_{j\rangle}}$$

is the universal point-particle source, while

$$Q_A^{ij} + Q_B^{ij}$$

is the finite-size internal worldtube correction sector.

The same statement can be written directly at the pair-frame level. For the orbital source quadrupole,

$$I_{ij}^{\mathrm{orb}} = \mu x_{\langle i} x_{j\rangle},$$

Newtonian order reduction yields

$$\ddot{I}_{ij}^{\mathrm{orb}} = 2\mu\left(v_{\langle i} v_{j\rangle} - \frac{GM}{r^3} x_{\langle i} x_{j\rangle}\right).$$

In the same STF basis used above, this becomes

$$\frac{1}{2\mu}\ddot{I}_{ij}^{\mathrm{orb}} = \sqrt{\frac{2}{3}}\left(\Pi^{(20)} - \frac{GM}{r}\right) E_{20,ij} + \Pi^{(21c)} E_{21c,ij} + \Pi^{(21s)} E_{21s,ij} + 2\Pi^{(22c)} E_{22c,ij} + 2\Pi^{(22s)} E_{22s,ij}.$$

This is the usable source-map formula. It shows that the full real quintet is already present, that the $|m| = 1, 2$ pieces are purely kinematic, and that only the axisymmetric $m = 0$ component carries the static source shift. The solved 2PN P_2 throat-support structure and the orbital/source STF quadrupole are therefore not competing stories. They are different views of the same real STF content.

7.5 Rotational invariance collapses the low-frequency $P_2 \to \mathrm{STF}$ map to one scalar function

Once the source map is known structurally, the quadrupole problem is no longer one of representation content but of matching:

$$\text{retarded throat/worldtube } P_2 \text{ basis} \longrightarrow \text{canonical STF source basis.}$$

Could the matching matrix be singular, anisotropic, or otherwise pathological at low frequency? Under rotational invariance, the answer is no.

The exact commutator test with the real $\ell = 2$ rotation generators shows that the commutant of the real STF $\ell = 2$ representation is scalar. Hence the low-frequency matching matrix must take the form

$$\mathcal{M}^R_{ab}(\omega) = \left[m_0 + m_2\omega^2 + m_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab}.$$

This eliminates an entire class of potential quadrupole failure modes at once. The map cannot selectively project out some STF components while keeping others, it cannot secretly rotate the source into an inequivalent tensor channel, and the retarded odd coefficient enters as a single scalar spectral datum.

On the natural passive/outgoing branch, the absorptive part must be positive semidefinite. In the present scalar-identity situation, that simply means

$$\Gamma_5 > 0.$$

So the low-frequency retarded quadrupole branch is already almost fully characterized:

$$\boxed{\mathcal{M}^R_{ab}(\omega) = \left[m_0 + m_2\omega^2 + m_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab}, \qquad \Gamma_5 > 0.}$$

At this point the quadrupole side is almost theorem-ready. Only one serious loophole remains: perhaps the map is isotropic and the odd coefficient is positive, but the static overlap vanishes so that the orbital quadrupole is projected away already at zero frequency.

7.6 The static overlap does not vanish on the natural branch

That last loophole is precisely the feared $m_0 = 0$ option. In the instantaneous pair frame, however,

$$x_{\langle i}x_{j\rangle} = \sqrt{\frac{2}{3}}\, r^2 E_{20,ij},$$

so the static orbital/source configuration already carries a nonzero canonical STF $m = 0$ component. Since the static low-frequency map has already been forced by rotational invariance to be

$$\mathcal{M}_0 = m_0\, \mathbf{1}_5,$$

one nonzero $m = 0$ matrix element forces the whole static overlap coefficient to be nonzero:

$$\boxed{m_0 \neq 0.}$$

On the natural small-body branch the result can be summarized more sharply as

$$\boxed{m_0 = 1 + O\!\left(\frac{a^2}{r^2}\right).}$$

The only obvious loophole would be an exact cancellation between the orbital STF source and the internal worldtube quadrupoles. But the same small-body reduction used throughout the lower-order papers already tells us that the first finite-size correction is quadrupolar and suppressed by

$$O\left(\frac{a^2}{r^2}\right).$$

So a natural total coefficient has the form

$$C_{20}^{\rm total} = C_{20}^{\rm orb}\Big[1 + \lambda(a/r)^2 + \cdots\Big], \qquad \lambda = O(1).$$

Exact cancellation would require

$$\lambda = -\left(\frac{r}{a}\right)^2,$$

which is parametrically huge and therefore lies outside the controlled small-body hierarchy. The $m_0 = 0$ branch is thus not a natural branch of the present reduction. It would require a tuned non-small-body cancellation, a radically anisotropic PDE branch, or a more serious breakdown of the hierarchy itself.

The quadrupole verdict is now short and stable:

The solved 2PN P_2 bundle already contains the full real STF $\ell = 2$ content;
the clean outgoing $\ell = 2$ branch supplies the correct positive $+i\omega^5$ slot;
the compact internal P_2 route is finite-size only;
and the surviving universal point-particle source is the orbital/worldtube STF quadrupole.

What remains open is no longer the existence of the branch, but its final normalization. On the natural isotropic/passive/outgoing small-body branch, the entire unresolved quadrupole question has been reduced to

$$\Gamma_5 \stackrel{?}{=} \Gamma_{5,\rm GR}.$$

That is the narrow PDE-side gap carried into the next section.

8 Conditional 2.5PN theorem and the remaining normalization gap

This section closes the channel audit begun in Section 4 and completed in Sections 5–7. Its purpose is not to introduce a new dynamical sector, but to compress the whole 2.5PN analysis into the strongest theorem statement that the present stack can honestly support. By this point the picture is much sharper than it was at the start of the campaign. The lower odd scalar and dipole channels are no longer generic universal spoilers on the natural compact/passive/outgoing small-body branch, the surviving universal point-particle route is the orbital/worldtube STF quadrupole, and the broad question "can the model host a GR-like dissipative gate at all?" has narrowed to one specific remaining issue: the final normalization of the passive/outgoing quadrupole branch.

8.1 Best current conditional theorem statement

The cleanest theorem statement is now conditional rather than unconditional. Assume that the completed moving-throat PDE satisfies all of the following on its physical low-frequency branch:

1. there is no new direct scalar source beyond the continuity-mediated and reciprocal-boundary ontology isolated in Sections 5 and 4,

2. the scalar and vector outlets are compact and outgoing at low frequency, so that the surviving scalar and dipole channels begin at the derivative/high-order thresholds established earlier,
3. the low-frequency defect/worldtube sector is rotationally invariant,
4. the surviving STF quadrupole branch is passive and outgoing,
5. the strict small-body hierarchy remains valid,
6. and the static orbital/source overlap remains on the natural branch,

$$\hat{m}_0 = 1 + O\left(\frac{a^2}{r^2}\right), \qquad \hat{m}_0 \neq 0.$$

Then the reduced nonspinning two-body dynamics has the structure

$$\mathbf{a} = \mathbf{a}_{\mathrm{N}} + \mathbf{a}_{1\mathrm{PN}} + \mathbf{a}_{2\mathrm{PN}} + \mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}} + \Delta\mathbf{a}^{>2.5\mathrm{PN}}_{\mathrm{fs}},$$

where $\mathbf{a}^{\mathrm{quad}}_{2.5\mathrm{PN}}$ is the standard local Burke–Thorne / Iyer–Will quadrupolar reaction term generated by the orbital/worldtube STF quadrupole, while all other currently derived odd channels are pushed above universal point-particle 2.5PN.

This is already a real theorem statement, but not yet an unconditional one. What remains unproved from first principles is not the whole architecture of the branch. What remains is the final normalization of the quadrupole coefficient on the actual completed PDE branch.

8.2 What is now effectively proven

The strongest positive result of the paper is that the lower odd channels are no longer the default theorem killers. Writing

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \frac{a}{r} \sim \epsilon^{\delta}, \qquad \delta > 0,$$

the continuity-derived breathing outlet and the clean outgoing dipole wake both enter only at

$$\epsilon^{5/2+3\delta},$$

while the compact internal P_2 mouth/support route enters only at

$$\epsilon^{9/2+5\delta}.$$

All of these therefore lie above universal point-particle 2.5PN on the strict small-body branch. The only branch left naturally in the universal slot is the orbital/worldtube STF quadrupole.

That surviving branch is no longer merely conjectural. The earlier sections already established that the solved 2PN P_2 package is exactly the full real STF $\ell = 2$ representation, that the orbital/worldtube source map exists and is nondegenerate on the natural branch, that rotational invariance collapses the low-frequency retarded matching matrix to the scalar form

$$\mathcal{M}^R_{ab}(\omega) = \left[\hat{m}_0 + \hat{m}_2\omega^2 + \hat{m}_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab}, \qquad \hat{m}_0 \neq 0, \qquad \Gamma_5 > 0,$$

and that the outgoing $\ell = 2$ branch has the correct positive $+i\omega^5$ parity/sign. In other words, the surviving universal branch is already fixed as a source-content, parity, and sign statement. What is no longer open is whether the model can support *some* quadrupolar odd term. What remains open is whether the actual moving-throat PDE realizes the branch with the correct normalization.

8.3 The remaining normalization gap

The final unresolved question can be stated in one line. If the worldtube/throat coordinate is normalized as

$$q_a^{\mathrm{toy}} = \hat{m}_0\, q_a^{\mathrm{can}},$$

and the retarded STF matching coefficient on the same branch is Γ_5, then the canonically normalized odd coefficient is controlled only by the product

$$\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \hat{m}_0^2 \Gamma_5.$$

Matching to Burke–Thorne is therefore equivalent to the scalar condition

$$\boxed{\hat{m}_0^2 \Gamma_5 = \frac{2G}{5c^5}.}$$

This is the sharpest form of the remaining theorem target.

The same statement can be rewritten in canonical invariant language. If one defines

$$\overline{K}_0 \equiv \hat{m}_0^2 K_0, \qquad \overline{K}_2 \equiv \hat{m}_0^2 K_2, \qquad \overline{K}_4 \equiv \hat{m}_0^2 K_4, \qquad \overline{\Gamma}_5 \equiv \hat{m}_0^2 \Gamma_5^{\mathrm{port}},$$

then the compact passive/outgoing isotropic $\ell = 2$ branch forces the exact low-frequency identities

$$\overline{K}_4 = \frac{4\overline{K}_2^2}{\overline{K}_0}, \qquad \overline{\Gamma}_5 = 9\, \frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}.$$

So once the canonical even pair $(\overline{K}_0, \overline{K}_2)$ is known on the same branch, the next even moment and the odd Burke–Thorne coefficient are no longer independent unknowns. They are fixed automatically. Equivalently, the current stack already determines the whole low-frequency quadrupole branch up to one remaining invariant scalar, which may be taken to be $\overline{K}_0$ or, in the directly dynamical language above, the product $\hat{m}_0^2 \Gamma_5$.

This is why the remaining theorem gap should be described as a normalization problem rather than a missing tensor-structure problem. The representation content is fixed, the sign is fixed, the static overlap is fixed on the natural branch, and even the allowed low-frequency branch relations are fixed. What is not yet fixed from the completed PDE is the final far-zone normalization that turns the surviving quadrupole structure into the correctly normalized point-particle 2.5PN coefficient.

8.4 Why the gap is serious but narrow

The remaining gap is serious because without it one cannot honestly claim an unconditional theorem. A merely structural quadrupole term is not enough. The coefficient must match the GR-like point-particle normalization. But the gap is also narrow in a way that was not true earlier in the session. The current paper has already eliminated, demoted, or strongly constrained nearly every other natural failure mode:

1. generic direct scalar-monopole failure routes,
2. generic same-order dipole contamination,
3. quadrupole representation mismatch,
4. vanishing static source-map overlap,
5. and wrong-sign passive/outgoing quadrupole behavior.

The remaining loopholes should therefore be treated as exceptional rather than natural branches. They include: a genuinely new direct scalar source outside the current ontology; a noncompact or effectively one-dimensional dissipative outlet; a strong anisotropy or singular $P_2 \to$ STF map; a breakdown of the strict small-body hierarchy itself; or a non-passive/non-outgoing quadrupole completion. None of these loopholes was derived on the natural branch isolated by the present audit.

The correct closing verdict is therefore the following. Within the declared closure hierarchy, a full GR-like point-particle 2.5PN theorem is now conditionally reachable from the present stack. The lower odd channels are no longer generic universal spoilers, the quadrupole route is the unique surviving universal branch, and the whole program has narrowed to the final determination of the passive/outgoing quadrupole normalization. That is the proper theorem endpoint of the present paper and the natural starting point for the next moving-throat calculation.

9 Fixed versus symbolic quantities

One of the main purposes of the present paper is to make the status of the 2.5PN theorem ledger explicit. Earlier stages of the program necessarily mix several logically different kinds of objects: imported lower-order constants, appendix-side support data, branch-conditioned structural results, and genuinely open dynamic quantities of the unresolved moving-throat completion. The point of this section is not to weaken the main theorem claim, but to sharpen it. The present 2.5PN audit does *not* fix a new long list of universal numerical coefficients in the same way the conservative 2PN paper did. Instead, it fixes the channel structure of the first dissipative PN gate, identifies which data are already frozen from the earlier hierarchy, and isolates what still has to remain symbolic.

The clean distinction is the following:

1. quantities frozen by the parent 4D/1PN/2PN derivation chain and therefore imported unchanged into the present audit,
2. quantities fixed by the present 2.5PN analysis once the natural compact/passive/outgoing small-body branch is declared,
3. and quantities that should still remain symbolic because they belong to the unresolved dynamic throat/DtN/PDE completion, to profile-dependent finite-size structure, or to sectors outside the present theorem.

9.1 Frozen carry-forward inputs

The lower-order conservative backbone is imported as frozen input. In particular,

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\rm add} = \frac{1}{2}, \qquad \kappa_{\rm PV} = \frac{3}{2}, \qquad \beta_{\rm 1PN} = 3,$$

with the understanding that $\kappa_{\rm PV} = 3/2$ and therefore $\beta_{\rm 1PN} = 3$ are carried within the declared adiabatic one-degree-of- freedom breathing closure rather than as assumption-free moving-throat theorems. The parity-even wake data are likewise frozen,

$$\alpha^2 = \frac{3}{4}, \qquad a_H = 0, \qquad K_{\rm vec} = \frac{2}{\pi^2}, \qquad \frac{L}{a} \approx 1.85.$$

These numbers are not retuned in the present paper. They define the inherited Newtonian+1PN+2PN viability branch against which the first dissipative gate is tested.

The same is true of the zero-frequency 2PN throat/support data. The carried odd wake residues are

$$R_{1\perp} = \frac{7}{2}, \qquad R_{10} = 4,$$

while the frozen even support/source layer satisfies

$$R_0 = R_{20} = R_{21} = R_{22} = 1, \qquad J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0\right), \qquad \Delta_{\rm geom} = \frac{281}{80}.$$

These are already fixed by the conservative 2PN package. The present paper uses them structurally; it does not reopen them as free parameters.

9.2 Quantities fixed by the present 2.5PN audit

What the present paper fixes is primarily *status*, not a new fit ledger. First, Section 4 establishes the decisive benchmark: a purely real-even conservative low-frequency kernel cannot generate a local 2.5PN reaction term, while the minimal retarded STF quadrupole benchmark lands exactly on the standard Burke–Thorne / Iyer–Will family member. So it is now fixed that any successful 2.5PN theorem must come from a genuinely retarded odd sector, and that the correct local target is quadrupolar rather than scalar or dipolar.

Second, Sections 5 and 6 fix the status of the lower odd channels on the natural compact/passive/outgoing small-body branch. Writing

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \frac{a}{r} \sim \epsilon^{\delta}, \qquad \delta > 0,$$

the continuity-derived scalar breathing outlet and the clean outgoing dipole wake both enter only at

$$\epsilon^{5/2+3\delta},$$

while the literal compact internal P_2 mouth/support route enters only at

$$\epsilon^{9/2+5\delta}.$$

Thus the present audit fixes that these channels are not universal point-particle 2.5PN drivers on the strict small-body branch. They survive only as higher-order finite-size or derivative-protected sectors.

Third, Section 7 fixes the surviving universal branch much more sharply than was possible at the start of the audit. The solved real P_2 bundle already exhausts the full STF $\ell = 2$ representation content, with the fixed diagonal basis conversion

$$q^{\rm can} = B\, q^{\rm port}, \qquad B = \mathrm{diag}\left(\sqrt{\frac{2}{3}}, 1, 1, 2, 2\right), \qquad \det B \neq 0,$$

so there is no remaining representation mismatch between the real throat ports and the canonical orbital/worldtube quadrupole. On the same natural branch the retarded matching problem collapses under isotropy to the scalar form

$$\mathcal{M}^R_{ab}(\omega) = \left[\hat{m}_0 + \hat{m}_2\omega^2 + \hat{m}_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab}, \qquad \hat{m}_0 \neq 0, \qquad \Gamma_5 > 0.$$

So the present audit fixes all of the following: the universal branch is the orbital/worldtube STF quadrupole; its passive/outgoing odd term has the correct $+i\omega^5$ parity/sign; and the static overlap is nonzero on the natural branch.

Fourth, once the same compact passive/outgoing isotropic $\ell = 2$ branch is assumed, the low-frequency quadrupole coefficients are no longer independent. Defining the canonical invariant coefficients

$$\overline{K}_0, \qquad \overline{K}_2, \qquad \overline{K}_4, \qquad \overline{\Gamma}_5,$$

the branch fingerprint is fixed to

$$\overline{K}_4 = \frac{4\overline{K}_2^2}{\overline{K}_0}, \qquad \overline{\Gamma}_5 = 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}.$$

This is a genuine fixed result of the present branch analysis. It means that once the canonical even pair $(\overline{K}_0, \overline{K}_2)$ is known on that same branch, the next even moment and the odd Burke–Thorne coefficient follow automatically.

9.3 Quantities that remain symbolic or open

What remains symbolic is now much narrower than it was before the 2.5PN push, but it is still real. The first open class is the dynamic throat/DtN completion already inherited from the 2PN paper. The zero-frequency operator is frozen, but the dynamical pole or compliance data are not:

$$\Omega_{1\perp},\ \Omega_{10},\ \Omega_0,\ \Omega_{20},\ \Omega_{21},\ \Omega_{22},\ \Omega_g.$$

Equivalently, the first genuinely new inner-throat dynamic observables remain the low-frequency even coefficients of the corresponding response functions. The present paper uses these objects structurally but does not derive them from the fully solved moving-throat PDE.

The second open class is the final quadrupole normalization itself. The present audit fixes the surviving branch shape, the sign, the representation content, and the invariant relations among the low-frequency coefficients. What it does not yet derive is the remaining invariant normalization scalar,

$$\mathcal{N}_Q \equiv \hat{m}_0^2 K_0, \qquad \gamma_{\text{quad}}^{\text{eff}} = \hat{m}_0^2 \Gamma_5^{\text{port}} = \mathcal{N}_Q \frac{a^5}{27 c_s^5}.$$

Equivalently, the unresolved theorem target is still

$$\hat{m}_0^2 \Gamma_5 = \frac{2G}{5c^5},$$

or, in canonical invariant language, the model-side determination of the pair $(\overline{K}_0, \overline{K}_2)$ that would force the corresponding $\overline{K}_4$ and $\overline{\Gamma}_5$ automatically. This is the narrow normalization gap already isolated in Section 8.

The third open class consists of broader symbolic data already outside the conservative 2PN and conditional 2.5PN theorem envelopes. These include the absolute dimensional normalization of the parent medium,

$$P(\rho) = K_{\text{EOS}} \rho^5,$$

with K_{EOS} still unfixed, the detailed projection and localization profiles

$$W(w), \qquad Z(w),$$

finite-size moments and higher worldtube multipoles beyond the universal point-particle limit, and more general dynamic response data such as damping, phase lag, multi-mode mixing, driven/open-system couplings, or noncompact outlet behavior outside the compact passive/outgoing branch adopted here.

Finally, several broader sectors remain outside the present paper altogether: spin couplings, strong-field completion, higher post-Newtonian orders beyond the present gate, and the fully solved moving-throat bulk PDE for arbitrary motion. The same caution applies to non-natural loopholes isolated in the audit, such as new direct scalar sources outside the continuity-mediated ontology, effectively one-dimensional dissipative outlets, or singular anisotropic $P_2 \to$ STF matching. None of these branches is derived on the natural channel selected by the present paper, but neither are they claimed away by a fully assumption-free microscopic theorem.

9.4 Interpretation

The cleanest summary of the section is therefore this:

> Within the declared closure hierarchy, what is fixed at 2.5PN is no longer a large free coefficient ledger but the universal channel structure itself. The lower odd scalar and dipole channels are demoted on the natural branch, the orbital/worldtube STF quadrupole is the unique surviving universal driver, and the remaining symbolic content is concentrated in the dynamic throat completion and the final invariant quadrupole normalization.

That is a materially sharper status statement than was available before the present audit. The current file stack does not yet give an unconditional first-principles 2.5PN theorem of the completed moving-throat PDE. But it does fix the correct theorem envelope very tightly: the broad structural question on the natural branch has been sharply narrowed, while the remaining open content has collapsed to one narrow normalization problem and the deeper symbolic observables from which that normalization must eventually be derived.

10 Verification and reproducibility ledger

The derivation in this paper is analytic rather than computational. The accompanying SymPy and Mathematica files are therefore not presented as black- box generators of the theorem. Their role is narrower and more auditable: they serve as a referee archive for the algebraic identities, low-frequency expansions, channel-audit statements, matching theorems, and normalization formulas used in the main text. This is the same stance already adopted in the conservative 2PN paper, but with a dual-CAS check at the present dissipative gate. The archive exists to make the present 2.5PN argument reproducible *within its declared closure hierarchy*; it does not replace any analytic step, and it does not upgrade the paper into an assumption-free theorem of a fully solved moving-throat bulk PDE.

At 2.5PN the verification story is naturally split into three layers. The first layer is the inherited lower-order archive carried forward from the 4D/1PN/2PN papers. That layer freezes the Newtonian+1PN+2PN backbone, its conventions, and its already-audited coefficient ledger. The second layer is the first 2.5PN master audit, implemented in parallel as `2_5pn_master_sympy_audit.py` and `2_5pn_master_mathematica_audit.wl`, whose scope is the core theorem chain behind Sections 4–7: the decisive Burke–Thorne benchmark, the low-frequency odd-channel framework, the scalar and dipole audits, and the core quadrupole representation/source-map checks. The third layer is the session-completion master audit, again implemented in parallel as `2_5pn_master_session_sympy_audit.py` and `2_5pn_master_session_mathematica_audit.wl`, which retain the earlier channel audit and extend it through the later quadrupole-normalization package: canonical normalization maps, outgoing $\ell = 2$ fingerprints, convention-invariant normalization products, grouped real P_2 formulas, the minimal isotropic quadrupole module, and the single-pole sufficiency theorem discussed in Section 8.

A practical consequence of this organization is that the 2.5PN archive is already rerunnable as a small number of named standalone scripts/harnesses. The SymPy and Mathematica master audits are written as direct referee-facing artifacts and do not depend on hidden notebook state or a local import tree. The first pair is the shortest executable route back through the channel selection theorems; the second pair is the shortest executable route back through the quadrupole-normalization algebra that remains after the channel audit has finished.

10.1 What the present archive verifies

The archive verifies the load-bearing symbolic statements of the paper sector by sector.

First, the benchmark module checks the exact STF two-body source decomposition, checks the closed-form Burke–Thorne reduction to the standard local basis, and verifies that the result lands on the Iyer–Will family member $(\alpha, \beta) = (4, 5)$. This is the decisive positive benchmark used in Section 4. The same audit also fixes the low- frequency odd-channel hierarchy $n = 1, 3, 5$ and the associated dissipation/Schott bookkeeping in explicit time- domain form.

Second, the scalar and dipole modules verify the demotion statements proved in Sections 5 and 6. On the scalar side, the first master audit checks the Gaussian projection-overlap counterexample, the exact continuity identity behind the projection-locking discussion, the derivative-coupled breathing-outlet classification, and the reciprocity/radiative-order result that pushes the compact mouth dissipation above the naive Ohmic threshold. On the dipole side, it checks the exact center-of-mass/relative split of the carried odd wake, the failure of the naive equal-and-opposite-force rescue, the outgoing $\ell = 1$ threshold law eliminating a dangerous linear odd term, the same-order matching obstruction against the standard 2.5PN family, and the final small- body scaling demotion.

Third, the first master audit verifies the surviving quadrupole branch itself. It checks that the solved real P_2 ports exhaust the full STF $\ell = 2$ representation content, that the port-to-canonical basis map is invertible, that the source map to the orbital/worldtube STF quadrupole is explicit, that the rotationally invariant retarded matching matrix collapses to a scalar multiple of the identity, that the clean outgoing $\ell = 2$ branch starts at a positive $+ik^5$ term, and that the static overlap gate passes with $\hat{m}_0 \neq 0$ on the natural low-frequency branch.

Fourth, the session-completion master audit verifies the normalization stack that lies behind Section 8. It checks the canonical quadrupole normalization map, the convention-invariant product controlling the physical odd coefficient, the grouped low-frequency trace/anomaly formulas for real $P_{20} \oplus P_{21} \oplus P_{22}$ data, the minimal isotropic quadrupole module, the single-pole sufficiency theorem, and the exact branch identities

$$\overline{K}_4 = \frac{4\overline{K}_2^2}{\overline{K}_0}, \qquad \overline{\Gamma}_5 = 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}},$$

which show that once the even conservative pair $(\overline{K}_0, \overline{K}_2)$ is fixed on the passive/outgoing isotropic branch, the next even moment and the Burke–Thorne odd coefficient follow automatically.

The archive therefore verifies exactly the kind of statements this paper needs: not a full PDE theorem, but explicit algebraic closure of the benchmark, selection-rule, demotion, source-map, and normalization identities used in the main text.

10.2 Reproducibility conventions

Four reproducibility conventions are worth recording explicitly.

Named symbolic checks. Every nontrivial statement in the master audits is attached to a human-readable section heading and then reduced to an explicit symbolic residual or solved coefficient relation. The helper routines make this audit trail visible rather than implicit. The point is that a referee can compare the manuscript's prose directly with a named symbolic check, instead of treating the archive as an opaque computational oracle. The same theorem chain is now carried in both SymPy and Mathematica so that agreement does not rely on a single CAS.

Local assumptions. The scripts do not bury the regime assumptions inside an unreported notebook state. Positivity domains, the passive/outgoing branch choice, the strict small-body ordering, isotropy hypotheses, and the static-overlap gate appear in the relevant modules where they are used. This keeps the distinction between exact identities and branch-conditioned statements aligned with the claim hierarchy used in the paper, and it makes SymPy/Mathematica mismatches easier to interpret if one ever appears.

Explicit bases and invariant ledgers. Reproducibility comes from exposing the actual representation bases and invariant quantities being solved on. In the first master audit this means, for example, the explicit real STF basis and the explicit port-to-canonical basis map. In the second audit it means the grouped real P_2 trace/anomaly variables, the minimal isotropic module, and the convention-invariant normalization products. The goal is not to hide the theorem behind a single "zero" output, but to show exactly which basis or invariant combination is closing.

Modular audit trail. The verification chain is intentionally modular. Lower-order freeze, decisive benchmark, scalar audit, dipole audit, quadrupole source-map work, and quadrupole-normalization completion are kept separate so that failure in one sector cannot be hidden by compensation in another. This separation is especially important at 2.5PN, where the logical distinction between lower- channel demotion and the final normalization gap must remain visible.

For readability, the main text reports only the archive structure, the theorem- ledger endpoints, and the scope limits of the scripts. A longer raw transcript can always be supplied as a repository note or supplementary appendix, but the named master audits already contain enough metadata for a referee to rerun the argument sector by sector.

10.3 What the archive does *not* verify

It is equally important to state the limits of the present archive.

First, neither master audit solves the fully dynamical moving-throat bulk PDE for arbitrary motion. The paper remains a reduced near-zone 2.5PN theorem attempt within the same declared closure hierarchy used by the earlier conservative papers. The scripts verify the algebra of that theorem attempt; they do not replace the missing microscopic completion.

Second, the archive does not verify a stronger claim than the paper itself makes. The first master audit verifies the channel theorem chain through the surviving quadrupole branch. The second master audit verifies that the remaining normalization problem has narrowed to the isotropic real P_2 quadrupole module, and that on that branch the odd Burke–Thorne coefficient is fixed once the even low-frequency conservative data are known. But the scripts do not, by themselves, derive the final physical normalization on the actual moving-throat branch. In the language of Section 8, the remaining scalar condition

$$\hat{m}_0^2\Gamma_5 = \frac{2G}{5c^5}$$

is isolated and organized, not yet derived from the completed PDE.

Third, the archive is intentionally limited to the sectors addressed in the present paper. It does not claim to cover spin couplings, strong-field completion, higher post-Newtonian orders beyond the present gate, non-passive or non-outgoing branches, noncompact outlet structures outside the natural compact branch, or the broader open-system physics left symbolic in Section 9.

Bottom line. The purpose of the verification archive is not to ask the reader to trust a computation instead of a derivation. It is to provide an auditable backstop showing that, under the stated assumptions, the load-bearing symbolic statements close exactly:

$$\begin{aligned}(\alpha, \beta) &= (4, 5),\\ \mathcal{M}^R_{ab}(\omega) &= \left[\hat{m}_0 + \hat{m}_2\omega^2 + \hat{m}_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab},\\ \overline{K}_4 &= \frac{4\overline{K}_2^2}{\overline{K}_0},\\ \overline{\Gamma}_5 &= 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}},\end{aligned}$$

while leaving the final physical normalization condition $\hat{m}_0^2\Gamma_5 = 2G/(5c^5)$ as the one narrow theorem gap still open on the actual moving-throat branch.

11 Conclusions

11.1 What has been established

The main result of this paper is that the 4D toy model now supports a *conditional point-particle* 2.5*PN theorem* within a clearly stated closure hierarchy. The point of the paper is not merely that one can write a term with the right dissipative scaling. Rather, once the derivation is organized by sector and the already-frozen Newtonian+1PN+2PN ledger is held fixed, the first dissipative gate reduces to a short named theorem chain with only one narrow normalization problem still open.

The proof line established in the main text is the following:

1. the exact 4D projection identities, projected continuity law with leakage, quasi-static Poisson hook, and controlled worldtube/coherent-defect particle reduction carried forward from the earlier papers;

2. the decisive low-frequency benchmark of Section 4, which shows that a purely real-even conservative kernel cannot generate a local 2.5PN reaction term, while the minimal retarded STF quadrupole benchmark lands exactly on the standard Burke–Thorne / Iyer–Will family member;

3. the scalar audit of Section 5, which shows that the continuity-derived breathing outlet is derivative-mediated, the compact mouth route is radiative/super-Ohmic on the natural branch, and the remaining scalar channels are therefore derivative-protected or finite-size suppressed rather than universal point-particle spoilers;

4. the dipole/vector audit of Section 6, which shows that the carried odd wake is real but that a clean outgoing $\ell = 1$ completion removes the dangerous linear odd term and pushes the surviving dipole contribution above universal point-particle 2.5PN on the strict small-body branch;

5. the quadrupole audit of Section 7, which proves that the solved real P_2 package already exhausts the full STF $\ell = 2$ representation content, that the orbital/worldtube source map is nondegenerate on the natural branch, that rotational invariance collapses the low-frequency retarded matching matrix to scalar form, that the passive/outgoing $\ell = 2$ channel has the correct positive $+i\omega^5$ parity/sign, and that the feared zero- overlap gate fails because $\hat{m}_0 \neq 0$ on that branch; and

6. the final theorem compression of Section 8, which shows that all currently derived lower odd channels are demoted above universal point-particle 2.5PN, while the surviving universal branch is the orbital/worldtube STF quadrupole with one remaining normalization target.

When these ingredients are assembled, the reduced nonspinning two-body dynamics has the structure

$$\mathbf{a} = \mathbf{a}_{\rm N} + \mathbf{a}_{\rm 1PN} + \mathbf{a}_{\rm 2PN} + \mathbf{a}^{\rm quad}_{\rm 2.5PN} + \Delta\mathbf{a}^{>2.5\rm PN}_{\rm fs},$$

where $\mathbf{a}^{\rm quad}_{\rm 2.5PN}$ is the standard local Burke–Thorne / Iyer–Will quadrupolar reaction term generated by the orbital/worldtube STF quadrupole, and the remaining currently derived odd channels are pushed above universal point-particle 2.5PN on the natural compact/passive/outgoing small-body branch.

Conceptually, the paper does three things at once. First, it upgrades the program from a full conservative 2PN derivation to the first dissipative post-Newtonian gate. Second, it keeps that upgrade readable by separating the proof line from the broader exploratory chronology: the main text isolates the load-bearing channel audit and the final theorem ledger, while the larger notes and scripts retain the wider constructive throat-response record. Third, it sharpens the bookkeeping of what is fixed, what is branch-conditioned, and what must still remain symbolic.

In that sharpened sense, the paper closes more than one conceptual gap. It shows that the lower odd scalar and dipole channels are not generic universal killers on the natural branch; it shows that the quadrupole route is not merely compatible with the target but is in fact the *unique* surviving universal point-particle branch; and it shows that the remaining unresolved problem is no longer broad "missing dissipative physics," but the final passive/outgoing quadrupole normalization. Once these points are separated cleanly, the overall 2.5PN theorem statement becomes much shorter than the raw session history from which it emerged.

As at 1PN and 2PN, the status labels matter. Some steps are exact identities, exact basis-conversion facts, or exact matching theorems. Some are controlled reductions, such as the Poisson hook, the worldtube particle limit, and the strict small-body ordering. Some are protocol or branch closures, such as the passive/outgoing compact low-frequency completion used to formulate the surviving quadrupole theorem. The final statement is therefore not an assumption-free theorem of a fully solved moving-throat bulk PDE. It is a conditional theorem within a stated closure hierarchy. Within that hierarchy, however, the bookkeeping is now materially sharper than before the present paper: the universal channel structure is fixed, and only one narrow invariant normalization problem remains open.

11.2 What remains open

The negative side of the scope should be kept equally explicit.

First, the paper does not solve the fully dynamical moving-throat PDE from first principles and then derive the point-particle 2.5PN dynamics as an automatic bulk theorem. The particle limit remains a controlled reduction, just as in the earlier 1PN and 2PN papers. The present work therefore proves the first dissipative theorem gate only within the declared reduction hierarchy, not as a microscopically completed defect theorem.

Second, the paper does not yet derive the final normalization of the surviving quadrupole branch. The remaining scalar target can be stated in one line:

$$\boxed{\hat{m}_0^2\Gamma_5 = \frac{2G}{5c^5}.}$$

Equivalently, in canonical invariant language one must determine the even pair $(\overline{K}_0, \overline{K}_2)$ on the physical passive/outgoing isotropic branch, after which the next even moment and the odd Burke–Thorne coefficient are fixed automatically by

$$\overline{K}_4 = \frac{4\overline{K}_2^2}{\overline{K}_0}, \qquad \overline{\Gamma}_5 = 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}.$$

So the surviving open issue is not tensor structure, sign, or representation content. It is the final far-zone normalization of the physical quadrupole branch.

Third, the paper does not claim that every conceivable lower odd channel has been ruled out in every possible completion of the model. The main theorem is tied to the natural compact/passive/outgoing small-body branch isolated by the present audit. A genuinely new direct scalar source outside the continuity plus reciprocal-boundary ontology, an effectively one-dimensional or noncompact low-frequency outlet, a strong anisotropy or singular $P_2 \to$ STF map, a breakdown of the strict small-body hierarchy, or a non-passive/non-outgoing quadrupole completion would all fall outside the present theorem envelope.

Fourth, the paper is restricted to the nonspinning two-body 2.5PN gate. It does not attempt a full treatment of spin couplings, strong-field completion, many-body dissipative dynamics, higher radiative post-Newtonian orders, or a replacement of the broader script-side throat-response record by a single closed microscopic PDE derivation. Those sectors remain future work, not hidden appendages of the present theorem.

These non-claims do not weaken the main result. They locate the remaining physics precisely. Before the present paper, it was still unclear whether the model could survive the first dissipative gate without running into an earlier scalar or dipole failure route. After the present paper, that structural question has been sharply narrowed on the natural branch. What remains open is the final normalization of an already identified surviving branch, not the existence of the branch itself.

11.3 Interface to the next calculation

This gives the project a much cleaner handoff than before. The next true job is no longer to reopen the scalar and dipole audits or to search again for a vague whole-theory dissipative mechanism. That work is substantially finished on the natural branch. The next job is the moving-throat quadrupole calculation itself: determine the passive/outgoing grouped real P_2 low-frequency data on the physical branch, fix the invariant even pair $(\overline{K}_0, \overline{K}_2)$ there, and thereby determine the remaining quantities $(\overline{K}_4, \overline{\Gamma}_5)$ automatically.

Put differently, future higher-order conservative or near-dissipative work does not need to rediscover a generic whole theory. The present paper has already reduced the universal point-particle route to the grouped real quadrupole bundle. The next calculation must therefore decide whether the physical moving-throat branch realizes exactly the minimal isotropic moment pattern isolated here, and whether its normalization yields $\hat{m}_0^2\Gamma_5 = 2G/(5c^5)$. That is the single place where the current conditional theorem must still be sharpened into an unconditional one.

The correct endpoint of the present paper is therefore also the correct starting point of the next one. The lower odd channels are no longer generic universal spoilers, the orbital/worldtube STF quadrupole is the unique surviving point-particle 2.5PN branch, and the only serious unresolved question lives in the final passive/outgoing quadrupole normalization. That is the strongest honest conclusion now supported by the present stack.

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A Low-frequency selection-rule framework

This appendix records the technical low-frequency bookkeeping used in Sec. 4. Its role is narrower than the full channel analysis of the main text. It does not decide which sector survives; that work is done later in the scalar, dipole, and quadrupole audits. Instead, it fixes four pieces of infrastructure that the later theorem chain repeatedly uses:

1. the Fourier-sign map that converts odd imaginary powers of ω into local time-domain forces,
2. minimal retarded kernels whose first odd contribution begins at ω, ω^3, or ω^5,
3. the separation of true dissipation from total-derivative Schott pieces, and
4. the model-specific low-frequency coefficient combinations inherited from the carried 2PN support package.

At quadratic order in a reduced collective coordinate $q(t)$, the retarded part of the influence functional may be written schematically as

$$S_{\mathrm{IF}}^{(2)} = \int \frac{d\omega}{2\pi}\, q_-(-\omega)\, K_R(\omega)\, q_+(\omega),$$

where $K_R(\omega)$ is the retarded kernel. The even real part organizes the conservative low-frequency response; the odd imaginary part classifies the first irreversible channel. For the 2.5PN audit, the decisive datum is therefore not the whole kernel but the location of its first nonvanishing odd term.

A.1 Fourier convention and odd-channel map

Throughout the paper we use the convention

$$q(t) = \int \frac{d\omega}{2\pi}\, e^{-i\omega t} q(\omega), \qquad \partial_t^n q(t) \longleftrightarrow (-i\omega)^n q(\omega).$$

Hence an odd imaginary kernel contribution

$$K_R^{\mathrm{odd}}(\omega) = i\gamma_n \omega^n, \qquad n = 2m+1,$$

induces the local time-domain operator

$$i\omega^{2m+1} \longleftrightarrow \frac{i}{(-i)^{2m+1}}\partial_t^{2m+1} = (-1)^{m+1}\partial_t^{2m+1}.$$

For the three channels relevant to the 2.5PN problem this gives

$$i\omega \longleftrightarrow -\partial_t, \qquad i\omega^3 \longleftrightarrow +\partial_t^3, \qquad i\omega^5 \longleftrightarrow -\partial_t^5.$$

If the reduced equation of motion is written schematically as

$$F^{\mathrm{odd}}(t) = K_R^{\mathrm{odd}}(\partial_t) q(t),$$

then the first three odd channels become

$$F_1 = -\gamma_1 \dot{q}, \qquad F_3 = +\gamma_3 q^{(3)}, \qquad F_5 = -\gamma_5 q^{(5)}.$$

This is the low-frequency selection rule used throughout the main text. The channel hierarchy is therefore

$$\boxed{\begin{gathered} n = 1 \text{ (scalar / monopole candidate)}, \\ n = 3 \text{ (dipole / vector candidate)}, \\ n = 5 \text{ (quadrupole candidate)}. \end{gathered}}$$

A GR-like point-particle 2.5PN theorem requires the lower odd scalar and vector channels to be absent, derivative-mediated, or finite-size demoted, while an $n = 5$ quadrupole branch must survive with the correct sign and source map.

A.2 Minimal retarded kernels and the first odd term

A convenient benchmark family that isolates the first odd term is

$$K_n(\omega) = \frac{g^2}{\Omega^2 - \omega^2 - i\sigma\omega^n}, \qquad n \in \{1, 3, 5\}, \qquad \sigma > 0.$$

This is not asserted to be the microscopic kernel of the completed throat PDE. Its role is only to show, in the cleanest possible way, how a passive retarded channel announces itself in low-frequency expansion. Expanding for small ω gives

$$K_1(\omega) = \frac{g^2}{\Omega^2} + i\frac{g^2\sigma}{\Omega^4}\,\omega + \left(\frac{g^2}{\Omega^4} - \frac{g^2\sigma^2}{\Omega^6}\right)\omega^2 + O(\omega^3),$$

$$K_3(\omega) = \frac{g^2}{\Omega^2} + \frac{g^2}{\Omega^4}\,\omega^2 + i\frac{g^2\sigma}{\Omega^4}\,\omega^3 + O(\omega^4),$$

$$K_5(\omega) = \frac{g^2}{\Omega^2} + \frac{g^2}{\Omega^4}\,\omega^2 + \frac{g^2}{\Omega^6}\,\omega^4 + i\frac{g^2\sigma}{\Omega^4}\,\omega^5 + O(\omega^6).$$

Two facts matter.

First, the leading odd contribution appears exactly at the advertised order: linear for the scalar/monopole candidate, cubic for the dipole/vector candidate, and quintic for the quadrupole candidate. Second, the passive sign choice $\sigma > 0$ produces a positive odd coefficient in every case. Once the Fourier-sign map of Appendix A.1 is applied, those positive coefficients become damping terms after subtraction of the corresponding Schott total derivatives.

This benchmark also makes clear why the conservative hierarchy by itself cannot reach 2.5PN. If the odd part is absent altogether, the kernel reduces to a real-even expansion in ω^{2m} only, which can renormalize conservative response coefficients but cannot generate a genuinely time-asymmetric local force.

A.3 Dissipation versus Schott pieces

The odd channels cannot be dismissed as “pure gauge” merely because they come with higher derivatives. What matters physically is the instantaneous power

$$P_n = \dot{q}\,F_n,$$

and its decomposition into a total derivative plus a negative semidefinite irreversible piece.

For the linear odd channel one finds immediately

$$P_1 = \dot{q}(-\gamma_1\dot{q}) = -\gamma_1\dot{q}^2.$$

So a positive γ_1 is directly dissipative.

For the cubic channel,

$$P_3 = \dot{q}\,(\gamma_3 q^{(3)}) = \gamma_3 \frac{d}{dt}(\dot{q}\,\ddot{q}) - \gamma_3 \ddot{q}^{\,2}.$$

The first term is the familiar Schott piece; the second is negative semidefinite. Thus the cubic odd term is not "only Schott" once the total derivative is removed.

For the quintic channel,

$$P_5 = \dot{q}\,(-\gamma_5 q^{(5)}) = -\gamma_5 \frac{d}{dt}\big(\dot{q}\,q^{(4)} - \ddot{q}\,q^{(3)}\big) - \gamma_5 \big(q^{(3)}\big)^2.$$

Again the first term is a total derivative while the second is negative semidefinite. Positive γ_5 therefore gives damping after the Schott term is separated off.

The practical consequence is the one used repeatedly in the main text: the *lowest surviving odd channel* is the dominant irreversible sector. A scalar or dipole odd term that survives on the universal point-particle branch cannot be dismissed by calling it "higher derivative" or "Schott-like." It must be shown absent, derivative-mediated, or finite-size demoted before the quadrupole branch can control the theorem.

A.4 Model-specific carried low-frequency coefficients

The abstract odd-channel hierarchy becomes useful only after it is translated into the carried support data inherited from the conservative 2PN files. On the scalar/geometry side, the relevant reduced combination is

$$S_{\rm eff}(\omega) = J_0^2 Y_0(\omega) + J_{20}^2 Y_{20}(\omega) - \Delta_{\rm geom} Y_g(\omega),$$

with the fixed static weights

$$J_0^2 = \frac{16}{5}, \qquad J_{20}^2 = \frac{25}{16}, \qquad \Delta_{\rm geom} = \frac{281}{80}.$$

If the scalar support and geometry sectors pick up low-frequency odd pieces

$$Y_0^{\rm ret} = Y_0^{\rm even} + i\delta_{01}\omega + \cdots, \qquad Y_g^{\rm ret} = Y_g^{\rm even} + i\delta_{g1}\omega + \cdots,$$

while the surviving quadrupole-support branch picks up

$$Y_{20}^{\rm ret} = Y_{20}^{\rm even} + i\delta_{205}\omega^5 + \cdots,$$

then the carried odd combinations are

$$\gamma_1^{\rm eff} = J_0^2 \delta_{01} - \Delta_{\rm geom}\,\delta_{g1} = \frac{16}{5}\,\delta_{01} - \frac{281}{80}\,\delta_{g1},$$

$$\gamma_5^{\rm eff} = J_{20}^2 \delta_{205} = \frac{25}{16}\,\delta_{205}.$$

These are not yet the final theorem coefficients; later sections still decide whether the scalar and dipole contributions are demoted and whether the quadrupole branch survives with the correct normalization. Their function is more basic. They translate the abstract selection-rule hierarchy into the carried 2PN scalar/geometry and grouped-P_2 support data that the model already fixes before the full moving-throat PDE is solved.

The dipole/vector channel is handled separately because it is not naturally packaged by the scalar weights above; instead it enters through the carried odd wake ports and their outgoing $\ell = 1$ spectral completion. But the appendix bookkeeping still fixes the basic logical structure of the whole 2.5PN audit:

$$\boxed{\gamma_1^{\rm eff} \text{ must vanish or be demoted,} \qquad \gamma_3 \text{ must vanish or be demoted,} \qquad \gamma_5^{\rm eff} > 0 \text{ must survive.}}$$

Once those statements hold on the natural compact/passive/outgoing small-body branch, the remaining theorem gap is reduced to the final normalization of the surviving quadrupole response.

B Scalar derivation ledger

This appendix records the technical scalar-side calculations that underlie Sec. 5. The main text states the theorem chain in manuscript form; the present appendix keeps the algebraic checkpoints visible in one place. Its role is therefore narrower than a full scalar PDE derivation. It fixes the precise low-frequency identities that the main text uses to demote most scalar dangers: the carried scalar odd coefficient, the projection-overlap counterexample and projection-locking criterion, the direct-versus-derivative coupling split, the continuity derivation of the breathing outlet, the exact subsystem source classification, and the compact reciprocal-mouth theorem that kills the dangerous z_2 term.

B.1 Outgoing scalar monopole and the carried odd coefficient

The first scalar audit begins from the outgoing three-dimensional monopole Green function

$$G_R(\omega, r) = \frac{e^{i\omega r/c_s}}{4\pi r}.$$

Its low-frequency expansion is

$$G_R(\omega, r) = \frac{1}{4\pi r} + \frac{i\omega}{4\pi c_s} - \frac{\omega^2 r}{8\pi c_s^2} - \frac{i\omega^3 r^2}{24\pi c_s^3} + \frac{\omega^4 r^3}{96\pi c_s^4} + O(\omega^5).$$

So any nonzero effective scalar monopole coupling to a passive gapless outlet naturally produces a linear odd term. In the carried scalar/geometry packaging, that danger is encoded in the combination

$$S_{\text{eff}}(\omega) = J_0^2 Y_0(\omega) + J_{20}^2 Y_{20}(\omega) - \Delta_{\text{geom}} Y_g(\omega), \qquad J_0^2 = \frac{16}{5}, \qquad J_{20}^2 = \frac{25}{16}, \qquad \Delta_{\text{geom}} = \frac{281}{80},$$

so if

$$Y_0^{\text{ret}} = Y_0^{\text{even}} + i\delta_{01}\omega + \cdots, \qquad Y_g^{\text{ret}} = Y_g^{\text{even}} + i\delta_{g1}\omega + \cdots,$$

then the carried scalar odd coefficient is

$$\boxed{\gamma_1^{\text{eff}} = \frac{16}{5}\,\delta_{01} - \frac{281}{80}\,\delta_{g1}.}$$

The first scalar theorem target is therefore the sharp condition

$$\gamma_1^{\text{eff}} = 0.$$

A useful refinement is the outgoing $\ell = 0$ completion of the scalar support/geometry port already carried by the conservative 2PN files. For the outgoing spherical Hankel branch,

$$h_0^{(1)}(z) = j_0(z) + iy_0(z) = -i\frac{e^{iz}}{z}, \qquad z = ka_0, \qquad k = \frac{\omega}{c_s},$$

one finds the exact logarithmic derivative

$$\Lambda_0(k) = k\,\frac{\partial_z h_0^{(1)}(z)}{h_0^{(1)}(z)}\bigg|_{z=ka_0} = ik - \frac{1}{a_0}.$$

Hence the normalized outgoing scalar admittance is

$$Y_{0,\text{norm}}(k) = \frac{1}{1 - ia_0 k} = 1 + ia_0 k - a_0^2 k^2 - ia_0^3 k^3 + a_0^4 k^4 + O(k^5),$$

and similarly the geometry branch gives

$$Y_{g,\mathrm{norm}}(k) = \frac{1}{1 - ia_g k} = 1 + ia_g k - a_g^2 k^2 - ia_g^3 k^3 + a_g^4 k^4 + O(k^5).$$

Inserting these into the carried scalar package gives

$$S_{\mathrm{eff}}^{\mathrm{ret}}(k) = \frac{5}{4} + ik\left(\frac{16}{5}a_0 - \frac{281}{80}a_g\right) + k^2\left(-\frac{16}{5}a_0^2 + \frac{281}{80}a_g^2\right) + ik^3\left(-\frac{16}{5}a_0^3 + \frac{281}{80}a_g^3\right) + O(k^4).$$

The effective scalar length on this branch is therefore

$$\boxed{\ell_{\mathrm{eff}} = \frac{16}{5}a_0 - \frac{281}{80}a_g.}$$

At equal scales one gets the nonvanishing residual

$$\ell_{\mathrm{eff}}\big|_{a_g = a_0} = -\frac{5}{16}a_0,$$

while exact cancellation would require

$$a_g = \frac{256}{281}a_0.$$

The main text uses this calculation only to demote the carried 2PN scalar/geometry branch to finite-size status on the strict small-body family; it does not treat the tuned cancellation as the main theorem.

B.2 Projection-overlap counterexample and projection-locking

Let a centered breathing family be written as

$$\rho(\mathbf{x}, w, t) = n(\mathbf{x}) g_a(w), \qquad \int d^3x\, n(\mathbf{x}) = N,$$

with fixed projection kernel $W(w)$. The brane-visible scalar content is then

$$N_{\mathrm{br}}(a) = N\, C(a), \qquad C(a) = \int dw\, W(w) g_a(w).$$

The exact integrated leakage identity reads

$$I_{\mathrm{leak}} = N\, \dot{a}\, \frac{dC}{da}.$$

So the leading scalar odd term vanishes only if the projection overlap is locked against the breathing coordinate.

A fully explicit counterexample is obtained from the Gaussian choice

$$W_\ell(w) = \frac{1}{\sqrt{\pi}\,\ell} e^{-w^2/\ell^2}, \qquad g_a(w) = \frac{1}{\sqrt{\pi}\, a} e^{-w^2/a^2}.$$

Then

$$C(a) = \int_{-\infty}^{\infty} dw\, W_\ell(w) g_a(w) = \frac{1}{\sqrt{\pi}\sqrt{a^2 + \ell^2}}, \qquad \frac{dC}{da} = -\frac{a}{\sqrt{\pi}\,(a^2 + \ell^2)^{3/2}} \neq 0.$$

The linearized continuity equation is also exact in this model. With

$$j_w(w, t) = \frac{\dot{a}}{a}\, w\, g_a(w),$$

one has

$$\dot{a}\,\partial_a g_a + \partial_w j_w = 0.$$

Projecting gives

$$I_{\rm leak} = \int_{-\infty}^{\infty} dw\, W'_\ell(w) j_w(w,t) = \dot{a}\,\frac{dC}{da} = -\frac{a\dot{a}}{\sqrt{\pi}\,(a^2+\ell^2)^{3/2}},$$

so for harmonic breathing $a(t) = a_0 + \varepsilon e^{-i\omega t}$,

$$\delta I_{\rm leak} = i\omega\varepsilon \frac{a_0}{\sqrt{\pi}\,(a_0^2+\ell^2)^{3/2}}.$$

This is the explicit centered counterexample showing that the current projection framework does not automatically enforce a scalar Ward identity.

The exact rescue target is projection-locking. For collective coordinates q^A one may write

$$I_{\rm leak}(t) = \dot{q}^A B_A(q), \qquad B_A(q) = \partial_A N_{\rm br}(q).$$

The leading scalar odd term vanishes if and only if

$$\boxed{B_A(q_0) v^A = 0}$$

for every dynamically allowed tangent vector v^A at the operating point. Equivalently, if the soft scalar tangent space is spanned by modes $\Psi_{(n)}$, then one needs

$$\int d^3x\, dw\, W(w) \Psi_{(n)}(\mathbf{x}, w) = 0 \qquad \text{for every soft scalar mode } \Psi_{(n)}.$$

For a two-coordinate tangent space, say $q^A = (a, L)$ with tangent vectors $v^A_{(1)}$ and $v^A_{(2)}$, a nondegenerate tangent matrix

$$\Delta = v^a_{(1)} v^L_{(2)} - v^L_{(1)} v^a_{(2)} \neq 0$$

forces the stronger statement

$$B_a = B_L = 0.$$

This is the exact linear-algebra form of the projection-locking criterion used in the main text.

B.3 Direct versus derivative coupling and the damped discrete-mode no-go

The next scalar gate is whether the breathing coordinate couples directly to a passive gapless scalar outlet or only through its time derivative. Write a low-frequency outlet kernel as

$$G_0^{\rm ret}(\omega) = A + iB\omega + C\omega^2 + iD\omega^3 + O(\omega^4),$$

with real A, B, C, D. A direct overlap gives

$$K_{\rm direct}(\omega) = g_0^2 G_0^{\rm ret}(\omega) = g_0^2 A + i g_0^2 B\omega + g_0^2 C\omega^2 + i g_0^2 D\omega^3 + \cdots,$$

so the leading odd coefficient is

$$\Im K_{\rm direct}(\omega) = g_0^2 B\,\omega + O(\omega^3).$$

A derivative coupling instead gives

$$K_{\rm deriv}(\omega) = g_d^2 \omega^2 G_0^{\rm ret}(\omega) = g_d^2 A\omega^2 + i g_d^2 B\omega^3 + g_d^2 C\omega^4 + O(\omega^5),$$

with
$$\Im K_{\rm deriv}(\omega) = g_d^2 B\,\omega^3 + O(\omega^5).$$
So derivative coupling kills the dangerous linear odd term automatically.

A separate no-go is that a discrete internal breathing mode does not rescue the problem if it inherits Ohmic leakage from a gapless outlet. Indeed,
$$\delta\kappa_{\rm PV}^{\rm ret}(\omega) = \frac{\lambda^2}{\Omega^2 - \omega^2 - i\eta\omega} = \frac{\lambda^2}{\Omega^2} + i\frac{\eta\lambda^2}{\Omega^4}\omega + O(\omega^2).$$
So a gapped internal mode still carries a linear odd term whenever its damping is Ohmic at low frequency. This is why the scalar theorem needed the continuity reanalysis of the breathing outlet rather than a simple appeal to discreteness.

B.4 Continuity derivation of the breathing-to-outlet vertex

Let the breathing family be parameterized as
$$q(t) = q_0 + \delta q(t), \qquad \rho = \rho_0 + \delta q\,\rho_q^{(1)} + O(\delta q^2).$$
A static displacement along the family remains a static configuration, so the linear scalar current cannot begin at $O(\delta q)$. The first allowed current is kinematic:
$$j^i = \dot{\delta q}\,\mathcal{J}_q^i + O(\delta q\dot{q}, \dot{q}^2, \ddot{q}).$$
Linearized continuity gives
$$\partial_i \mathcal{J}_q^i = -\rho_q^{(1)}.$$
Projecting then yields
$$\delta S_{\rm leak}(\mathbf{x},t) = \dot{\delta q}(t) B_q(\mathbf{x}),$$
with
$$B_q(\mathbf{x}) = -\left[W\mathcal{J}_q^w\right]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} dw\, W'(w)\mathcal{J}_q^w.$$
Therefore in frequency space,
$$\boxed{\delta S_{\rm leak}(\mathbf{x},\omega) = (-i\omega)\,\delta q(\omega)\,B_q(\mathbf{x}).}$$
This is the decisive classification theorem: the breathing-to-outlet vertex derived from continuity is $\dot{q}$-like or zero, not direct q-like.

The earlier Gaussian counterexample must therefore be reinterpreted. It still shows that the derivative monopole coefficient can be nonzero, since
$$\delta I_{\rm leak}(t) = \dot{\delta q}(t)\,\partial_q N_{\rm br}\big|_{q_0},$$
but it no longer shows a direct static overlap source.

If one now integrates out a low-frequency scalar outlet with retarded kernel
$$\Pi_q^R(\omega) = A + iB\omega + C\omega^2 + iD\omega^3 + \cdots,$$
then the induced breathing kernel becomes
$$K_{qq}^R(\omega) = \omega^2\Pi_q^R(\omega) = A\omega^2 + iB\omega^3 + C\omega^4 + iD\omega^5 + \cdots.$$
Hence the first derived odd dissipative term is cubic:
$$\boxed{\Im K_{qq}^R(\omega) \propto \omega^3.}$$
This is the algebraic reason the breathing channel is demoted in the main text.

B.5 No extra linear scalar source beyond leakage and boundary

Integrating the exact projected continuity law over a moving defect-centered control volume $D(t)$ gives

$$\frac{d}{dt}\int_{D(t)} \rho_{\rm br}\, d^3x = -\oint_{\partial D(t)} (\mathbf{J}_{\rm br} - \rho_{\rm br}\mathbf{u}_b)\cdot\mathbf{n}\, dA + \int_{D(t)} S_{\rm leak}\, d^3x.$$

So the exact scalar-outlet ledger contains only two linear channels:

$$\boxed{\text{worldtube/mouth boundary flux}} \qquad \text{and} \qquad \boxed{\text{projected leakage}}.$$

Around a static background the projected momentum-leakage and projection-induced stress channels are quadratic in velocities, schematically

$$\rho v_a v_w, \qquad \rho v_a v_b,$$

so they cannot generate an independent linear scalar source.

A convenient frequency-space restatement of the same point is the local product expansion

$$Z_\sigma(\omega) = iz_1\omega + z_2\omega^2 + iz_3\omega^3 + O(\omega^4), \qquad u_\sigma(\omega) = u_0 q + iu_1\omega q + u_2\omega^2 q + O(\omega^3 q),$$

which gives

$$j_\sigma(\omega) = Z_\sigma(\omega)\,\nu_\sigma(\omega) = iu_0 z_1\,\omega q + (u_0 z_2 - u_1 z_1)\omega^2 q + i(u_0 z_3 + u_1 z_2 + u_2 z_1)\omega^3 q + O(\omega^4 q).$$

Because $Z_\sigma(0) = 0$ on the static no-flux branch, there is no independent $O(q)$ mouth source. The leading mouth source is derivative-like, in agreement with the control-volume classification.

B.6 Mouth admittance, reciprocity, and the compact radiative-order theorem

The remaining scalar loophole is a genuinely dynamic mouth/worldtube boundary admittance. A naive low-frequency parameterization would be

$$Z_\sigma^{\rm eff}(\omega) = iz_1\omega + z_2\omega^2 + iz_3\omega^3 + \cdots,$$

where $iz_1\omega$ is reactive and $z_2\omega^2$ is the first apparently dissipative term. If one estimates

$$z_2 \sim \frac{a^2}{c_s^3}, \qquad \ell_{\rm mouth} \sim \frac{a^2}{L},$$

then this would place the old breathing channel at the dangerous order tracked in the main text.

The compact branch admits a stronger theorem. Assume:

1. the static mouth problem is a compliance problem rather than a static flux law,
2. the linear scalar mouth port is reciprocal and kinematic,

$$j = i\omega\beta q,$$

3. and the only linear dissipation is compact outgoing scalar radiation.

Then the loaded scalar coordinate obeys

$$u_\sigma(\omega) = \Big[K - (M + M_{\rm add})\omega^2 + iR_2\omega^3 + O(\omega^4)\Big] q(\omega).$$

Hence the effective admittance is

$$Y(\omega) = \frac{j}{u_\sigma} = \frac{i\beta\omega}{K - (M + M_{\rm add})\omega^2 + iR_2\omega^3 + O(\omega^4)}.$$

Expanding gives

$$Y(\omega) = i\frac{\beta}{K}\omega + i\frac{\beta(M + M_{\text{add}})}{K^2}\omega^3 + \frac{\beta R_2}{K^2}\omega^4 + O(\omega^5).$$

Therefore

$$\boxed{z_2 = 0.}$$

The first dissipative mouth term is not $O(\omega^2)$ but $O(\omega^4)$.

The contrast with a genuinely Ohmic one-dimensional outlet is immediate. For

$$Y_{1\text{D}}(\omega) = \frac{i\beta\omega}{K - M\omega^2 + iZ_1\omega},$$

one finds

$$\Re Y_{1\text{D}}(\omega) = \frac{\beta Z_1}{K^2}\omega^2 + O(\omega^4),$$

so the dangerous z_2 term is present precisely when the outlet behaves as a semi-infinite Ohmic line. The compact branch therefore differs qualitatively, not merely quantitatively, from the transmission-line benchmark.

After subsystem reduction the surviving mouth contribution takes the schematic form

$$\delta\kappa^{\text{ret}}_{\text{mouth}}(\omega) = \delta\kappa^{(0)}_{\text{mouth}} + \kappa_2\omega^2 - i\Lambda_3\left(\frac{\omega}{c_s}\right)^3 + O(\omega^4),$$

so the first odd mouth term is cubic. This is the exact low-frequency statement used in the main scalar demotion theorem.

Finally, the same calculation exposes the last scalar no-go. Within the current ontology plus a local reversible finite-throat completion, no derived linear Ohmic scalar mouth bleed appears. To recover one, the future PDE would need to introduce genuinely new structure outside the present continuity-plus-reciprocity framework, such as a semi-infinite one-dimensional outlet, an explicitly non-Hamiltonian resistive boundary layer, or a direct q-like scalar overlap source that bypasses the continuity derivation.

B.7 Scalar ledger summary

The technical results recorded in this appendix can be summarized as follows. First, the carried scalar danger is real enough to be written as the explicit coefficient

$$\gamma_1^{\text{eff}} = \frac{16}{5}\delta_{01} - \frac{281}{80}\delta_{g1},$$

and neither projected continuity nor static no-flux alone forces it to vanish. Second, the projection framework admits an explicit centered counterexample, which sharpens the exact rescue target to projection-locking. Third, the continuity derivation reclassifies the breathing outlet: the linear vertex is $\dot{q}$-like or zero, not direct q-like, so the first derived odd breathing term is cubic. Fourth, the exact subsystem scalar ledger contains no third independent linear outlet beyond projected leakage and worldtube/mouth boundary flux. Fifth, the compact reciprocal-mouth theorem forces $z_2 = 0$ and thereby eliminates the dangerous Ohmic $O(\omega^2)$ mouth dissipation on the natural branch.

Those are precisely the technical ingredients needed for the scalar demotion statement quoted in Sec. 5. They do not yet prove that every future moving-throat PDE completion will respect the same ontology, but they do show that within the present derived framework the scalar side is not a generic direct $i\omega$ kill-switch for the universal point-particle 2.5PN theorem.

C Dipole derivation ledger

This appendix records the technical dipole/vector-side calculations that underlie Sec. 6. The main text states the theorem chain in manuscript form; the present appendix keeps the algebraic checkpoints visible in one place. Its role is therefore narrower than a full moving-throat PDE analysis. It fixes the exact content of the carried odd wake ports, the center-of-mass/relative split, the minimal low-frequency odd completion and its jerk-type force, the failure of the obvious vector Ward rescues, the outgoing $\ell = 1$ threshold law, the same-order matching obstruction against the standard Iyer–Will family, and the small-body scaling argument that demotes the clean outgoing dipole branch above universal point-particle 2.5PN.

C.1 Carried odd wake ports and exact CM/relative split

The constructive 2PN throat-response package already contains a concrete odd wake sector. In the instantaneous pair frame,

$$\mathbf{v} = \mathbf{u} + d\,\mathbf{n}, \qquad \mathbf{u}\cdot\mathbf{n} = 0, \qquad \mathbf{n} = \frac{\mathbf{x}}{r}, \qquad d = \dot{r},$$

with $\mathbf{u}$ the tangential velocity, the carried odd ports are

$$\Pi^{(1\perp)} = \sqrt{\frac{7}{2}}\,\mathbf{u}, \qquad \Pi^{(10)} = 2d.$$

These are velocity-like objects, not static internal moments, which is already a warning sign: if they acquire a retarded odd completion, they can feed directly into dissipative relative dynamics.

To test whether the wake is secretly a center-of-mass artifact, write

$$\mathbf{x}_A = \mathbf{X} + \eta_B\mathbf{x}, \qquad \mathbf{x}_B = \mathbf{X} - \eta_A\mathbf{x}, \qquad \eta_A = \frac{m_A}{M}, \qquad \eta_B = \frac{m_B}{M}, \qquad M = m_A + m_B,$$

so that

$$\mathbf{v}_A = \mathbf{V} + \eta_B\mathbf{v}, \qquad \mathbf{v}_B = \mathbf{V} - \eta_A\mathbf{v}, \qquad \mathbf{V} = \dot{\mathbf{X}}, \qquad \mathbf{v} = \dot{\mathbf{x}}.$$

Decomposing both velocities relative to $\mathbf{n}$,

$$\mathbf{V} = \mathbf{U} + D\,\mathbf{n}, \qquad \mathbf{v} = \mathbf{u} + d\,\mathbf{n},$$

one obtains the exact body-local port split

$$\Pi_A^{(1\perp)} = \sqrt{\frac{7}{2}}\,(\mathbf{U} + \eta_B\mathbf{u}), \qquad \Pi_B^{(1\perp)} = \sqrt{\frac{7}{2}}\,(\mathbf{U} - \eta_A\mathbf{u}),$$

$$\Pi_A^{(10)} = 2(D + \eta_B d), \qquad \Pi_B^{(10)} = 2(D - \eta_A d).$$

Therefore the difference ports are exactly

$$\boxed{\Pi_A^{(1\perp)} - \Pi_B^{(1\perp)} = \sqrt{\frac{7}{2}}\,\mathbf{u},} \qquad \boxed{\Pi_A^{(10)} - \Pi_B^{(10)} = 2d.}$$

This is the first theorem-level result of the dipole audit. The carried odd wake difference channel is pure relative velocity, not pure center-of-mass motion. The harmless mass-dipole identity

$$D_i^{\text{mass}} = MX_i$$

therefore does not rescue the carried wake sector.

C.2 Minimal retarded completion and the first failure signal

The minimal low-frequency odd completion is

$$Y_{1\perp}^{\text{ret}}(\omega) = R_{1\perp} + i\sigma_\perp\omega + \cdots, \qquad Y_{10}^{\text{ret}}(\omega) = R_{10} + i\sigma_\parallel\omega + \cdots,$$

with the carried static residues

$$R_{1\perp} = \frac{7}{2}, \qquad R_{10} = 4.$$

At this stage the point is not that this is the correct completion; the point is to ask whether *any* nonzero linear odd term would already be dangerous. In the relative variables, the local odd part of the influence functional reduces to

$$S_{\text{odd,loc}}^{(1)} = \frac{1}{2}\int dt \left[\frac{7}{2}\,\sigma_\perp\,\mathbf{u}_-\cdot\dot{\mathbf{u}}_+ + 4\,\sigma_\parallel\,d_-\dot{d}_+\right].$$

The Euler–Lagrange derivative then gives the relative jerk-type force

$$F_x = -\frac{7\sigma_\perp}{4}x^{(3)}, \qquad F_y = -\frac{7\sigma_\perp}{4}y^{(3)}, \qquad F_z = -2\sigma_\parallel z^{(3)}.$$

So any nonzero linear odd coefficient in either dipole channel produces a genuine relative dissipative force. At this stage of the audit, the dipole wake looked worse than the scalar side had begun to look, because the wake appeared to give a direct lower-order relative obstruction with no obvious cancellation partner.

C.3 Why the obvious vector Ward rescues fail

The first rescue attempt is a vector Ward-identity argument: perhaps total momentum conservation forces the dangerous odd coefficient to vanish. That fails. At the body-action level,

$$L_{\text{body}} = a\,(\dot{x}_{A-} - \dot{x}_{B-})(\ddot{x}_{A+} - \ddot{x}_{B+})$$

gives

$$F_A = -a\,(x_A^{(3)} - x_B^{(3)}), \qquad F_B = +a\,(x_A^{(3)} - x_B^{(3)}),$$

so that

$$F_A + F_B = 0.$$

Thus total momentum conservation is perfectly compatible with a nonzero odd dipole wake in the relative sector.

The same term remains dangerous after the center-of-mass reduction. In CM/relative variables it becomes simply

$$L_{\text{body}} = a\,\dot{x}\,\ddot{x},$$

with no dependence on the center-of-mass coordinate X. So translational invariance also does not kill the term. Nor does the old vanishing worldtube-dipole theorem apply here: that theorem suppresses a static first moment of the defect profile, whereas the carried odd wake ports are velocity-like auxiliary channels.

The pure-Schott reinterpretation also fails generically. The power associated with the jerk-type force is

$$P = -\frac{d}{dt}\left[\frac{7\sigma_\perp}{4}(\dot{x}\,\ddot{x} + \dot{y}\,\ddot{y}) + 2\sigma_\parallel\dot{z}\,\ddot{z}\right] + \frac{7\sigma_\perp}{4}(\ddot{x}^2 + \ddot{y}^2) + 2\sigma_\parallel\ddot{z}^2.$$

Unless the odd coefficients vanish, there is a genuine non-total-derivative irreversible piece. So the carried odd wake is not generically “just Schott.” The interim theorem gate is therefore sharp:

$$\sigma_\perp = 0, \qquad \sigma_\parallel = 0,$$

or else the dipole wake contaminates the reduced relative dynamics.

C.4 Outgoing $\ell = 1$ threshold law

The first major rescue comes from abandoning the generic Ohmic auxiliary picture and asking instead what happens if the dipole ports arise from a clean outgoing three-dimensional partial-wave channel. For the outgoing spherical Hankel branch,

$$j_1(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z}, \qquad y_1(z) = -\frac{\cos z}{z^2} - \frac{\sin z}{z}, \qquad h_1^{(1)}(z) = j_1(z) + iy_1(z),$$

with $z = ka$ and $k = \omega/c$, the exact logarithmic derivative is

$$\Lambda_1(k) = k\,\frac{\partial_z h_1^{(1)}(z)}{h_1^{(1)}(z)}\bigg|_{z=ka} = \frac{ia^2k^2 - 2ak - 2i}{a(ak+i)}.$$

Its low-frequency expansion is

$$\Lambda_1(k) = -\frac{2}{a} + ak^2 + ia^2k^3 - a^3k^4 - ia^4k^5 + a^5k^6 + O(k^7).$$

So the absorptive part begins at

$$\operatorname{Im}\Lambda_1 \sim k^3,$$

not at k. After inversion and normalization to a fixed static residue R, one obtains

$$Y_{1,\text{norm}}(k) = R\left[1 + \frac{1}{2}(ak)^2 + i\frac{1}{2}(ak)^3 - \frac{1}{4}(ak)^4 + O((ak)^5)\right].$$

Thus the first odd dipole term is

$$\boxed{\operatorname{Im} Y_1^{\text{ret}}(\omega) \propto \left(\frac{a\omega}{c}\right)^3,}$$

with no linear $O(\omega)$ contribution. This materially changes the meaning of the earlier jerk-type failure. It becomes a warning about *generic Ohmic auxiliary completion*, not the necessary behavior of a clean outgoing dipole channel.

The rescue is real but not complete. Because the ports are velocity-like, an $i\omega^3$ odd term still produces a fifth-derivative local force,

$$F_x = -\frac{7\chi_\perp}{4}x^{(5)}, \qquad F_y = -\frac{7\chi_\perp}{4}y^{(5)}, \qquad F_z = -2\chi_\parallel z^{(5)}.$$

So the dipole channel is no longer an earlier-than-2.5PN killer, but it still remains a possible *same-order* modifier of the local 2.5PN sector.

C.5 Order reduction and the same-order matching obstruction

After Newtonian order reduction, an isotropic fifth-derivative force lies in the same local basis as the standard 2.5PN reaction term,

$$\{\dot r\,\mathbf{n}, \mathbf{v}\}.$$

A direct symbolic reduction gives

$$\mathbf{x}^{(5)} = \frac{15GM\dot r}{r^6}\Big(2GM + 7r\dot r^2 - 3rv^2\Big)\mathbf{n} + \frac{GM}{r^6}\Big(-8GM - 45r\dot r^2 + 9rv^2\Big)\mathbf{v},$$

so the isotropic fifth-derivative structure really does sit in the standard local 2.5PN basis.

Let $\lambda_\perp$ and $\lambda_\parallel$ denote the normalized transverse and longitudinal dipole coefficients after factoring out the standard overall 2.5PN prefactor. Then the dipole correction may be written as

$$\Delta\mathbf{a}_{\rm dip} = \frac{8G^2M^2\nu}{5c^5r^3}\left[\Delta A\,\dot r\,\mathbf{n} + \Delta B\,\mathbf{v}\right],$$

with

$$\Delta A = (9\lambda_\perp + 36\lambda_\parallel)v^2 + (-8\lambda_\perp - 22\lambda_\parallel)\frac{GM}{r} + (-45\lambda_\perp - 60\lambda_\parallel)\dot r^2,$$

$$\Delta B = -9\lambda_\perp v^2 + 8\lambda_\perp\frac{GM}{r} + 45\lambda_\perp \dot r^2.$$

It is convenient to package the coefficients in the six-slot basis

$$\mathcal{B} = (A_{v^2}, A_{GM/r}, A_{\dot r^2}; B_{v^2}, B_{GM/r}, B_{\dot r^2}).$$

In this basis, the standard Iyer–Will family is

$$F(\alpha,\beta) = \left(3(1+\beta), \frac{23+6\alpha-9\beta}{3}, -5\beta; -(2+\alpha), \alpha-2, 3(1+\alpha)\right),$$

while the Burke–Thorne benchmark is

$$F_{\rm BT} = \left(18, \frac{2}{3}, -25; -6, 2, 15\right)$$

and the dipole deformation is

$$D(\lambda_\perp,\lambda_\parallel) = \left(9\lambda_\perp + 36\lambda_\parallel, -8\lambda_\perp - 22\lambda_\parallel, -45\lambda_\perp - 60\lambda_\parallel; -9\lambda_\perp, 8\lambda_\perp, 45\lambda_\perp\right).$$

The same-order matching problem is therefore

$$F_{\rm BT} + D(\lambda_\perp,\lambda_\parallel) = s\,F(\alpha,\beta).$$

The symbolic solve gives the unique solution

$$\boxed{\alpha = 4, \qquad \beta = 5, \qquad s = 1, \qquad \lambda_\perp = 0, \qquad \lambda_\parallel = 0.}$$

This is stronger than saying that the dipole term "merely shifts the coefficients a little." It means that if a nonzero dipole wake survives at the same formal 2.5PN order, it cannot be absorbed into the standard GR local 2.5PN gauge family at all. A same-order surviving dipole contribution is therefore a genuine universality-breaking deformation of the point-particle sector.

Two physical readings are especially clean. First, a nonzero transverse piece $\lambda_\perp$ changes the circular-orbit tangential damping coefficient itself, so it is immediately observable and not a gauge reshuffle. Second, even if the circular limit looked less dramatic, a nonzero longitudinal piece $\lambda_\parallel$ still breaks the generic-orbit structure and cannot be absorbed into the remaining Iyer–Will freedom. The dipole gate is therefore fully sharp: same-order survival is fatal for GR-like universality.

C.6 Small-body scaling rescue and conditional demotion

After the matching test there is no remaining gauge-theoretic rescue route for a same-order dipole contribution. The only rescue is scaling suppression, and on the clean outgoing branch that rescue is available.

The carried odd dipole wake sits on top of a conservative 1PN cross-sector ingredient, so it carries one explicit factor

$$\epsilon \equiv \left(\frac{v}{c}\right)^2.$$

The outgoing $\ell = 1$ odd factor contributes

$$\left(\frac{a\omega}{c}\right)^3.$$

Using the orbital near-zone scaling

$$\omega \sim \frac{v}{r} \sim \frac{c\sqrt{\epsilon}}{r},$$

one gets

$$\frac{a\omega}{c} \sim \sqrt{\epsilon}\,\frac{a}{r}.$$

Hence the dipole odd correction scales as

$$\epsilon\left(\sqrt{\epsilon}\frac{a}{r}\right)^3 = \epsilon^{5/2}\left(\frac{a}{r}\right)^3.$$

If the small-body family obeys

$$\frac{a}{r} \sim \epsilon^{\delta}, \qquad \delta > 0,$$

then the dipole term enters at

$$\boxed{\epsilon^{5/2+3\delta},}$$

which lies strictly above universal point-particle 2.5PN. Representative cases are

$$\frac{a}{r} \sim \epsilon^{1/2} \Rightarrow 4\text{PN}, \qquad \frac{a}{r} \sim \epsilon \Rightarrow 5.5\text{PN}.$$

This is the main dipole rescue theorem of the appendix:

With a clean outgoing $\ell = 1$ completion, the carried dipole wake is naturally a finite-size correction rather than a universal point-particle 2.5PN obstruction.

The verdict is therefore conditional but meaningful. The dipole wake is real and structurally dangerous if treated naively. It is not removed by center-of-mass bookkeeping, momentum conservation, translational invariance, or other algebraic Ward-type arguments. But on the clean outgoing branch the absorptive part begins at ω^3, not ω, and in the strict small-body family the resulting correction is demoted above universal point-particle 2.5PN. What is *not* proven here is equally important: the current stack does not yet show from first principles that the moving-throat PDE enforces this clean outgoing $\ell = 1$ branch, excludes non-outgoing or Ohmic auxiliary completions, or makes the dipole coefficients vanish exactly rather than merely pushing them above the point-particle sector.

D Quadrupole representation and source-map ledger

This appendix records the technical quadrupole-side calculations that underlie Sec. 7. The main text states the theorem chain in manuscript form; the present appendix keeps the algebraic checkpoints visible in one place. Its role is narrower than the later quadrupole-normalization package and much narrower than a full moving-throat PDE analysis. It fixes six ingredients that the main text uses repeatedly:

1. the exact identification of the solved real P_2 ports with the full real STF $\ell = 2$ representation,
2. the outgoing $\ell = 2$ threshold law and its positive $+i\omega^5$ retarded parity,

3. the damping sign of the local odd quadrupole kernel,
4. the finite-size demotion of the literal compact internal P_2 route,
5. the worldtube/orbital source split and exact pair-frame source map,
6. and the rotational-invariance/static-overlap gates that leave one scalar low-frequency matching function rather than a generic 5×5 matrix problem.

The only issue intentionally deferred is the final same-basis quadrupole normalization. This appendix proves that the branch exists and is not naturally projected away; the later normalization package is what fixes its final canonical coefficient.

D.1 The solved P_2 ports already are the full real STF $\ell = 2$ bundle

In the instantaneous pair frame write

$$\mathbf{v} = \mathbf{u} + d\,\mathbf{n}, \qquad \mathbf{u} \cdot \mathbf{n} = 0, \qquad \mathbf{u} = (u_x, u_y, 0), \qquad \mathbf{v} = (u_x, u_y, d),$$

with $\mathbf{n} = \mathbf{x}/r$ and $d = \dot{r}$. The solved real P_2 ports are

$$\Pi^{(20)} = \frac{1}{2}(3d^2 - v^2), \qquad \Pi^{(21c)} = \sqrt{2}\, du_x, \qquad \Pi^{(21s)} = \sqrt{2}\, du_y,$$

$$\Pi^{(22c)} = \frac{u_x^2 - u_y^2}{2\sqrt{2}}, \qquad \Pi^{(22s)} = \frac{2u_x u_y}{2\sqrt{2}}.$$

Now form the STF tensor

$$Q_{ij} = v_i v_j - \frac{1}{3}\delta_{ij} v^2.$$

Choose the canonical real orthonormal STF basis

$$E_{20} = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad E_{21c,ij} = \frac{e_{x,i}e_{z,j} + e_{z,i}e_{x,j}}{\sqrt{2}}, \qquad E_{21s,ij} = \frac{e_{y,i}e_{z,j} + e_{z,i}e_{y,j}}{\sqrt{2}},$$

$$E_{22c} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E_{22s} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The exact pair-frame decomposition is then

$$q_{20} = \sqrt{\frac{2}{3}}\, \Pi^{(20)}, \qquad q_{21c} = \Pi^{(21c)}, \qquad q_{21s} = \Pi^{(21s)},$$

$$q_{22c} = 2\Pi^{(22c)}, \qquad q_{22s} = 2\Pi^{(22s)},$$

and therefore

$$Q_{ij} = \sqrt{\frac{2}{3}}\, \Pi^{(20)} E_{20,ij} + \Pi^{(21c)} E_{21c,ij} + \Pi^{(21s)} E_{21s,ij} + 2\Pi^{(22c)} E_{22c,ij} + 2\Pi^{(22s)} E_{22s,ij}.$$

This is the first theorem-level positive result on the quadrupole side: the solved 2PN P_2 package is not merely quadrupole-like, but exactly the full real STF $\ell = 2$ representation content. The missing PDE does not need to create a new tensor channel. It only needs to determine how the already isolated real P_2 bundle matches onto the actual source/radiative quadrupole of the reduced two-body problem.

D.2 Outgoing $\ell = 2$ threshold law and positive $+i\omega^5$ parity

For the outgoing three-dimensional quadrupole partial wave,

$$j_2(z) = \left(\frac{3}{z^3} - \frac{1}{z}\right)\sin z - \frac{3\cos z}{z^2}, \qquad y_2(z) = -\left(\frac{3}{z^3} - \frac{1}{z}\right)\cos z - \frac{3\sin z}{z^2}, \qquad h_2^{(1)}(z) = j_2(z) + iy_2(z),$$

with $z = ka$ and $k = \omega/c_s$, define the logarithmic derivative

$$\Lambda_2(k) = k\,\frac{\partial_z h_2^{(1)}(z)}{h_2^{(1)}(z)}\bigg|_{z=ka}.$$

Its low-frequency expansion is

$$\Lambda_2(k) = -\frac{3}{a} + \frac{a}{3}k^2 + \frac{a^3}{9}k^4 + i\frac{a^4}{9}k^5 - \frac{2a^5}{27}k^6 + O(k^7).$$

After inversion and normalization to fixed static residue R_2, one gets

$$Y_2^{\rm ret}(k) = R_2\left[1 + \frac{a^2k^2}{9} + \frac{4a^4k^4}{81} + i\frac{a^5k^5}{27} + O(k^6)\right].$$

Therefore

$$\Im Y_2^{\rm ret}(\omega) = \frac{R_2 a^5}{27c_s^5}\,\omega^5 + O(\omega^7),$$

with *positive* sign on the outgoing branch. This is the exact quadrupole analogue of the lower partial-wave threshold laws used earlier in the scalar and dipole audits:

$$\ell = 0 \Rightarrow \omega, \qquad \ell = 1 \Rightarrow \omega^3, \qquad \ell = 2 \Rightarrow \omega^5.$$

So the clean outgoing $\ell = 2$ branch lands exactly in the universal local 2.5PN slot.

D.3 The local odd quadrupole kernel is dissipative, not antidamping

For one quadrupole component with local odd kernel

$$L_{\rm rr}^{(2)} = -\frac{\Gamma}{2}\,q_- q_+^{(5)}, \qquad \Gamma > 0,$$

the physical force is

$$F = -\frac{\Gamma}{2}q^{(5)}.$$

The instantaneous power becomes

$$P = F\dot q = -\frac{\Gamma}{2}\dot q\,q^{(5)}.$$

Integrating by parts once gives the standard Schott decomposition

$$P = \frac{dE_{\rm Schott}}{dt} - \frac{\Gamma}{2}\big(q^{(3)}\big)^2, \qquad E_{\rm Schott} = -\frac{\Gamma}{2}\Big(\dot q\,q^{(4)} - \ddot q\,q^{(3)}\Big).$$

So the positive $+i\omega^5$ coefficient is exactly the desired dissipative sign. At this point the quadrupole branch has already passed three major filters: the representation exists, the odd term sits at the correct order, and the sign is damping rather than antidamping.

D.4 Why the literal compact internal P_2 route is finite-size only

The previous subsection does *not* imply that a literal compact internal mouth/support quadrupole is the universal point-particle source. Suppose one tries to use a compact internal P_2 support mode of size a as the direct carrier of the theorem. The conservative support sector already enters with a 2PN prefactor,

$$\epsilon \equiv \left(\frac{v}{c}\right)^2, \qquad \epsilon^2,$$

while the outgoing threshold factor contributes

$$\left(\frac{a\omega}{c_s}\right)^5 \sim \left(\sqrt{\epsilon}\,\frac{a}{r}\right)^5.$$

The total scaling is therefore

$$\epsilon^2\left(\sqrt{\epsilon}\,\frac{a}{r}\right)^5 = \epsilon^{9/2}\left(\frac{a}{r}\right)^5.$$

This lies above universal point-particle 2.5PN even before any extra small-body suppression is imposed. The compact internal P_2 branch is thus real, but it is not the universal theorem driver. It belongs to the higher finite-size throat-response sector.

This is a clarification, not a defeat. The theorem is actually cleaner once one separates the two roles:

1. the compact internal P_2 mouth/support branch remains a real higher-order finite-size signature,
2. while the universal point-particle 2.5PN source is carried by the orbital/worldtube STF quadrupole.

The rest of the appendix records that source-map statement explicitly.

D.5 Worldtube decomposition and the universal orbital/source split

For one compact defect A, in defect-centered coordinates $\boldsymbol{\xi}$, the source quadrupole is

$$I_{ij}^{(A)} = \int d^3\xi\,\rho_A(\boldsymbol{\xi},t)\big(X_A^i+\xi^i\big)\big(X_A^j+\xi^j\big)^{\mathrm{STF}}.$$

Expanding gives

$$I_{ij}^{(A)} = m_A X_A^{\langle i}X_A^{j\rangle} + 2X_A^{\langle i}d_A^{j\rangle} + Q_A^{ij},$$

with

$$d_A^i \equiv \int d^3\xi\,\rho_A\xi^i, \qquad Q_A^{ij} \equiv \int d^3\xi\,\rho_A\xi^{\langle i}\xi^{j\rangle}.$$

The carried worldtube reduction already imposes vanishing defect-centered dipole,

$$d_A^i = 0,$$

so the source simplifies to

$$I_{ij}^{(A)} = m_A X_A^{\langle i}X_A^{j\rangle} + Q_A^{ij}.$$

Summing the two bodies and going to the center-of-mass frame yields

$$I_{ij}^{\mathrm{eff}} = \mu x_{\langle i}x_{j\rangle} + Q_A^{ij} + Q_B^{ij} + O(c^{-2}, a^3\nabla^3\Phi).$$

This is the decisive source split:

$$\boxed{I_{ij}^{\mathrm{orb}} = \mu x_{\langle i}x_{j\rangle}}$$

is the universal point-particle source, while

$$Q_A^{ij} + Q_B^{ij}$$

is the finite-size internal worldtube correction sector. The internal quadrupole bundle therefore survives without being mistaken for the universal source itself.

D.6 Exact pair-frame source map to the solved P_2 quintet

For the orbital source quadrupole,

$$I_{ij}^{\rm orb} = \mu x_{\langle i} x_{j\rangle},$$

Newtonian order reduction gives

$$\ddot{I}_{ij}^{\rm orb} = 2\mu \left(v_{\langle i} v_{j\rangle} - \frac{GM}{r^3} x_{\langle i} x_{j\rangle} \right).$$

In the instantaneous pair frame, choose the relative position along the local z-axis,

$$\mathbf{x} = (0,0,r), \qquad \mathbf{a} = -\frac{GM}{r^3}\mathbf{x},$$

so that

$$X_{ij}^{\rm STF} \equiv x_{\langle i} x_{j\rangle} = \sqrt{\frac{2}{3}}\, r^2 E_{20,ij},$$

and the purely kinematic STF velocity piece is

$$V_{ij}^{\rm STF} \equiv v_{\langle i} v_{j\rangle} = \sqrt{\frac{2}{3}}\, \Pi^{(20)} E_{20,ij} + \Pi^{(21c)} E_{21c,ij} + \Pi^{(21s)} E_{21s,ij} + 2\Pi^{(22c)} E_{22c,ij} + 2\Pi^{(22s)} E_{22s,ij}.$$

Since only the axisymmetric position quadrupole survives in this frame,

$$X_{21c} = X_{21s} = X_{22c} = X_{22s} = 0, \qquad X_{20} = \sqrt{\frac{2}{3}}\, r^2.$$

Therefore the order-reduced orbital source map is

$$\frac{1}{2\mu}\ddot{I}_{ij}^{\rm orb} = \sqrt{\frac{2}{3}} \left(\Pi^{(20)} - \frac{GM}{r} \right) E_{20,ij} + \Pi^{(21c)} E_{21c,ij} + \Pi^{(21s)} E_{21s,ij} + 2\Pi^{(22c)} E_{22c,ij} + 2\Pi^{(22s)} E_{22s,ij}.$$

Equivalently, the STF coefficients obey

$$I_{21c} = 2\mu\, \Pi^{(21c)}, \qquad I_{21s} = 2\mu\, \Pi^{(21s)}, \qquad I_{22c} = 4\mu\, \Pi^{(22c)}, \qquad I_{22s} = 4\mu\, \Pi^{(22s)},$$

$$I_{20} = 2\mu \sqrt{\frac{2}{3}} \left(\Pi^{(20)} - \frac{GM}{r} \right).$$

This is the source-map formula in usable handoff form. It shows that the full real quintet is already present, that the $|m| = 1,2$ pieces are purely kinematic, and that only the axisymmetric $m = 0$ channel carries the static source shift. The solved 2PN P_2 throat-support structure and the orbital STF source are therefore not competing stories. They are two views of the same real STF $\ell = 2$ content.

D.7 Rotational invariance collapses the low-frequency matching to one scalar function

Once the source map is known structurally, the remaining representation-level question is the matching

$$\text{retarded throat/worldtube } P_2 \text{ basis} \longrightarrow \text{canonical STF source basis.}$$

There are two distinct facts here.

First, the port basis and the canonical STF basis are equivalent. The exact basis map is diagonal,

$$B = \operatorname{diag}\left(\sqrt{\frac{2}{3}}, 1, 1, 2, 2 \right), \qquad q^{\rm can} = B\, q^{\rm port}, \qquad \det B = \frac{4\sqrt{6}}{3} \neq 0.$$

So the solved port description is not secretly singular.

Second, rotational invariance collapses the whole low-frequency matching problem from a generic 5×5 matrix to one scalar function. Let J_x, J_y, J_z be the real $\ell = 2$ rotation generators acting on the STF quintet. The exact commutator test shows that the commutant of this representation is scalar:

$$[M, J_i] = 0 \quad \forall i \qquad \Longrightarrow \qquad M = m\,\mathbf{1}_5.$$

Therefore the low-frequency retarded matching matrix must take the form

$$\mathcal{M}^R_{ab}(\omega) = \left[m_0 + m_2\omega^2 + m_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab}.$$

Under the same passive/outgoing assumptions used elsewhere in the audit, the absorptive part is positive semidefinite, so the quadrupole odd coefficient obeys

$$\Gamma_5 > 0.$$

This is the decisive simplification of the quadrupole side: once isotropy is admitted, the entire $P_2 \to$ STF problem collapses to one scalar low-frequency matching function rather than an anisotropic tensor-valued obstruction.

D.8 Static-overlap gate: the branch is not naturally projected away

After the matching theorem, the only serious representation-level loophole would be a vanishing static overlap: perhaps the map is isotropic and the odd coefficient is positive, but the orbital STF source is projected away already at zero frequency. The pair-frame source map rules this out on the natural small-body branch.

Indeed, the static orbital/source configuration already carries a nonzero canonical $m = 0$ component,

$$x_{\langle i}x_{j\rangle} = \sqrt{\frac{2}{3}}\,r^2 E_{20,ij}, \qquad \frac{1}{2\mu}\ddot{I}^{\rm orb}_{ij}\Big|_{\rm static} = -\sqrt{\frac{2}{3}}\,\frac{GM}{r}\,E_{20,ij}.$$

Because the static low-frequency map is already forced to be

$$\mathcal{M}_0 = m_0\,\mathbf{1}_5,$$

one nonzero $m = 0$ matrix element forces the whole static overlap coefficient to be nonzero:

$$\boxed{m_0 \neq 0.}$$

In the raw audit, the solved static port amplitude is

$$q^{\rm static}_{20,\rm port} = \frac{5}{4},$$

so the mixed-basis overlap extracted from the $m = 0$ component is manifestly nonzero. That raw number is *not* yet the final canonical quadrupole normalization, because it mixes a port-basis numerator with a canonical STF source denominator. Its only role in the present appendix is to prove the gate statement that the branch is not projected away. The basis-clean canonical normalization is deferred to the later quadrupole-normalization appendix.

What matters here is the physical consequence. Finite-size worldtube quadrupoles enter only as

$$Q^{ij}_A + Q^{ij}_B = O\!\left(\frac{a^2}{r^2}\right) I^{\rm orb}_{ij},$$

so a natural total overlap has the form

$$\hat{m}_0 = 1 + \lambda\left(\frac{a}{r}\right)^2 + \cdots, \qquad \lambda = O(1).$$

Exact cancellation would require

$$\lambda = -\left(\frac{r}{a}\right)^2,$$

which is parametrically huge and therefore outside the controlled small-body hierarchy. The feared zero-overlap option is thus not a natural branch of the present reduction. It would require a tuned non-small-body cancellation, a radically anisotropic low-frequency branch, or a more serious breakdown of the hierarchy itself.

The appendix-level quadrupole verdict is therefore already strong:

$$\boxed{\begin{array}{c}\text{the solved 2PN } P_2 \text{ bundle is exactly the real STF } \ell = 2 \text{ content,}\\ \text{the clean outgoing } \ell = 2 \text{ branch supplies the correct positive } +i\omega^5 \text{ slot,}\\ \text{the literal compact internal } P_2 \text{ route is finite-size only,}\\ \text{the surviving universal point-particle source is the orbital/worldtube STF quadrupole,}\\ \text{and the static overlap gate passes: the branch is not naturally projected away.}\end{array}}$$

What remains open is no longer the existence of the quadrupole branch, but its final same-basis normalization on the passive/outgoing small-body branch.

E Quadrupole normalization package

This appendix records the technical normalization ledger behind the narrowing stated in Sec. 8. Appendix D already fixed three things that are more basic than normalization itself: the solved real P_2 ports already exhaust the full real STF $\ell = 2$ representation, the clean outgoing $\ell = 2$ branch carries the correct positive $+i\omega^5$ parity/sign, and the literal compact internal P_2 route is a finite-size correction rather than the universal point-particle source. The present appendix starts from that point and packages the remaining low-frequency quadrupole problem as cleanly as possible.

The main result is that the entire unresolved theorem gap can be reduced to one invariant scalar normalization. The branch shape, parity, and sign are already fixed; what remains open is the final passive/outgoing normalization on the actual moving-throat branch. To make that statement usable in later conservative work, the appendix also gives the minimal isotropic conservative quadrupole module, the grouped-P_2 extraction ledger through $O(\omega^4)$, and the axisymmetric anisotropy diagnostics that any future higher-order calculation will have to pass.

E.1 Canonical normalization target

Let

$$I_{ij} = \mu\, x_{\langle i} x_{j\rangle}$$

be the canonical orbital STF quadrupole of the relative two-body motion, and let q_a denote its components in a canonical real STF basis E_a,

$$q_a \equiv I_{ij} E_a^{ij}.$$

At quadratic order, the unique local odd action with the correct STF structure is

$$L_{\rm rr}^{(2)} = -\frac{\gamma}{2} \sum_a q_{a,-}\, q_{a,+}^{(5)}.$$

In the physical limit this gives

$$a_i = -\gamma\, x^j I_{ij}^{(5)}.$$

Comparison with the Burke–Thorne reaction force,

$$a_i^{\mathrm{BT}} = -\frac{2G}{5c^5}\, x^j I_{ij}^{(5)},$$

therefore fixes the exact canonical GR target

$$\gamma_{\mathrm{GR}} = \frac{2G}{5c^5}.$$

The toy-model quadrupole sector is not written directly in that canonical basis. If the working worldtube/throat source coordinate is normalized as

$$q_a^{\mathrm{toy}} = \hat{m}_0\, q_a^{\mathrm{can}},$$

and if the retarded matching matrix on the same branch has low-frequency form

$$\mathcal{M}_{ab}^{R}(\omega) = \Big[\hat{m}_0 + \hat{m}_2\omega^2 + \hat{m}_4\omega^4 + i\Gamma_5\omega^5 + \cdots\Big]\delta_{ab},$$

then the canonically normalized odd coefficient depends only on the product

$$\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \hat{m}_0^2\Gamma_5.$$

Matching to Burke–Thorne is therefore equivalent to the scalar condition

$$\boxed{\hat{m}_0^2\Gamma_5 = \frac{2G}{5c^5}.} \tag{1}$$

On the natural orbital/worldtube branch used throughout the paper,

$$\hat{m}_0 = 1 + O\!\left(\frac{a^2}{r^2}\right), \qquad \hat{m}_0 \neq 0,$$

so the final theorem gap is the passive/outgoing quadrupole normalization itself rather than the existence of the branch.

E.2 Outgoing $\ell = 2$ PDE fingerprint

For the STF quadrupole sector the exterior radial problem is the outgoing $\ell = 2$ Helmholtz equation

$$u'' + \frac{2}{r}u' + \left(k^2 - \frac{6}{r^2}\right)u = 0, \qquad u(r) \propto h_2^{(1)}(kr), \qquad k = \omega/c_s.$$

At the matching radius a, the exact outgoing Dirichlet-to-Neumann map is

$$\Lambda_2(k) = k\,\frac{h_2^{(1)\prime}(ka)}{h_2^{(1)}(ka)}.$$

Its low-frequency expansion is

$$\Lambda_2(k) = -\frac{3}{a} + \frac{a}{3}k^2 + \frac{a^3}{9}k^4 + i\frac{a^4}{9}k^5 - \frac{2a^5}{27}k^6 + O(k^7).$$

After inversion and normalization by the static residue, the outgoing quadrupole admittance becomes

$$\widehat{Y}_2(k) = 1 + \frac{a^2k^2}{9} + \frac{4a^4k^4}{81} + i\frac{a^5k^5}{27} + O(k^6),$$

or, equivalently,

$$\widehat{Y}_2(\omega) = 1 + \frac{a^2\omega^2}{9c_s^2} + \frac{4a^4\omega^4}{81c_s^4} + i\frac{a^5\omega^5}{27c_s^5} + O(\omega^6).$$

If the low-frequency retarded kernel is written as

$$\mathcal{M}^R_{ab}(\omega) = \Big[m_0 + m_2\omega^2 + m_4\omega^4 + i\Gamma_5^{\mathrm{PDE}}\omega^5 + \cdots\Big]\delta_{ab},$$

then the outgoing branch enforces

$$m_2 = m_0\frac{a^2}{9c_s^2}, \qquad m_4 = m_0\frac{4a^4}{81c_s^4}, \qquad \Gamma_5^{\mathrm{PDE}} = m_0\frac{a^5}{27c_s^5},$$

and therefore the invariant relations

$$\boxed{m_4 = \frac{4m_2^2}{m_0}, \qquad \Gamma_5^{\mathrm{PDE}} = 9\,\frac{m_2^{5/2}}{m_0^{3/2}}.} \tag{2}$$

Two points matter immediately. First, the odd term carries the correct passive outgoing sign,

$$+i\Gamma_5^{\mathrm{PDE}}\omega^5, \qquad \Gamma_5^{\mathrm{PDE}} > 0,$$

on the natural branch. Second, this is only the raw PDE fingerprint. By itself it is not yet the Burke–Thorne worldline coefficient, because the source-to-port normalization still has to be fixed.

E.3 Source/port normalization cleanup

The solved conservative throat-support package is organized in the real port basis used by the P_2 analysis, while the worldline theorem is written in a canonical STF basis. Let q^{port} denote the real P_2 port vector and q^{can} the canonical STF coefficients. The fixed basis conversion is

$$q^{\mathrm{can}} = B\,q^{\mathrm{port}}, \qquad B = \mathrm{diag}\left(\sqrt{\frac{2}{3}}, 1, 1, 2, 2\right), \qquad \det B = \frac{4\sqrt{6}}{3} \neq 0.$$

So the port and canonical descriptions are equivalent; the normalization issue is not one of missing tensor content but of keeping numerator and denominator in one and the same basis.

The raw static audit illustrates the bookkeeping point. The canonical orbital source coefficient is

$$C_{20}\left[\frac{\ddot{I}_{\mathrm{orb}}}{2\mu}\right] = -\frac{\sqrt{6}\,GM}{3r},$$

while the same-basis port coefficient is

$$P_{20}\left[\frac{\ddot{I}_{\mathrm{orb}}}{2\mu}\right] = -\frac{GM}{r}.$$

Using the solved static port amplitude

$$q^{\mathrm{static}}_{20,\mathrm{port}} = \frac{5}{4},$$

one finds the mixed-basis ratio

$$m_0^{\mathrm{mixed}} = \frac{q^{\mathrm{static}}_{20,\mathrm{port}}}{C_{20}[\ddot{I}_{\mathrm{orb}}/(2\mu)]} = -\frac{5\sqrt{6}\,r}{8GM},$$

whereas the same-basis raw overlap is

$$m_{0,\mathrm{raw}} = \frac{q_{20,\mathrm{port}}^{\mathrm{static}}}{P_{20}[\ddot{I}_{\mathrm{orb}}/(2\mu)]} = -\frac{5r}{4GM}.$$

If the numerator is first converted into canonical STF units, the same-basis ratio is recovered. The printed mixed-basis number is therefore a convention artifact rather than the physical quadrupole normalization.

The natural cure is to normalize the pure orbital point-particle branch to unit static overlap. Define

$$\mathcal{N}_{\mathrm{orb}} = \frac{1}{m_{0,\mathrm{raw}}} = -\frac{4GM}{5r}, \qquad \hat{m}_0 \equiv \mathcal{N}_{\mathrm{orb}}\, m_{0,\mathrm{raw}}.$$

Then on the pure orbital point-particle branch,

$$\hat{m}_0 = 1,$$

while worldtube finite-size corrections can only perturb this as

$$\hat{m}_0 = 1 + O\left(\frac{a^2}{r^2}\right).$$

This is the same natural branch assumption used in Sec. 8.

E.4 Convention-invariant normalization product

It is helpful to separate the source normalization from the port-basis retarded kernel before combining them again into the physical coefficient. Write the port variable as

$$p_a = \mathfrak{m}_0\, q_a^{\mathrm{can}},$$

and let the port-basis kernel be

$$K_{ab}^{R}(\omega) = \Big[K_0 + K_2\omega^2 + K_4\omega^4 + i\Gamma_5^{\mathrm{port}}\omega^5 + \cdots\Big]\delta_{ab}.$$

The physical Burke–Thorne coefficient is not either object separately. It is obtained only after pulling the port kernel back to the canonical STF basis:

$$\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \mathfrak{m}_0^2\Gamma_5^{\mathrm{port}}.$$

This combination is basis-invariant. Under a port rescaling

$$p'_a = \lambda p_a,$$

one has

$$\mathfrak{m}'_0 = \lambda\mathfrak{m}_0, \qquad K'_0 = \frac{K_0}{\lambda^2}, \qquad \Gamma_5^{\mathrm{port}\,\prime} = \frac{\Gamma_5^{\mathrm{port}}}{\lambda^2},$$

so the invariant products are

$$\mathfrak{m}_0'^2 K'_0 = \mathfrak{m}_0^2 K_0, \qquad \mathfrak{m}_0'^2\Gamma_5^{\mathrm{port}\,\prime} = \mathfrak{m}_0^2\Gamma_5^{\mathrm{port}}.$$

Now insert the outgoing $\ell = 2$ fingerprint,

$$\Gamma_5^{\mathrm{port}} = K_0\frac{a^5}{27c_s^5}.$$

Then the full physical odd coefficient becomes

$$\gamma_{\rm quad}^{\rm eff} = \mathfrak{m}_0^2 K_0 \, \frac{a^5}{27c_s^5}.$$

It is therefore natural to define the single invariant normalization product

$$\boxed{\mathcal{N}_Q \equiv \mathfrak{m}_0^2 K_0.} \tag{3}$$

The quadrupole coefficient is then simply

$$\gamma_{\rm quad}^{\rm eff} = \mathcal{N}_Q \, \frac{a^5}{27c_s^5}.$$

Burke–Thorne matching gives the clean target

$$\boxed{\mathcal{N}_Q^{\rm target} = \frac{54Gc_s^5}{5a^5c^5}.} \tag{4}$$

This is the most compact statement of the unresolved normalization problem. The open theorem target is not separately "what is $\mathfrak{m}_0$?" and "what is K_0?" It is the single invariant product $\mathcal{N}_Q = \mathfrak{m}_0^2 K_0$.

E.5 Canonical low-frequency fingerprint and the single-prefactor obstruction

Define the canonical invariant coefficients

$$\overline{K}_0 \equiv \mathfrak{m}_0^2 K_0, \qquad \overline{K}_2 \equiv \mathfrak{m}_0^2 K_2, \qquad \overline{K}_4 \equiv \mathfrak{m}_0^2 K_4, \qquad \overline{\Gamma}_5 \equiv \mathfrak{m}_0^2 \Gamma_5^{\rm port}.$$

The outgoing branch then forces

$$\boxed{\overline{K}_2 = \overline{K}_0 \frac{a^2}{9c_s^2}, \qquad \overline{K}_4 = \overline{K}_0 \frac{4a^4}{81c_s^4}, \qquad \overline{\Gamma}_5 = \overline{K}_0 \frac{a^5}{27c_s^5}.} \tag{5}$$

Eliminating the geometric ratio a/c_s gives the fully invariant branch relations

$$\boxed{\overline{K}_4 = \frac{4\overline{K}_2^2}{\overline{K}_0}, \qquad \overline{\Gamma}_5 = 9\, \frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}.} \tag{6}$$

So once the even pair $(\overline{K}_0, \overline{K}_2)$ is known on the same branch, the next even moment and the odd Burke–Thorne coefficient are no longer independent unknowns.

Imposing exact Burke–Thorne matching,

$$\overline{\Gamma}_5 = \frac{2G}{5c^5},$$

fixes the target invariant pair and the next even moment:

$$\boxed{\overline{K}_0^{\rm target} = \frac{54Gc_s^5}{5a^5c^5}, \qquad \overline{K}_2^{\rm target} = \frac{6Gc_s^3}{5a^3c^5}, \qquad \overline{K}_4^{\rm target} = \frac{8Gc_s}{15ac^5}.} \tag{7}$$

If the asymptotic propagation speed closes to $c_s = c$, these simplify to

$$\overline{K}_0^{\rm target} = \frac{54G}{5a^5}, \qquad \overline{K}_2^{\rm target} = \frac{6G}{5a^3c^2}, \qquad \overline{K}_4^{\rm target} = \frac{8G}{15ac^4}.$$

The obstruction is therefore extremely narrow: the current stack already fixes the entire low-frequency quadrupole branch up to one scalar prefactor. What it does not yet derive from the completed moving-throat PDE is that prefactor.

E.6 What the frozen 2PN files already fix

The dynamic normalization remains open, but the static and representation-side quadrupole content is already strongly constrained by the frozen 2PN files. At the kinematic level, the pair-frame STF map is exact:

$$\frac{\ddot{I}_{ij}}{2\mu} = \mathrm{STF}(v_i v_j - U n_i n_j), \qquad \mathbf{n} = \hat{z}, \qquad \mathbf{v} = (u_x, u_y, d), \qquad U = \frac{GM}{r}.$$

Projecting this onto the canonical real STF basis reproduces exactly the real P_2 package used by the conservative support hierarchy, with the same fixed basis map

$$q^{\mathrm{can}} = B\, q^{\mathrm{port}}, \qquad B = \mathrm{diag}\left(\sqrt{\frac{2}{3}}, 1, 1, 2, 2\right).$$

So the solved P_2 channels are already the full real STF $\ell = 2$ representation content.

The zero-frequency support/source data are frozen as well. In the static even support layer,

$$R_0 = R_{20} = R_{21} = R_{22} = 1,$$

while the scalar source vector is

$$J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0\right).$$

The same static ledger also identifies the pure-potential geometry-closure deficit

$$\Delta_{\mathrm{geom}} = \frac{281}{80}.$$

What the frozen files *do not* fix is equally clear: the important dynamic pole scales, including the quadrupole poles

$$\Omega_{20},\ \Omega_{21},\ \Omega_{22},$$

remain genuine inner-throat PDE observables rather than solved conservative numbers. The frozen 2PN files therefore already fix the representation content and the static support/source layer, but they do not yet determine the final dynamic quadrupole normalization.

E.7 One-pole insufficiency and the minimal positive conservative precursor

Write

$$A \equiv \frac{a^2}{9c_s^2}.$$

The normalized outgoing branch is then

$$\widehat{Y}_2^{\mathrm{out}}(\omega) = 1 + A\omega^2 + 4A^2\omega^4 + i\frac{a^5}{27c_s^5}\omega^5 + \cdots.$$

A normalized one-pole conservative proxy,

$$\widehat{Y}_Q^{(1\mathrm{pole})}(\omega) = \frac{1}{1 - \omega^2/\Omega_1^2},$$

reproduces the quadratic moment by choosing

$$\Omega_1 = \frac{1}{\sqrt{A}} = \frac{3c_s}{a},$$

but then its ω^4 coefficient is only A^2, whereas the outgoing branch requires $4A^2$. So a single conservative pole by itself is already too small.

The smallest isotropic conservative module with the correct even moments is

$$\widehat{Y}_Q^{\rm cons}(\omega) = c_0 + \frac{c_1}{1-\omega^2/\Omega_Q^2}.$$

Matching the exact $O(\omega^0)$, $O(\omega^2)$, and $O(\omega^4)$ moments produces the unique positive-real solution

$$\boxed{c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}.} \tag{8}$$

Hence the minimal conservative quadrupole module is

$$\boxed{\widehat{Y}_{Q,\min}^{\rm cons}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}.} \tag{9}$$

Equivalently,

$$\widehat{Y}_{Q,\min}^{\rm cons}(\omega) = \frac{3(a^2\omega^2 - 3c_s^2)}{4a^2\omega^2 - 9c_s^2}.$$

This is the smallest positive conservative precursor compatible with the exact outgoing $\ell = 2$ moments. The old one-pole-per-channel conservative scaffold is therefore ruled out before the radiative odd term is even added.

E.8 Minimal isotropic module in canonical and grouped-P_2 language

In canonical invariant language, the same conservative branch reads

$$\overline{K}_Q^{\rm cons}(\omega) = \overline{K}_0\left[\frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}\right].$$

Its even coefficients are

$$\overline{K}_2 = \frac{\overline{K}_0}{4\Omega_Q^2}, \qquad \overline{K}_4 = \frac{\overline{K}_0}{4\Omega_Q^4},$$

so the exact conservative branch identity is

$$\boxed{\overline{K}_0\,\overline{K}_4 = 4\overline{K}_2^2.} \tag{10}$$

On the same outgoing branch the odd coefficient is then fixed automatically:

$$\overline{\Gamma}_5 = 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}} = \frac{9\overline{K}_0}{32\Omega_Q^5}.$$

This is the single-pole sufficiency statement: once the conservative pole and static normalization are known on the minimal isotropic branch, the full low-frequency quadrupole ledger is fixed.

In grouped real-channel language, isotropy means

$$Y_{20} = Y_{21} = Y_{22} = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}, \qquad \Omega_{20} = \Omega_{21} = \Omega_{22} = \Omega_Q.$$

So the next conservative calculation no longer needs to be described vaguely as "do 3PN and 5PN somehow." It has a very concrete target: extract the common grouped P_{20}, P_{21}, and P_{22} coefficients through $O(\omega^4)$ and test whether they coincide and satisfy the minimal-branch identity.

E.9 Explicit extraction ledger

Suppose a future conservative calculation yields the isotropic low-frequency series

$$Y_Q^{\text{cons}}(\omega) = c_0 + c_2\omega^2 + c_4\omega^4 + O(\omega^6).$$

On the minimal isotropic branch,

$$c_0 = \overline{K}_0, \qquad c_2 = \overline{K}_2, \qquad c_4 = \overline{K}_4,$$

and the effective conservative pole may be reconstructed in two equivalent ways,

$$\Omega_Q^2 = \frac{c_0}{4c_2}, \qquad \Omega_Q^2 = \frac{c_2}{c_4}, \qquad \overline{K}_0 = \frac{4c_2^2}{c_4}.$$

The corresponding odd coefficient is then fixed automatically:

$$\boxed{\overline{\Gamma}_5 = \frac{9}{8}\frac{c_4^{3/2}}{\sqrt{c_2}}.} \tag{11}$$

Burke–Thorne matching,

$$\overline{\Gamma}_5 = \frac{2G}{5c^5},$$

is therefore equivalent to the scalar GR gate

$$\boxed{\frac{c_4^3}{c_2} = \frac{256G^2}{2025c^{10}}.} \tag{12}$$

So the full 2.5PN closure on the minimal branch reduces to a compact five-step ledger:

1. show that the conservative real P_2 bundle is isotropic through $O(\omega^4)$,
2. extract the common coefficients (c_0, c_2, c_4),
3. verify the branch identity $c_0c_4 = 4c_2^2$,
4. reconstruct Ω_Q and $\overline{\Gamma}_5$ from the conservative data,
5. and finally test the GR gate $\overline{\Gamma}_5 = 2G/(5c^5)$.

This is the sharpest reduced theorem target available before the full moving-throat PDE is solved.

E.10 Normalized-support and axisymmetric grouped-P_2 formulas

Because the static support residues are already frozen to one,

$$R_{20} = R_{21} = R_{22} = 1,$$

it is often cleaner to work in the normalized-support convention

$$Y_{20}(\omega) = 1 + u_2^{(20)}\omega^2 + u_4^{(20)}\omega^4 + O(\omega^6),$$

$$Y_{21}(\omega) = 1 + u_2^{(21)}\omega^2 + u_4^{(21)}\omega^4 + O(\omega^6),$$

$$Y_{22}(\omega) = 1 + u_2^{(22)}\omega^2 + u_4^{(22)}\omega^4 + O(\omega^6).$$

The genuinely new conservative payload is then just the six moments

$$\boxed{u_2^{(20)},\ u_2^{(21)},\ u_2^{(22)},\ u_4^{(20)},\ u_4^{(21)},\ u_4^{(22)}.}$$

On the minimal isotropic branch,

$$u_2^{(20)} = u_2^{(21)} = u_2^{(22)} \equiv u_2, \qquad u_4^{(20)} = u_4^{(21)} = u_4^{(22)} \equiv u_4,$$

and the normalized-support branch identity is simply

$$\boxed{u_4 = 4u_2^2.} \tag{13}$$

The effective pole is

$$\Omega_Q^2 = \frac{1}{4u_2}, \qquad \Omega_Q = \frac{1}{2\sqrt{u_2}},$$

and the normalized odd coefficient becomes

$$\boxed{\Gamma_{5,\mathrm{norm}} = 9u_2^{5/2}.} \tag{14}$$

The physical odd coefficient still carries the overall quadrupole normalization,

$$\overline{\Gamma}_5 = \overline{K}_0 \Gamma_{5,\mathrm{norm}},$$

so Burke–Thorne matching fixes

$$\boxed{\overline{K}_0^{\mathrm{target}} = \frac{2G}{45c^5 u_2^{5/2}} = \frac{64G}{45c^5}\Omega_Q^5.} \tag{15}$$

For practical grouped calculations, it is useful to package the channel moments in the axisymmetric five-component form

$$Y^{(2)}_{\mathrm{cons}}(\omega) = I_5 + \omega^2 U_2 + \omega^4 U_4 + O(\omega^6),$$

with

$$U_2 = \bar{u}_2 I_5 + a_2 T + b_2 S, \qquad U_4 = \bar{u}_4 I_5 + a_4 T + b_4 S,$$

where

$$T = \mathrm{diag}(4, -1, -1, -1, -1), \qquad S = \mathrm{diag}(0, 1, 1, -1, -1).$$

Then the grouped channels are

$$u_2^{(20)} = \bar{u}_2 + 4a_2, \qquad u_2^{(21)} = \bar{u}_2 - a_2 + b_2, \qquad u_2^{(22)} = \bar{u}_2 - a_2 - b_2,$$

and similarly at $O(\omega^4)$,

$$u_4^{(20)} = \bar{u}_4 + 4a_4, \qquad u_4^{(21)} = \bar{u}_4 - a_4 + b_4, \qquad u_4^{(22)} = \bar{u}_4 - a_4 - b_4.$$

The inverse map for any grouped triple (x_{20}, x_{21}, x_{22}) is

$$\boxed{\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}.} \tag{16}$$

The anisotropy norms are

$$\mathcal{A}_2^2 = \frac{1}{5}\mathrm{tr}(U_2 - \bar{u}_2 I_5)^2 = 4a_2^2 + \frac{4}{5}b_2^2,$$

$$\mathcal{A}_4^2 = \frac{1}{5}\mathrm{tr}(U_4 - \bar{u}_4 I_5)^2 = 4a_4^2 + \frac{4}{5}b_4^2.$$

So the isotropy gate is simply

$$\boxed{a_2 = b_2 = a_4 = b_4 = 0.} \tag{17}$$

If that gate passes, the grouped sector collapses to one common branch with

$$\bar{u}_4 = 4\bar{u}_2^2, \qquad \Omega_Q^2 = \frac{1}{4\bar{u}_2}, \qquad \Gamma_{5,\mathrm{norm}} = 9\bar{u}_2^{5/2}, \qquad \overline{K}_0^{\mathrm{target}} = \frac{2G}{45c^5\bar{u}_2^{5/2}}.$$

This is the exact axisymmetric template that a future conservative grouped $P_{20} \oplus P_{21} \oplus P_{22}$ calculation will have to hit if the minimal isotropic quadrupole branch is to remain viable.

E.11 What the package fixes and what it still leaves open

The quadrupole normalization package therefore does five things at once. First, it rewrites the remaining theorem gap in the clean canonical form (1). Second, it shows that the outgoing $\ell = 2$ PDE already fixes the full low-frequency branch shape through (2) and (6). Third, it isolates the normalization problem as the single invariant product $\mathcal{N}_Q$ in (3). Fourth, it identifies the smallest admissible conservative precursor, namely the minimal isotropic module (9). Fifth, it turns the vague higher-order quadrupole task into a concrete grouped-P_2 extraction ledger through $O(\omega^4)$, with explicit isotropy, branch, and GR gates.

What it does *not* do is derive the remaining invariant prefactor directly from the completed moving-throat PDE. That is the one narrow obstruction left at the end of the present 2.5PN program. The branch itself is structurally in place; the unresolved issue is its final passive/outgoing normalization on the actual moving-throat solution.

F Imported lower-order anchors

This appendix collects the specific formulas and status statements imported from the parent 4D paper and the full conservative 1PN and 2PN papers and used without rederivation in the present 2.5PN audit. Its role is technical rather than historical. We do not repeat the full parent action, the full 1PN or 2PN derivations, or the longer appendix-side calculations that fixed the lower-order response data. Instead, we record only the load-bearing identities, reduced equations, and frozen coefficients that organize the scalar, dipole, and quadrupole audits in the main text.

The claim-status boundary is the same one used throughout the earlier papers. Exact projection identities from the parent 4D theory are kept visibly exact. The near-zone Poisson hook, the zero-mode Maxwell law, and the worldtube point-particle reduction are carried only as controlled reductions. The conservative 1PN and 2PN ledgers are treated as frozen outputs of the earlier program within its declared closure hierarchy, not as live fit parameters to be retuned in the present paper.

F.1 Exact 4D projection identities and controlled brane reductions

The parent theory lives on a $(4+1)$-dimensional spacetime with coordinates

$$X^M = (t, x, y, z, w), \qquad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \qquad \mathbf{x} = (x, y, z) \in \mathbb{R}^3.$$

The bulk variables carried into the present paper are the matter field $\psi(\mathbf{X}, t)$, its density

$$\rho(\mathbf{X}, t) = |\psi(\mathbf{X}, t)|^2,$$

the optional localized gauge field $A_M(\mathbf{X}, t)$ with localization profile $Z(w)$, and the effective throat geometry variables

$$a(t), \qquad L(t).$$

When electric charge is named explicitly in the inherited downstream notation, it is the corrected branch quantity

$$q_* = \eta_Q e_*, \qquad q_{\rm eff} = \frac{q_*}{\sqrt{Z_{\rm int}}}, \qquad Z_{\rm int} = \int_{-\infty}^{\infty} Z(w)\,dw,$$

not the gravity-side scalar coefficient κ_ρ used in the post-Newtonian ledger.

Brane observables are defined by a fixed normalized projection kernel $W(w)$,

$$\int_{-\infty}^{\infty} W(w)\,dw = 1, \qquad \mathcal{P}_W[Q](t,\mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w)Q(t,\mathbf{x},w)\,dw.$$

In particular,

$$\rho_{\rm br} = \mathcal{P}_W[\rho], \qquad \mathbf{J}_{\rm br} = \mathcal{P}_W[(j^x, j^y, j^z)], \qquad \mathbf{v}_{\rm br} = \frac{\mathbf{J}_{\rm br}}{\rho_{\rm br}} \quad (\rho_{\rm br} > 0).$$

Projection is not the same as reduction. Projection defines what a brane observer measures; reduction refers to the further elimination of the extra coordinate under an explicit ansatz or scaling regime.

From exact bulk continuity,

$$\partial_t \rho + \partial_i j^i = 0, \qquad i \in \{x, y, z, w\},$$

one obtains the exact open-system brane continuity law

$$\partial_t \rho_{\rm br} + \nabla_3 \cdot \mathbf{J}_{\rm br} = S_{\rm leak},$$

with leakage source

$$S_{\rm leak} = -\left[W(w) j^w\right]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w) j^w\,dw.$$

Under fast decay of $W j^w$ at infinity this simplifies to

$$S_{\rm leak} = \int_{-\infty}^{\infty} W'(w) j^w\,dw.$$

After Helmholtz decomposition of the brane velocity,

$$\mathbf{v}_{\rm br} = \nabla_3 \phi_{\rm br} + \mathbf{v}_T, \qquad \nabla_3 \cdot \mathbf{v}_T = 0,$$

projection continuity becomes the exact longitudinal identity

$$\rho_{\rm br}\,\nabla_3^2 \phi_{\rm br} = S_{\rm leak} - \partial_t \rho_{\rm br} - (\nabla_3 \rho_{\rm br}) \cdot (\nabla_3 \phi_{\rm br} + \mathbf{v}_T).$$

This identity is not yet a Poisson equation. The Poisson equation appears only after the usual quasi-static near-zone simplifications, in which case one obtains the controlled Poisson hook

$$\nabla_3^2 \phi_{\rm br} \simeq \frac{S_{\rm eff}}{\rho_0}.$$

For a localized effective source,

$$S_{\rm eff}(\mathbf{x}) \sim \mathcal{S}\,\delta^{(3)}(\mathbf{x}),$$

this gives the inverse-square longitudinal profile

$$\phi_{\rm br}(\mathbf{x}) \sim -\frac{\mathcal{S}}{4\pi\rho_0}\frac{1}{r}, \qquad \nabla_3 \phi_{\rm br}(\mathbf{x}) \sim \frac{\mathcal{S}}{4\pi\rho_0}\frac{\hat{\mathbf{r}}}{r^2}.$$

The measured Newtonian potential is then introduced by normalization of this scalar channel,

$$\Phi_N(\mathbf{x},t) = \lambda_\Phi\, \phi_{\rm br}(\mathbf{x},t),$$

so that an isolated compact source B has the near-zone exterior field

$$\Phi_B(\mathbf{x},t) = -\frac{Gm_B}{|\mathbf{x}-\mathbf{X}_B(t)|} + O\!\left(\frac{Gm_B a_B^2}{|\mathbf{x}-\mathbf{X}_B|^3}\right).$$

The second carried reduction is the coherent-defect / small-body worldtube limit. For a compact defect of size $a \ll r$ in a smooth external field, the center-of-mass reduction gives

$$M\ddot{\mathbf{X}}_{\rm cm} = -M\nabla_3\Phi_N(\mathbf{X}_{\rm cm},t) + O\!\left(Q\,\nabla^3\Phi_N\right),$$

with the first finite-size correction quadrupolar and suppressed by

$$\frac{|\mathbf{F}_{\rm quad}|}{|\mathbf{F}_{\rm mono}|} = O\!\left(\frac{a^2}{r^2}\right).$$

The corresponding frozen Newtonian two-body block is therefore

$$L_0 = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 + \frac{Gm_A m_B}{r}.$$

For completeness, the parent 4D paper also supplies a controlled zero-mode Maxwell reduction. Under the far-field brane assumptions

$$A_w \approx 0, \qquad \partial_w A_\mu \approx 0, \qquad J^w \approx 0, \qquad F_{\mu w} \approx 0,$$

integration over w yields the effective brane Maxwell law

$$\partial_\mu F^{\mu\nu} = \mu_0^{\rm eff} J^\nu_{\rm eff}, \qquad \mu_0^{\rm eff} = \frac{\mu_0}{Z_{\rm int}}.$$

This reduction suppresses the mixed channels only in the controlled far-field limit. It does not erase $(A_w, J^w, F_{\mu w})$ from the microscopic ontology, which matters later whenever hidden-core transport or mixed response channels are discussed.

F.2 Frozen Newtonian and 1PN conservative ledger

The present 2.5PN paper imports the Newtonian+1PN conservative ledger as fixed lower-order input. At the one-body level, the scalar/optical worldline ansatz carried from the full 1PN paper is

$$L_{\rm sc} = -M_0\left(1+\kappa_\rho\frac{\Phi_N}{c^2}\right)c^2\sqrt{1-\frac{v^2/c^2}{1+(n-1)\Phi_N/c^2}}.$$

Its weak-field expansion is

$$L_{\rm sc} = -M_0c^2 - \kappa_\rho M_0\Phi_N + \frac{1}{2}M_0v^2 + \frac{1}{8}M_0\frac{v^4}{c^2} + M_0\left(\frac{\kappa_\rho}{2} - \frac{n-1}{2}\right)\frac{\Phi_N v^2}{c^2} + O(c^{-4}).$$

Newtonian matching fixes

$$\kappa_\rho = 1,$$

while weak-field optical consistency for the barotropic law

$$P(\rho) = K_{\rm EOS}\rho^n$$

fixes

$$n = 5.$$

With these values, the free-particle quartic coefficient is frozen to 1/8 in the Lagrangian, and the self-sector pair term becomes

$$L_{\text{self}}^{(AB)} = \frac{Gm_A m_B}{c^2 r_{AB}} \frac{3}{2} (v_A^2 + v_B^2).$$

The static nonlinear 1PN term follows from local mass scaling and exact pair counting. The frozen two-body result is

$$L_{\text{stat}}^{(AB)} = -\frac{G^2 m_A m_B (m_A + m_B)}{2c^2 r_{AB}^2}.$$

The response-side self ledger is carried in the form

$$\beta_{1\text{PN}} = \kappa_\rho + \kappa_{\text{add}} + \kappa_{\text{PV}},$$

with the already frozen values

$$\kappa_{\text{add}} = \frac{1}{2}, \qquad \kappa_{\text{PV}} = \frac{3}{2}, \qquad \beta_{1\text{PN}} = 3.$$

Within the declared adiabatic one-degree-of-freedom breathing closure, the same response module also carries the static energy partition and density slope

$$E_w : E_f : E_{\text{PV}} = 11 : 2 : 5, \qquad \frac{d \ln a_*}{d \ln \rho} = -\frac{57}{64}.$$

The counterfactual compact 4D bubble would instead give

$$\kappa_{\text{add}}^{(B^4)} = \frac{1}{3},$$

so $\kappa_{\text{add}} = 1/2$ remains a topology discriminator rather than a tunable fit.

The parity-even wake data are also frozen by the full conservative 1PN assembly. The pairwise cross sector is written as

$$L_{\text{vec}}^{(AB)} = \frac{Gm_A m_B}{c^2 r_{AB}} \left[C_\parallel \, \mathbf{v}_A \cdot \mathbf{v}_B + C_L (\mathbf{v}_A \cdot \hat{\mathbf{n}}_{AB})(\mathbf{v}_B \cdot \hat{\mathbf{n}}_{AB}) \right],$$

with

$$C_\parallel = K_{\text{vec}} \pi^2 (-1 + a_H^2 - \alpha^2), \qquad C_L = K_{\text{vec}} \pi^2 (-1 + a_H^2 + \alpha^2).$$

The thermodynamic wake closure gives

$$\alpha^2 = 1 - \frac{1}{n-1} = \frac{3}{4},$$

and the real even solution collapses to

$$a_H = 0, \qquad K_{\text{vec}} = \frac{2}{\pi^2},$$

so the exact EIH cross coefficients are

$$C_\parallel = -\frac{7}{2}, \qquad C_L = -\frac{1}{2}.$$

Putting the self, static, and wake sectors together gives the frozen conservative Newtonian+1PN two-body block

$$L_1 = \frac{1}{8}(m_A v_A^4 + m_B v_B^4) + \frac{Gm_A m_B}{r} \left[\frac{3}{2}(v_A^2 + v_B^2) - \frac{7}{2} v_{AB} - \frac{1}{2} v_{An} v_{Bn} \right] - \frac{G^2 m_A m_B (m_A + m_B)}{2r^2}.$$

Nothing in the present paper is allowed to retune these lower-order values. They are carry-forward anchors rather than open coefficients.

F.3 Frozen 2PN support/response hierarchy

The full conservative 2PN paper closes the comparable-mass c^{-4} ledger against the standard generic-frame ADM target within the same declared closure hierarchy, but the present 2.5PN paper imports from it only the load-bearing response-side structure. At the one-body level, the exact isotropic Schwarzschild gate through 2PN is packaged by

$$D_{\text{target}}(u) = 1 - 4u + 8u^2 + O(u^3), \qquad u \equiv \frac{U}{c^2},$$

and the frozen static 2PN gate is

$$\lambda_\rho = \frac{1}{2}, \qquad L_{2,\text{static}} = \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}.$$

The natural 2PN self/static seed is therefore

$$\begin{aligned} L_{2,\text{self+static}} = {}& \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{7G m_A m_B}{8r}(v_A^4 + v_B^4) \\ &+ \frac{2G^2 m_A m_B}{r^2}(m_B v_A^2 + m_A v_B^2) + \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}. \end{aligned} \tag{18}$$

The exact two-body coefficient list is not repeated here because the present 2.5PN audit uses mainly the constructive response packaging that the 2PN theorem exposes.

The decisive ontology change at 2PN is that the defect is no longer described adequately as a pure scalar monopole. The conservative cross sector already splits into three response lanes:

1. a carried odd dipole wake sector,
2. a genuine even $P_0 \oplus P_2$ mouth/support layer,
3. and a separate geometry-closure channel.

The appendix-side zero-frequency throat data make this explicit. The odd dipole residues are frozen to

$$R_{1\perp} = \frac{7}{2}, \qquad R_{10} = 4,$$

while the canonical zero-frequency even support residues are

$$R_0 = R_{20} = R_{21} = R_{22} = 1.$$

The same solved static even operator fixes the source-weight vector

$$J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0\right), \qquad J \cdot J = \frac{381}{80},$$

and the pure-potential geometry-closure deficit

$$\Delta_{\text{geom}} = \frac{281}{80}.$$

The monopole wall add-on

$$\Delta K_{00} = \frac{109}{280}$$

is kept distinct from Δ_{geom}; they are not the same object and should not be merged in later bookkeeping.

What remains open on the inner-throat side after the conservative 2PN theorem is not these zero-frequency data but the fully dynamical completion. In particular, the pole or compliance scales

$$\Omega_{1\perp},\ \Omega_{10},\ \Omega_0,\ \Omega_{20},\ \Omega_{21},\ \Omega_{22},\ \Omega_g$$

are not claimed as already derived from the completed moving-throat PDE. This status boundary is central for the present paper. The scalar, dipole, and quadrupole audits are allowed to use the fixed 2PN residues and source data, but they are not allowed to treat the full retarded completion as already solved.

F.4 What is fixed, what is reduced, and what remains open

The imported lower-order anchors should therefore be read in four layers.

First, the parent 4D projection map, projected continuity law with leakage, and the longitudinal identity are exact structural formulas.

Second, the Poisson hook, the zero-mode Maxwell law, and the worldtube point-particle limit are controlled reductions that define the near-zone particle dictionary used by the post-Newtonian papers.

Third, the Newtonian+1PN and conservative 2PN ledgers are frozen lower-order outputs within the declared closure hierarchy. The present paper inherits their constants, residues, and response packaging as fixed input rather than as free data.

Fourth, several sectors remain explicitly open: the full moving-throat PDE, the fully dynamical inner-throat pole data, dissipative leakage and radiation reaction, spin couplings, and strong-field or higher-PN completions beyond the solved conservative scope.

That is the exact sense in which this appendix provides anchors rather than a fresh foundation. It records the formulas that the present 2.5PN audit is allowed to use, and it keeps visible the boundary between what the earlier papers already fixed and what the current paper is still trying to determine.